



Metal-Organic Frameworks (MOFs)

- from basic research to potential applications

Christoph Janiak

University of Düsseldorf, Germany

Email: janiak@uni-duesseldorf.de



Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis
6. Selected applications

MOFs: Prototypical examples

- HKUST-1
- MOF-5
- IR-MOF- n

- MIL-53
- Al-fumarate
- MIL-100
- MIL-101
- MIL-125

- UiO-66

- CAU-1
- CAU-10-H

- ZIF-8

MOF-terminology

Acronyms:

MOF = metal organic framework (Yaghi) - general, generic term
i.e., without implying any special metal or ligand.

Other terms more implications:

Yaghi-MOF: Zn^{2+} and di-carboxylate ligand (O. M. Yaghi)

MIL = Materials Institute Lavoisier/Material of Institute Lavoisier (G. Férey)
mostly $\text{M(III)} = \text{M}^{3+}$, di- or tri-carboxylate ligand, zeolite topology

PCP = porous coordination polymer (S. Kitagawa)
from the Kitagawa group, often dynamic, breathing MOFs

ZIF = zeolitic imidazolate frameworks

$\text{M} = \text{Zn}^{2+}$ and imidazolate ligands

MOF-terminology

and **university acronyms**, e.g.

CAU = Christians-Albrecht-University (Kiel, Germany) (N. Stock)

NU = Northwestern University (Michigan, USA) (J. T. Hupp, O. K. Farha)

DUT = Dresden University of Technology (Germany) (S. Kaskel)

UiO = University in Oslo (**M** = Zr^{4+} , K. P. Lillerud)

compare to zeolite acronyms:

ZSM-5 - Zeolite Socony Mobil – 5, patented by Mobil Oil Company in 1975

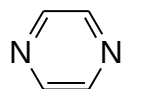
MCM = Mobil Crystalline Materials, produced at Mobil Corporation laboratories

MCM-41 etc. = Mobil Composition of Matter No. 41

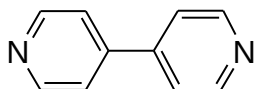
SBA-15 = Santa Barbara Amorphous type material

MOFs – mostly based on rigid linkers:

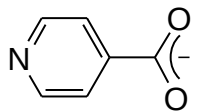
Examples of linear, trigonal and tetrahedral ligands



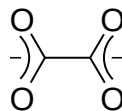
pyrazine



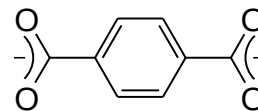
4,4'-bipyridine



isonicotinate



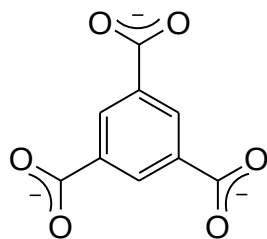
oxalate



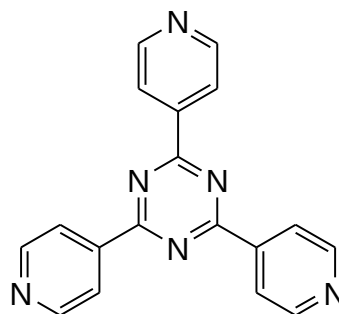
benzene-1,4-dicarboxylate,
terephthalate

linear rigid

(also
linear flexible
possible)

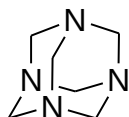


benzene-1,3,5-tricarboxylate,
trimesate

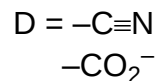
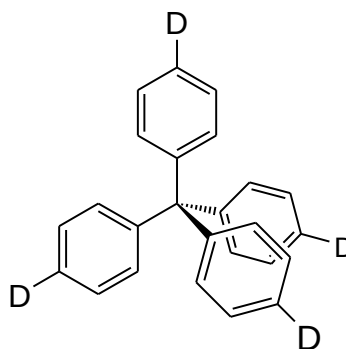


2,4,6-tris(4-pyridyl)-
1,3,5-triazine

trigonal



hexamethylene-
tetramine



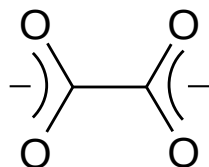
tetra(4-cyanophenyl)methane
tetra(4-carboxyphenyl)methane

tetrahedral

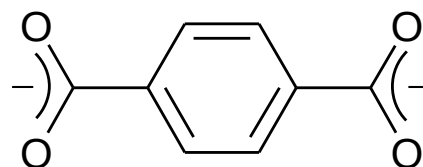
Oxtoby et al., *PNAS* **2002**, 99, 4905; Kitagawa, Masaoka, *Coord. Chem. Rev.* **2003**, 246, 73;
Barnett, Champness, *Coord. Chem. Rev.* **2003**, 246, 145; Zheng et al., *Coord. Chem. Rev.* **2003**, 246, 185;
Ye et al., *Coord. Chem. Rev.* **2005**, 249, 545; Zhang, Chen, *Chem. Commun.* **2006**, 1689

MOFs – mostly based on rigid linkers:

MOFs based on di- and tri-carboxylate ligands:

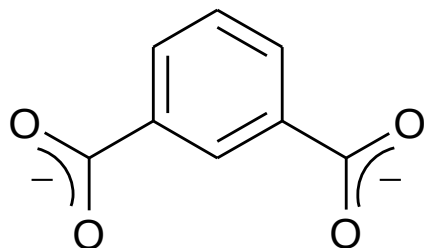


oxalate
ox

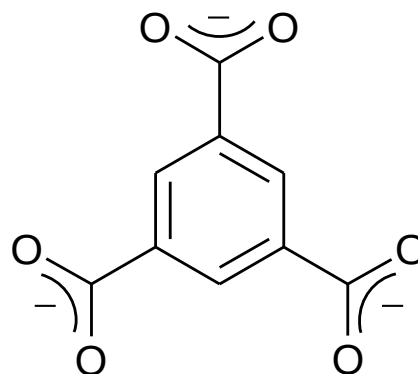


benzene-1,4-dicarboxylate,
terephthalate
1,4-bdc (or simply bdc)

- rigid ligands



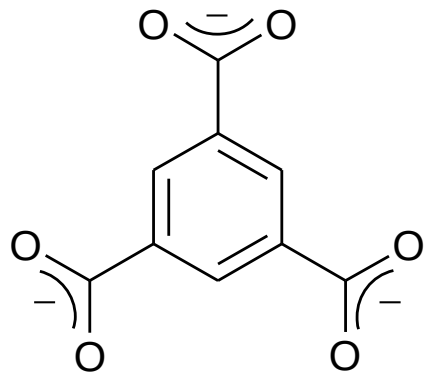
benzene-1,3-dicarboxylate,
isophthalate
1,3-bdc



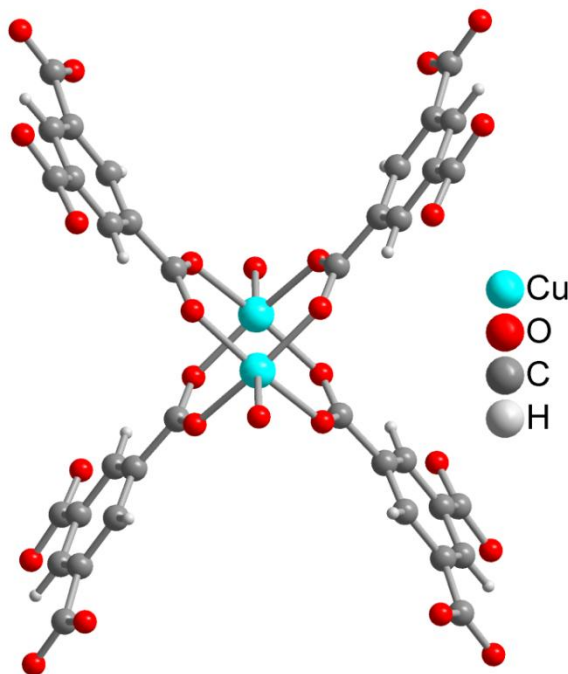
benzene-1,3,5-tricarboxylate,
trimesate
btc

MOF example:
 $3\text{D}[\text{Cu}_3(\text{btc})_2(\text{H}_2\text{O})_3]$ (also called **HKUST-1** or **Cu-btc**)

HKUST = Hongkong University of Science and Technology



benzene-1,3,5-tricarboxylate,
trimesate
btc

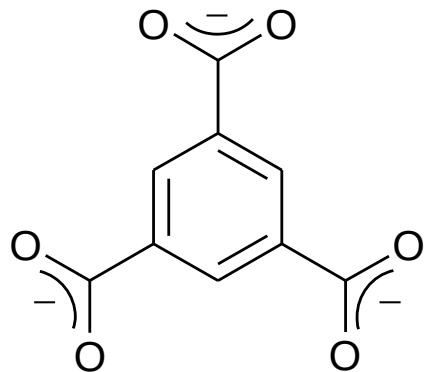


One of the most studied MOFs is copper trimesate, (“HKUST-1”) because of easy formation, high porosity and known properties associated with this compound.

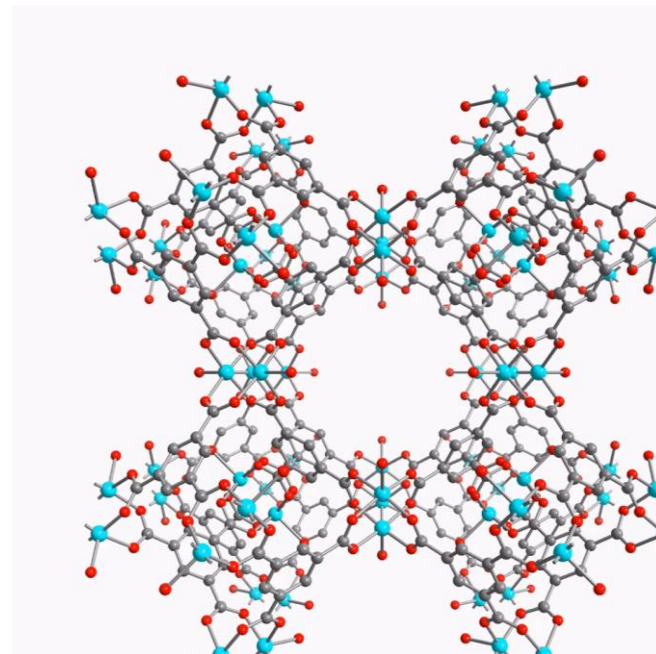
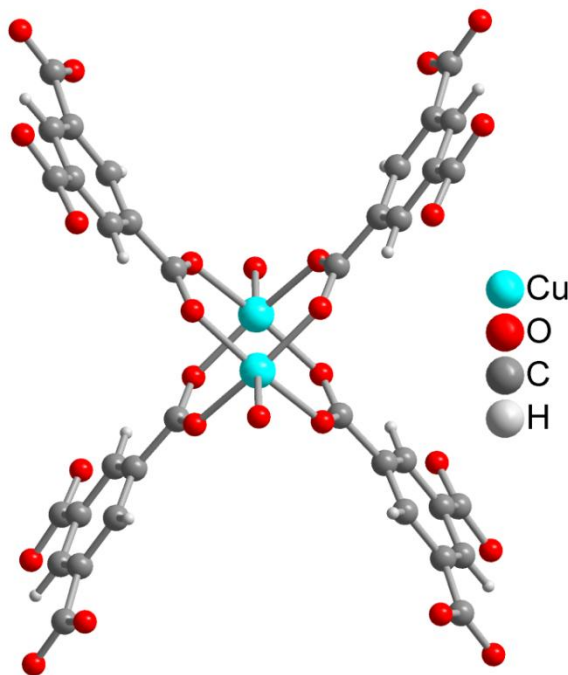
$3\text{D}[\text{Cu}_3(\text{btc})_2(\text{H}_2\text{O})_3]$ (also called HKUST-1 or Cu-btc) is one of the first 3-dimensional porous coordination polymers that has been studied intensively. It contains $\{\text{Cu}_2\}$ units coordinated by four carboxylate groups in the well-known **paddle-wheel structure** of copper acetate.

MOF example:
 $3\text{D}-[\text{Cu}_3(\text{btc})_2(\text{H}_2\text{O})_3]$ (also called **HKUST-1** or **Cu-btc**)

HKUST = Hongkong University of Science and Technology

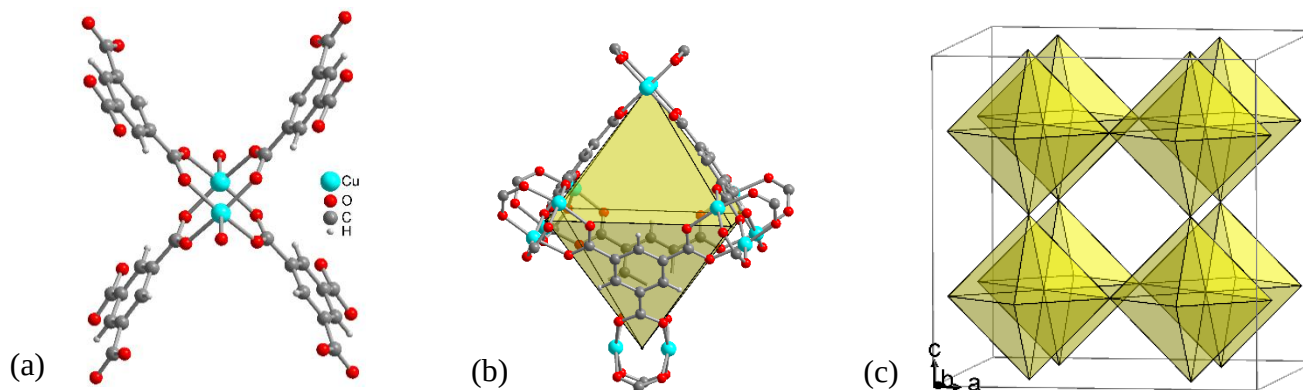


benzene-1,3,5-tricarboxylate,
trimesate
btc

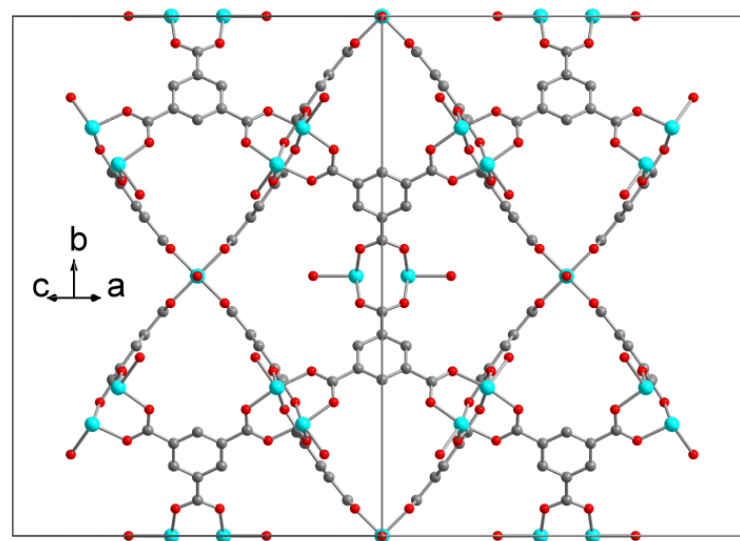
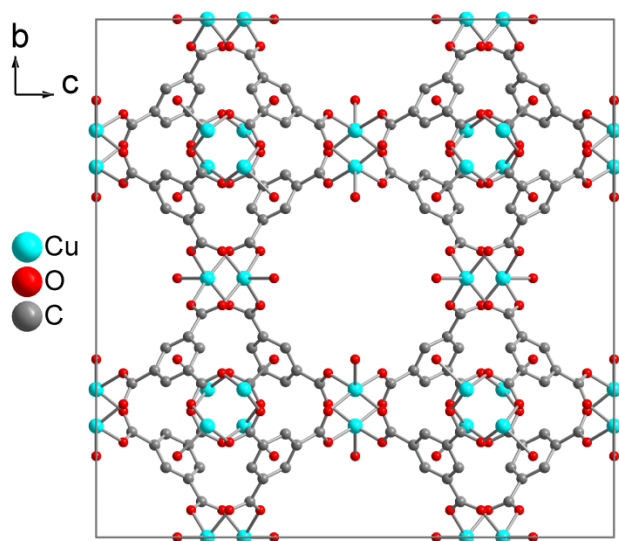


$\{\text{Cu}_2(\text{btc})_4\}$ building unit and two views of the packing diagram with the **cubic unit cell** of $3\text{D}-[\text{Cu}_3(\text{btc})_2(\text{H}_2\text{O})_3] \cdot \sim 10\text{H}_2\text{O}$ (HKUST-1, $a = 26.34 \text{ \AA}$) to illustrate the zeolite analogy of this highly symmetric and porous framework. The disordered water molecules in the pores are not shown, nor are the H atoms on the aqua ligands and on carbon in the packing diagrams. The different objects are not drawn to scale.

MOF example:
 $3\text{D}-[\text{Cu}_3(\text{btc})_2(\text{H}_2\text{O})_3]$ (also called **HKUST-1** or **Cu-btc**)



MOF example:
3D-[Cu₃(btc)₂(H₂O)₃] (also called **HKUST-1** or **Cu-btc**)



The 3D-coordination polymer [Cu₃(btc)₂(H₂O)₃] crystallizes with formation of a highly porous cubic structure with a complicated 3D network of channels. Along the *a*-axis there are large square channels of 9x9 Å (btc = benzene-1,3,5-tricarboxylate).

Cu-btc is **stable up to 240 °C**. Both the water of crystallization and the **aqua ligands** coordinating the copper atoms **can be removed** thermally without loss of the structural integrity and exchanged by pyridine ligands.

BET surface $S_{BET} = 1500 \text{ m}^2/\text{g}$

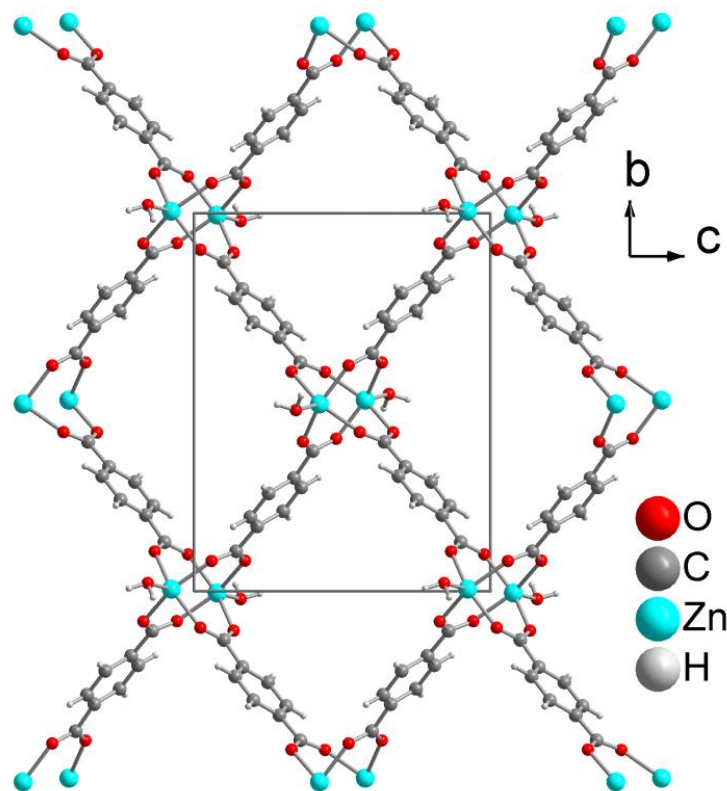
Synthesis

3D-[Cu₃(btc)₂(H₂O)₃] (also called HKUST-1 or Cu-btc)

Place copper nitrate trihydrate (0.875 g), 1,3,5-benzenetricarboxylic acid (0.42 g) and 25 mL of ethanol in a 50 mL round-bottom flask equipped with a magnetic stirring bar and an air-condenser. Heat the mixture at reflux under thorough stirring using an oil bath for 48 h and after this time, cease the stirring of the solution, and let the bath cool down to room temperature. Filter-off the formed precipitate and wash it with water (2 x 10 mL) followed by ethanol (2 x 10 mL). Dry the solid by heating *in vacuo* of a membrane pump (10 Torr) at a temperature of 80 °C for 1 h.

further MOF with paddle-wheel structure:
 $2\text{D}[\text{Zn}_2(\text{bdc})_2(\text{H}_2\text{O})_2]$

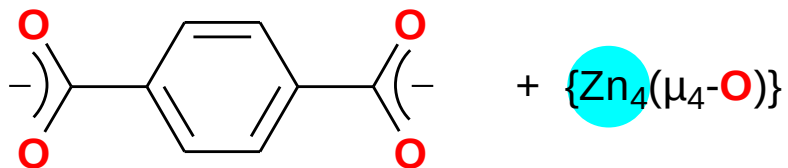
The paddle-wheel coordination in a MOF is also found for zinc with a 2D-framework of $\{\text{Zn}_2\}$ - and terephthalate units:



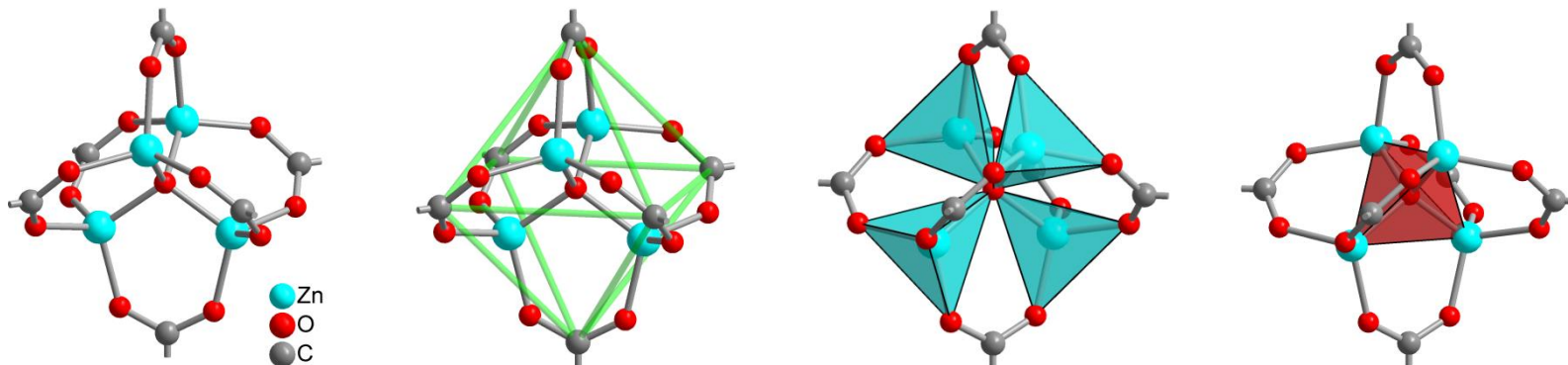
Packing diagram for 2D-
 $[\text{Zn}_2(\text{bdc})_2(\text{H}_2\text{O})_2]$ (bdc = benzene-
1,4-dicarboxylate, terephthalate).

Dimethylformamide guest
molecules in the channels are
omitted for clarity.

MOF example:
3D-[Zn₄O(bdc)₃] or **MOF-5** or **IRMOF-1**



benzene-1,4-dicarboxylate,
terephthalate, bdc

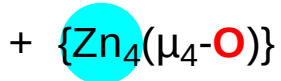
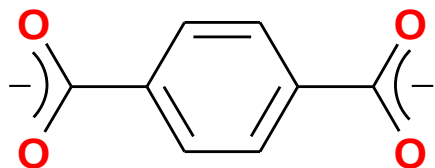


MOF-5, [Zn₄O(BDC)₃] BDC = benzene-1,4-dicarboxylate, terephthalate) is **one of the best known and prototypical MOFs**, belonging to the IRMOF-*n* series. MOF-5 is also named IRMOF-1. The structure of MOF-5 with the terephthalate linker (BDC) is probably one of the most widely recognized. Activated MOF-5 has a high **surface area (3000m²/g)** and **high thermal stability (up to 400 °C)**.

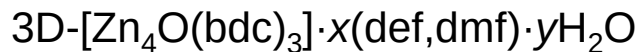
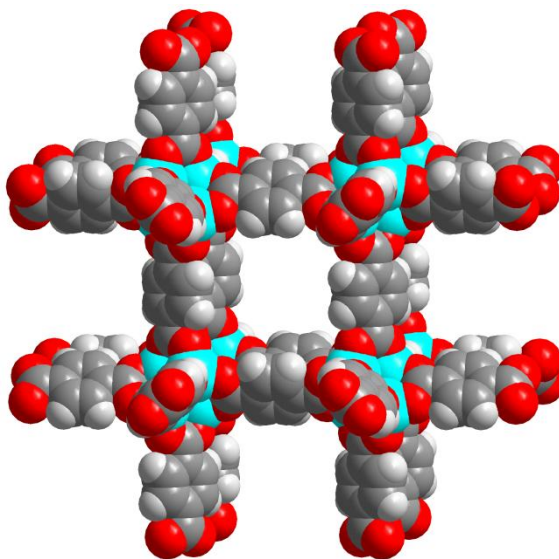
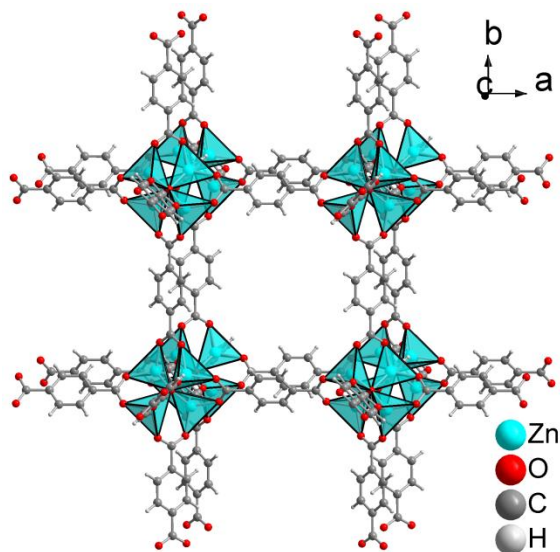
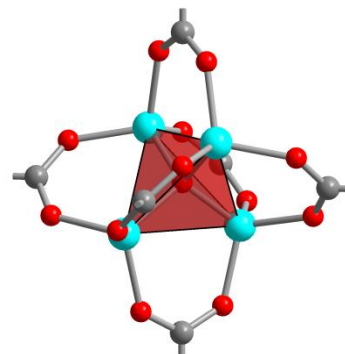
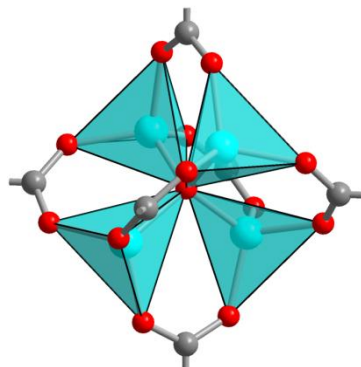
Six carboxylate groups span the six edges of the {Zn₄O} secondary building unit, a tetrahedron (depicted at left in polyhedral presentation) in an octahedral fashion

MOF examples:

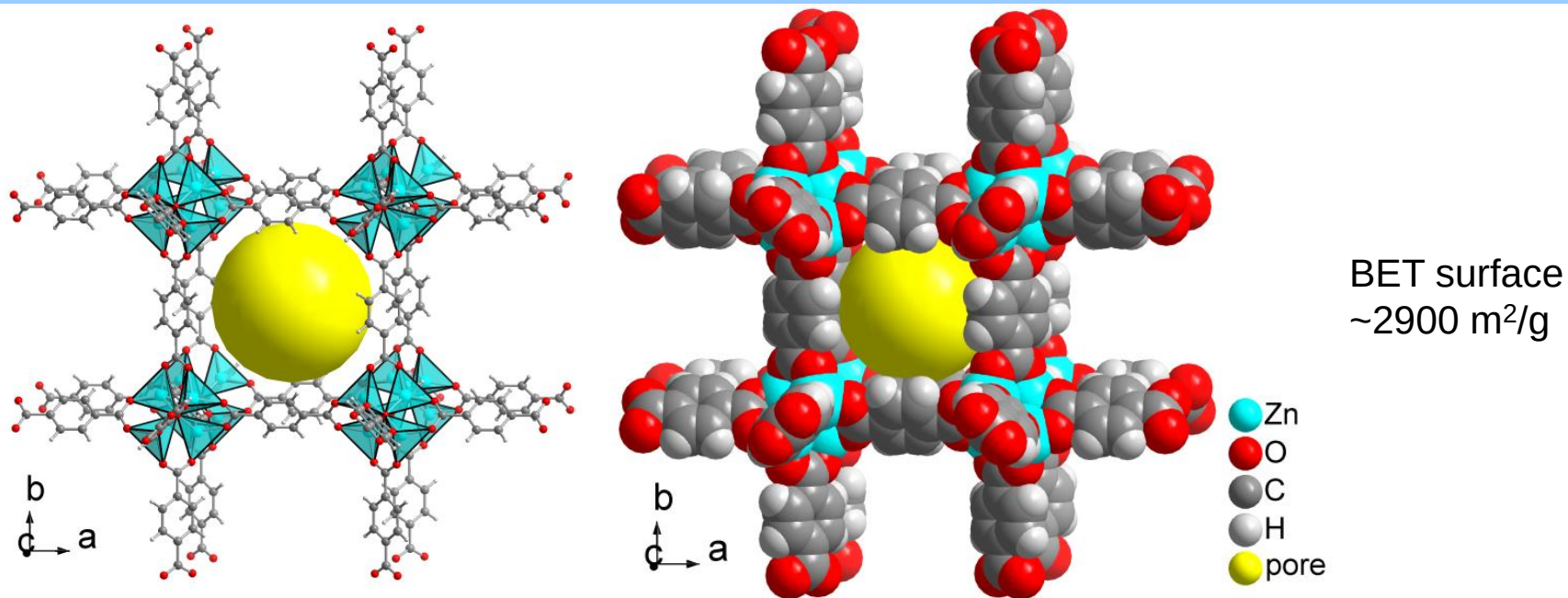
3D-[Zn₄O(bdc)₃] or **MOF-5** or **IRMOF-1**



benzene-1,4-dicarboxylate,
terephthalate, bdc



MOF examples: 3D-[Zn₄O(bdc)₃] or MOF-5



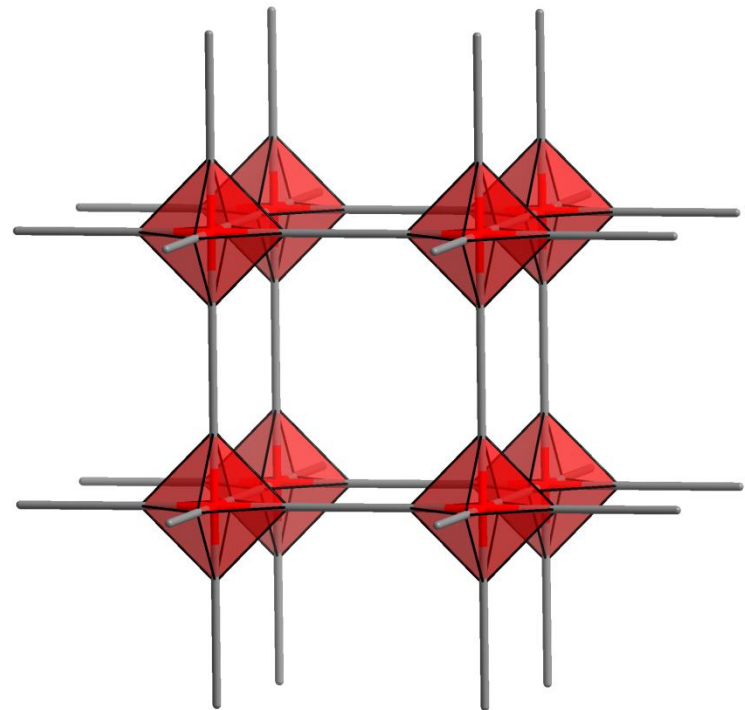
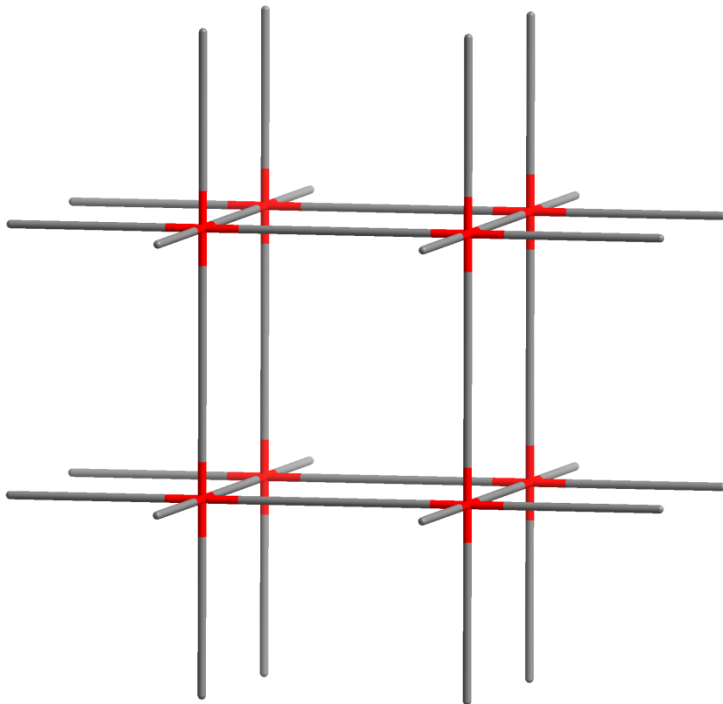
Section of the crystal packing diagram of MOF-5 (IRMOF-1). Six carboxylate groups span the six edges of the {Zn₄O} secondary building unit, a tetrahedron (depicted at left in polyhedral presentation) in an octahedral fashion to yield the **three-dimensional primitive cubic pcu-a or cab₁ network** shown here as ball-and-stick and as space-filling representation. The **yellow sphere with a radius of 5 Å** takes into account the van-der-Waals radii of the framework walls.

MOF examples: 3D-[Zn₄O(bdc)₃] or MOF-5

three-dimensional **primitive cubic pcu-a or cab network**.

The **augmented "-a" notation** refers to a net where the simple point vertices of the original pcu net are replaced by **vertex figures** (polyhedron or polygon), here by an octahedron.

The B atoms in CaB₆ form such octahedra which as vertices are connected to a pcu-a net. Hence, cab is an alternative symbol for pcu-a.



Synthesis:

3D-[Zn₄O(bdc)₃] or MOF-5

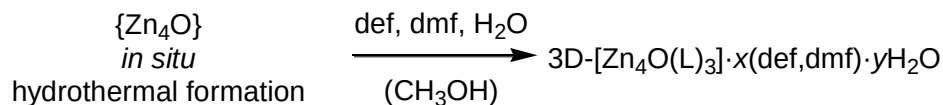
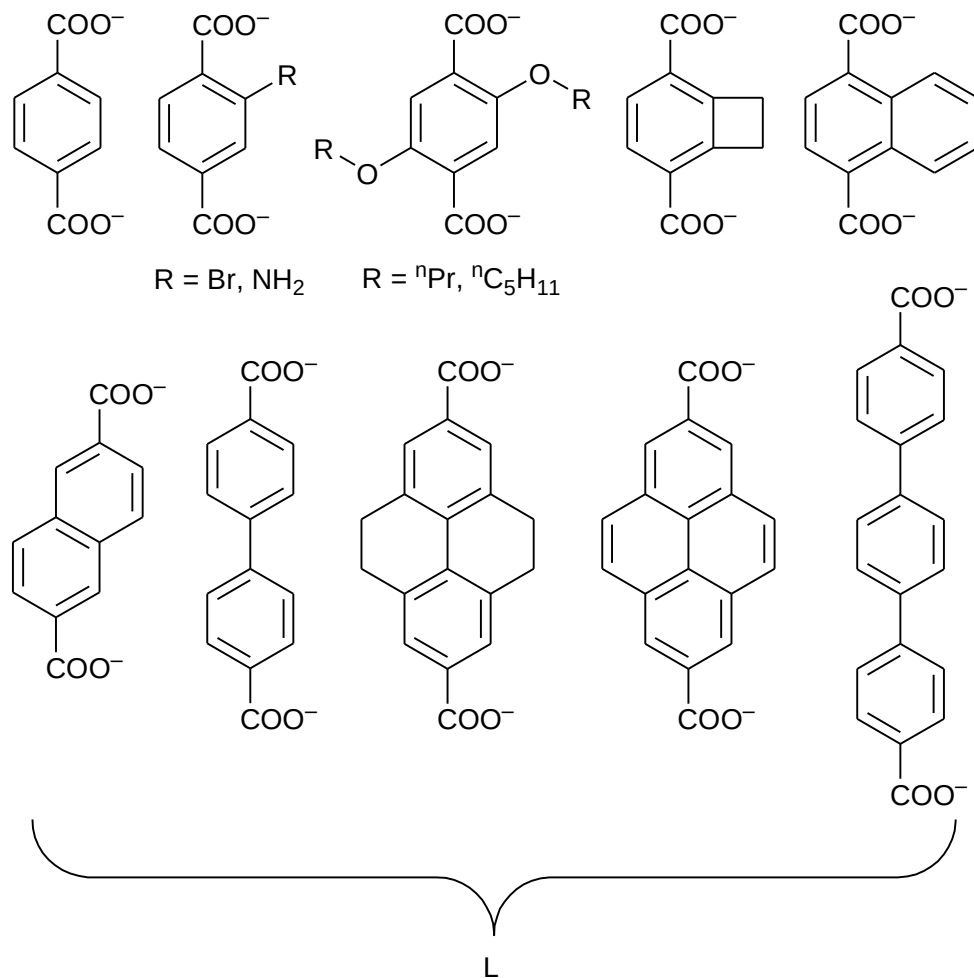
Place zinc nitrate hexahydrate (0.36 g, 1.22 mmol), terephthalic acid (0.066 g, 0.40 mmol), and *N,N*-diethylformamide (DEF; 10 mL) into a dried and inert gas purged 20 mL Schlenk flask.² After final purging, close the flask, sonicate the solution for 10 min and place it in an oven set to 95 °C for at least 10 h.

Large colorless cubic crystals should grow on the walls of the flask during the specified time. Remove the vessel from the oven, and allow the reaction mixture to cool down to r.t. Look at your crystals under the microscope. Filtrate off the mother liquor under inert conditions and add 4 mL of DEF (perform all the operations under inert gas). Close the flask and sonicate the contents for 15 min using an ultrasonic bath. Repeat the washing procedure with DEF once more. Similarly, wash the sample four times with CHCl₃ (8 mL per wash). After the final CHCl₃ wash, draw off the solvent and remove the residual CHCl₃ in vacuo. Store the material under inert gas.

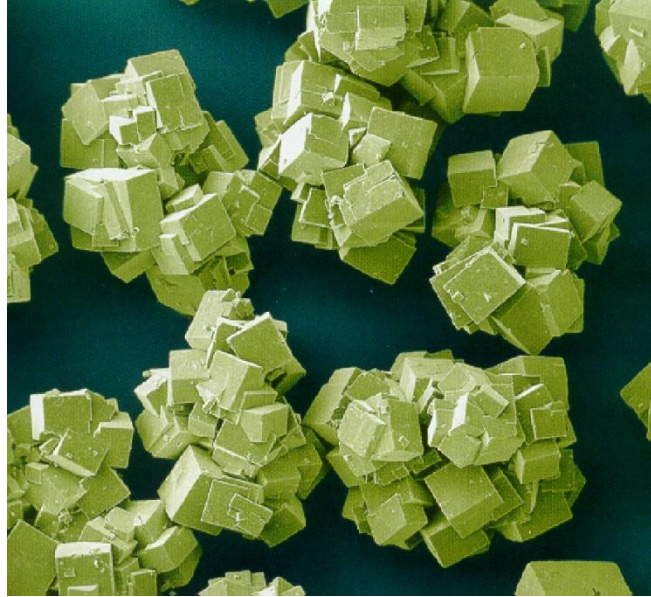
MOF examples: IR-MOF-*n* series

Based on hydrothermally obtained tetranuclear and tetrahedral $\{Zn_4O\}$ building units a series of **cubic iso-reticular** MOF structures, **IR-MOF-*n*** (*n* = 1-16), with larger tunable pore sizes ranging from 3.8 to 29 Å, specific pore volumes up to 1 cm³/g and, thus, very high porosity were synthesized. These materials were obtained using the geometric concept of **iso-reticular synthesis** with coordinated metal ions in the so-called secondary building unit (SBU) $\{Zn_4O\}$ and the following organic carboxylate linkers:

M. Eddaoudi, J. Kim, N. Rosi, D. Vodak, J. Wachter, M. O'Keeffe and O. M. Yaghi, *Science*, **2002**, 295, 469-472.



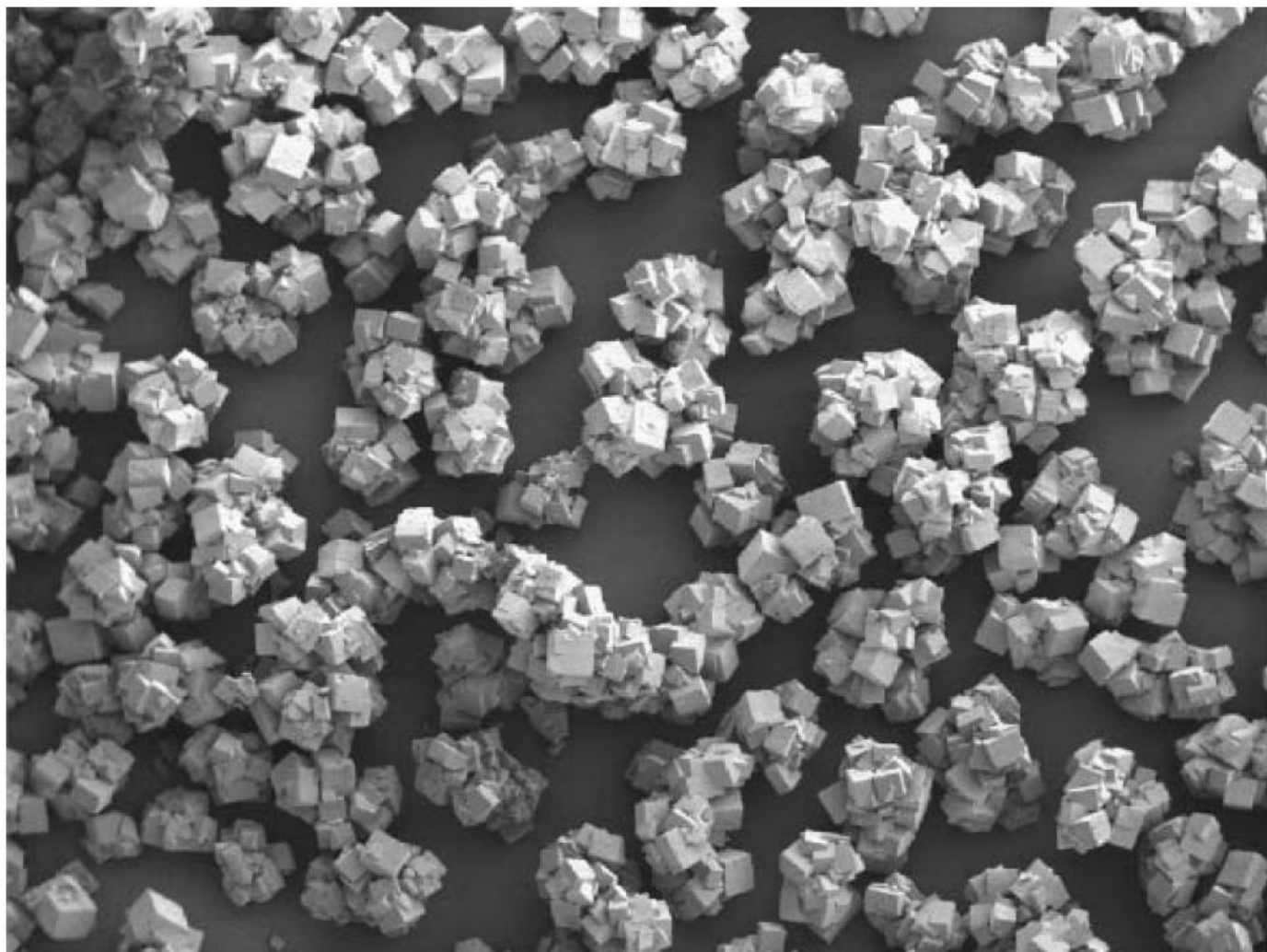
MOF examples: IR-MOF-*n*



SEM IR-MOF-1 = MOF5 picture from BASF AG

Nanocubes of industrially prepared MOFs, also noted in connotation to nanomaterials and infinite coordination polymer particles (ICPs).

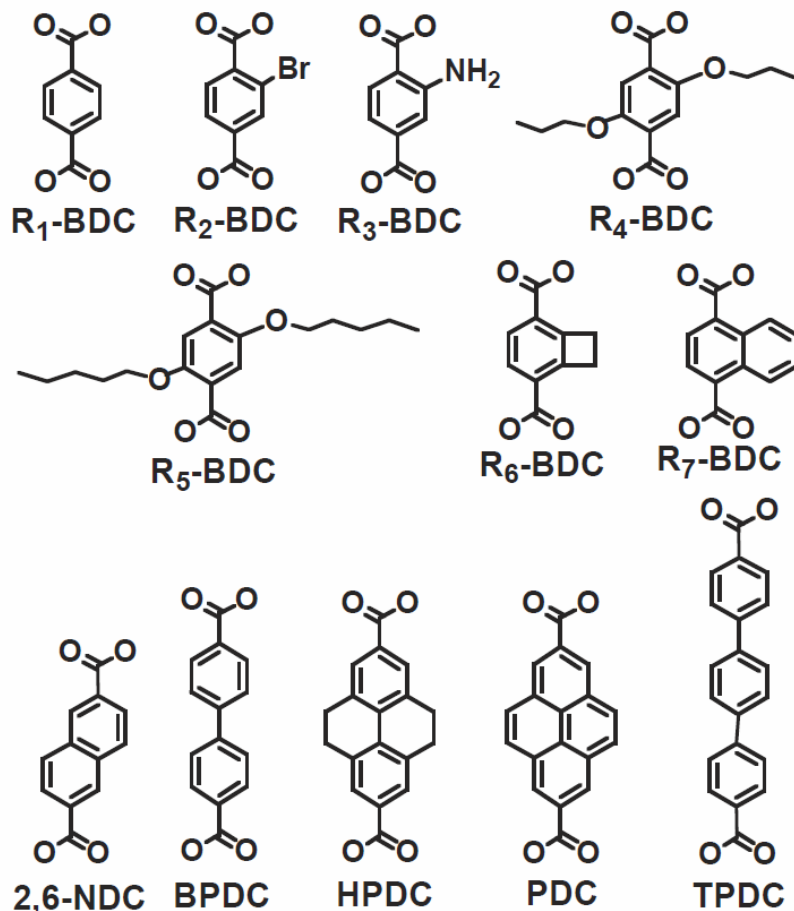
MOF examples:
IR-MOF-*n*



 1 mm

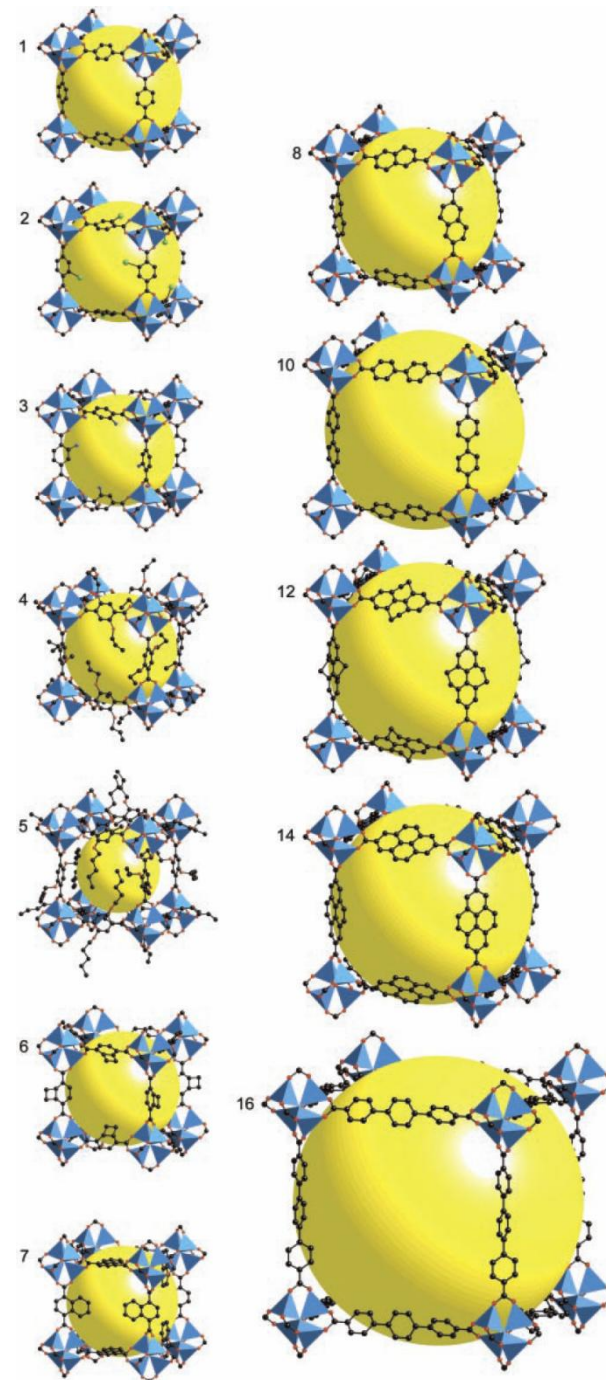
SEM-picture of large MOF-5 crystals from laboratory preparation (scale bar: 1 mm)

Isoreticular MOFs, **IRMOF-*n***

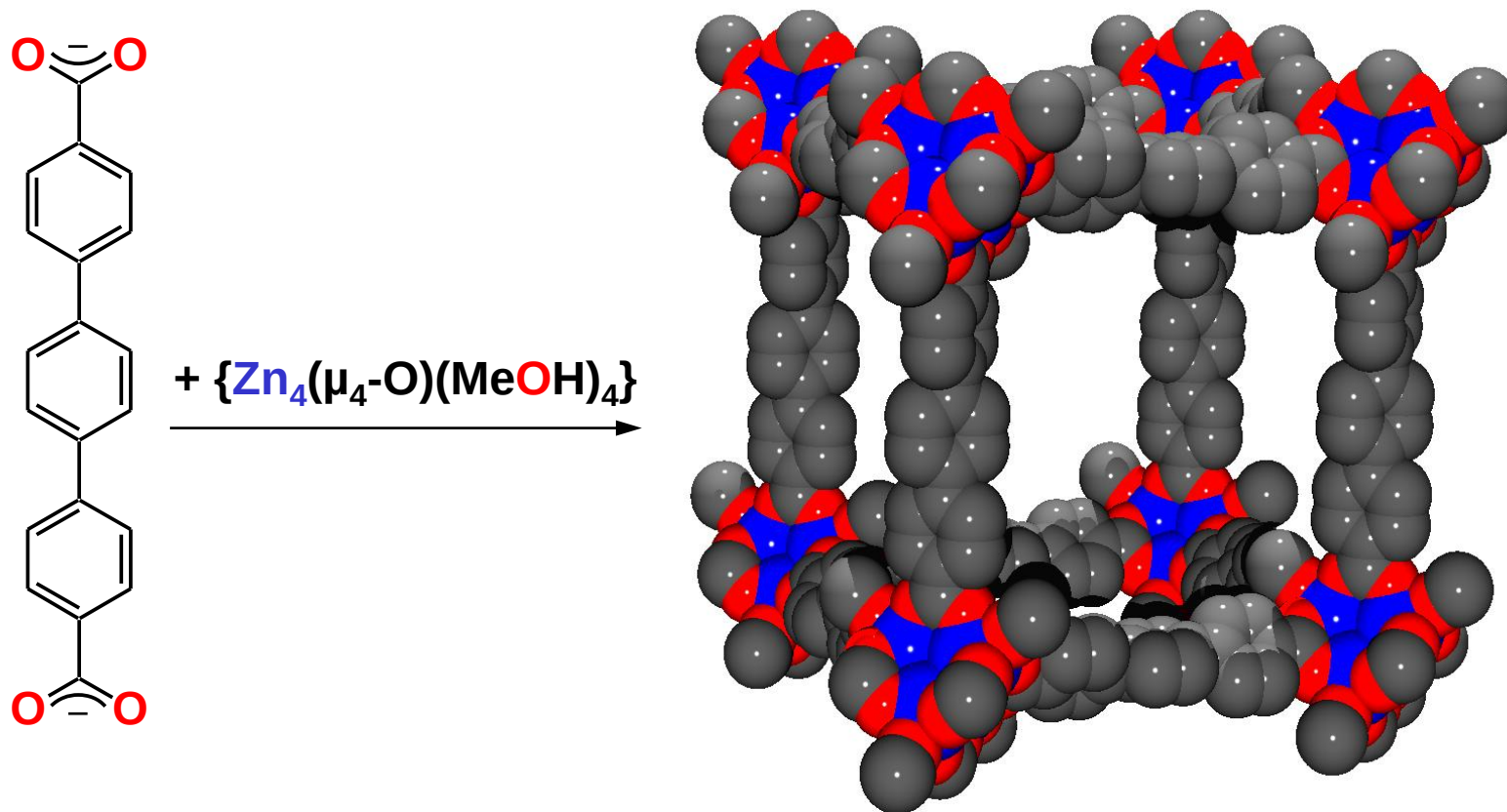


(IRMOF-1 = MOF-5)

M. Eddaoudi, J. Kim, N. Rosi, D. Vodak, J. Wachter, M. O'Keeffe and O. M. Yaghi, *Science*, **2002**, 295, 469-472.



MOF examples: IR-MOF-*n*



Building blocks for **IR-MOF-15**, **-16** and space-filling packing diagram of the resulting 3D- $[\text{Zn}_4\text{O}(\text{L})_3]$ (differing in space group and solvent-guest molecules). IR-MOF-15, -16 has **91% pore volume** in the absence of the solvent guest molecules (not shown). The distances between the centers of the $\{\text{Zn}_4\text{O}\}$ nodes are 21.5 Å. Not shown is the **crystallographically induced disorder** where the linkers and the nodes are rotated by 90° along the linker axis. In the crystal structure of IRMOF-16 the Zn atoms are five-coordinated with methanol as a fifth ligand

M. Eddaoudi, J. Kim, N. Rosi, D. Vodak, J. Wachter, M. O'Keeffe, O. M. Yaghi, *Science*, **2002**, 295, 469.

N. L. Rosi, J. Eckert, M. Eddaoudi, D. T. Vodak, J. Kim, M. O'Keeffe, O. M. Yaghi, *Science*, **2003**, 300, 1127

MOF-terminology

Acronyms:

MOF = metal organic framework (Yaghi) - general, generic term
i.e., without implying any special metal or ligand.

Other terms more implications:

Yaghi-MOF: Zn^{2+} and di-carboxylate ligand (O. M. Yaghi)

MIL = Materials Institute Lavoisier/Material of Institute Lavoisier (G. Férey)
mostly $\text{M(III)} = \text{M}^{3+}$, di- or tri-carboxylate ligand, zeolite topology

PCP = porous coordination polymer (S. Kitagawa)
from the Kitagawa group, often dynamic, breathing MOFs

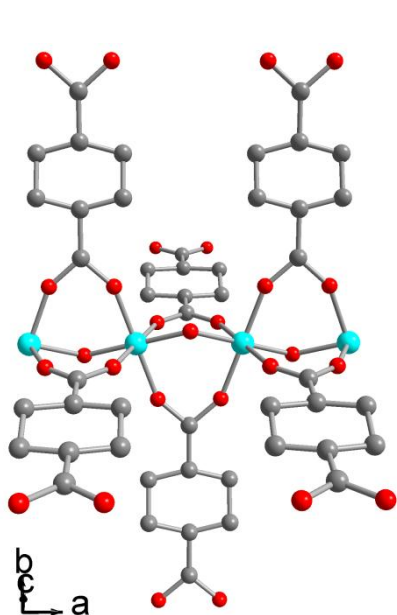
ZIF = zeolitic imidazolate frameworks

$\text{M} = \text{Zn}^{2+}$ and imidazolate ligands

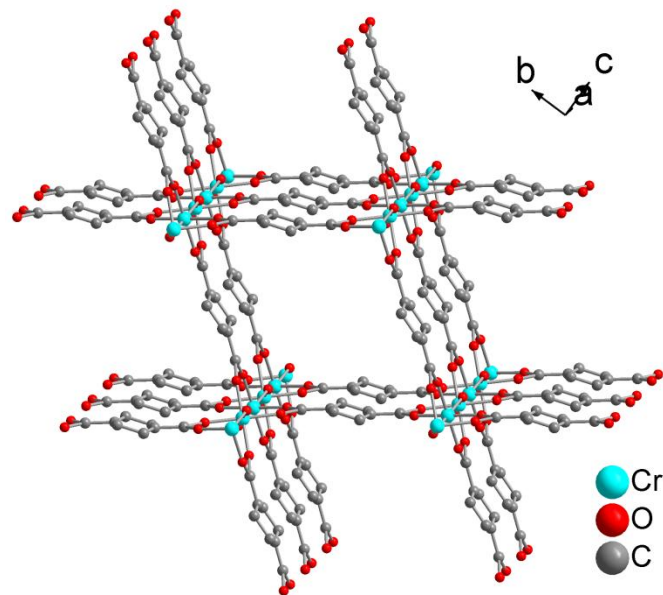
MIL example:

$3\text{D}[\text{M}(\mu_4\text{-bdc})(\mu\text{-OH})]$ (M = Cr, Fe, Al) or MIL-53

The important MIL-type structure $[\text{M}(\text{bdc})(\mu\text{-OH})]$, MIL-53 can contain M = Al(III), Cr(III) or Fe(III) with benzene-1,4-dicarboxylate (bdc, terephthalate) as the linker. The **framework** of MIL-53 is a **very flexible**, **'breathing'-type network**, that is, it can assume different shapes and porosities depending on the guest's presence or absence interaction



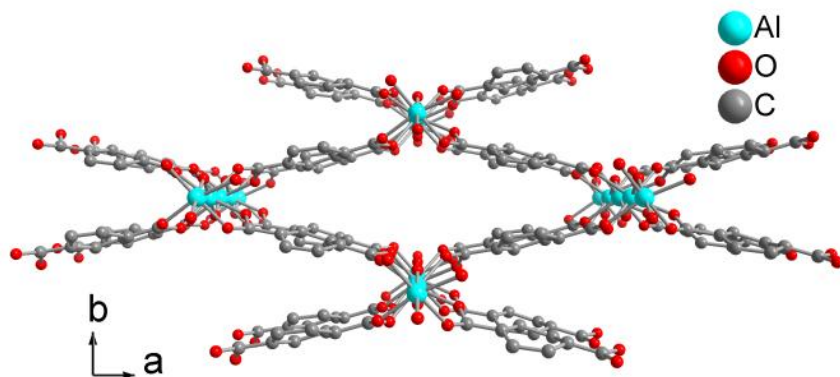
Section of the bdc- and OH⁻ bridged Al-strand



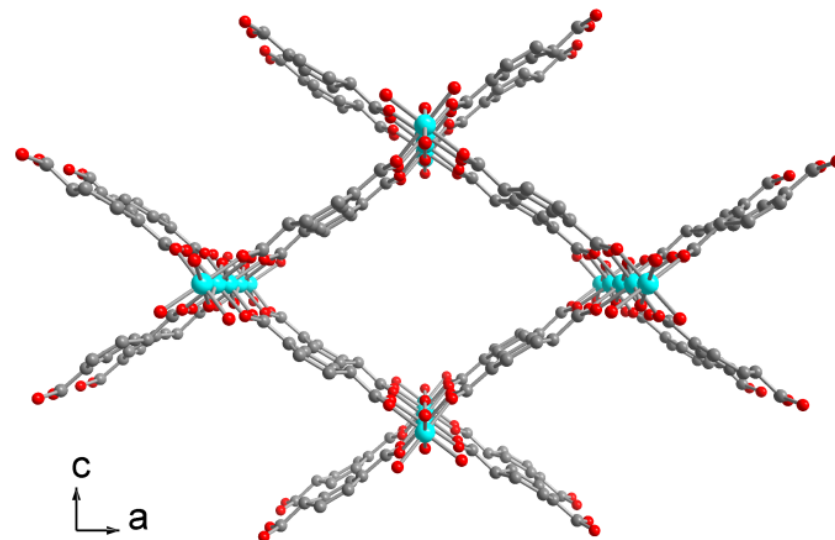
3D network

Cr: C. Serre, F. Millange, C. Thouvenot, M. Nogues, G. Marsolier, D. Louer and G. Férey, *J. Am. Chem. Soc.*, **2002**, 124, 13519.
Fe: F. Millange, N. Guillou, R. I. Walton, J.-M. Greneche, I. Margiolaki and G. Férey, *Chem. Commun.*, **2008**, 4732.
Al: L. Alaerts, M. Maes, L. Giebeler, P. A. Jacobs, J. A. Martens, J. F. M. Denayer, C. E. A. Kirschhock and D. E. De Vos, *J. Am. Chem. Soc.*, **2008**, 130, 14170-14178.

MIL example:
 $3\text{D}[\text{M}(\mu_4\text{-bdc})(\mu\text{-OH})]$ (M = Cr, Fe, Al) or MIL-53



MIL-53/t



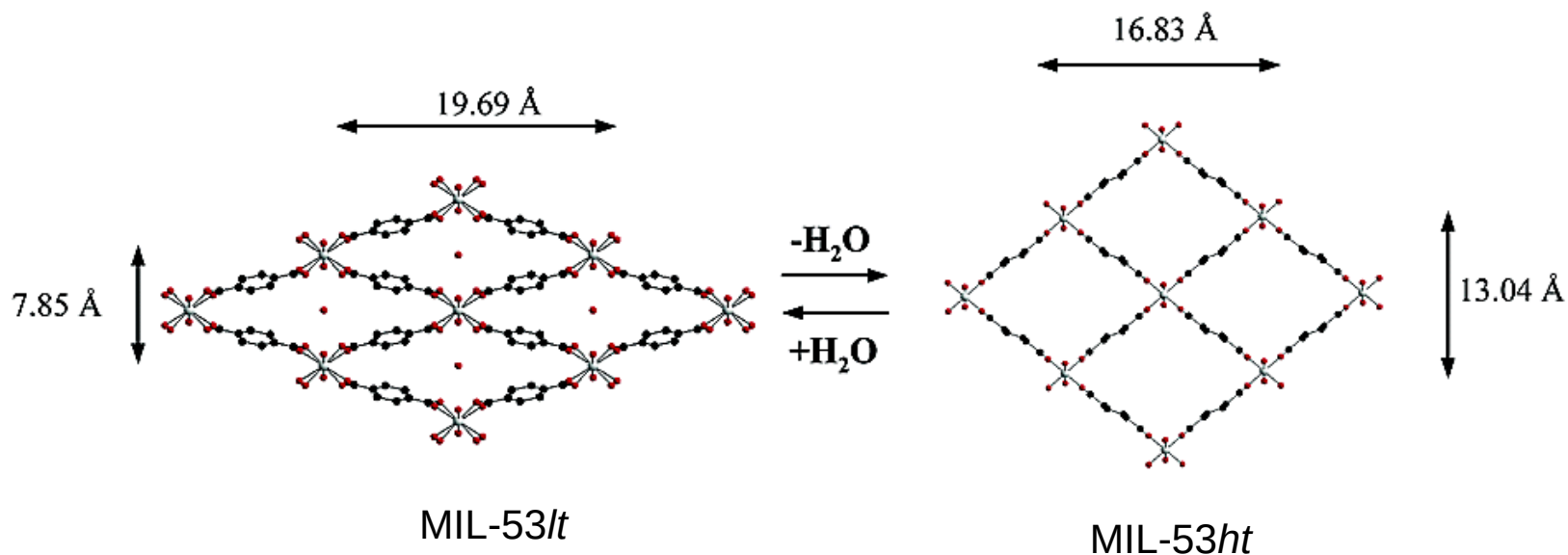
MIL-53as = MIL-53ht

Section of the packing diagram of $[\text{Al}(\text{bdc})(\mu\text{-OH})]$, MIL-53 with a flexible, 'breathing' network adapting to guest molecules. Each benzene-1,4-dicarboxylate ligand bridges between four Al atoms. The hydroxido-bridging occurs along the metal chains in the *b* direction. The channels can contain **guest molecules in the as-synthesized high-temperature (*ht*) structure** (MIL-53as) (right, guest molecules not shown) **or be empty** after a thermal guest removal in the identical activated high-temperature, large pore structure MIL-53ht. Cooling down to room temperature with adsorption of water from air then **transforms the structure into the low-temperature, narrow pore form MIL-53/t** (left).

MIL example:
 $3\text{D}[\text{M}(\mu_4\text{-bdc})(\mu\text{-OH})]$ (M = Cr, Fe, Al) or **MIL-53**

When MIL-53 $_{ht}$ is subjected to cooling and exposed to polar molecules, the channels shrink:

free pores in the ht -network have a diameter of 8.5 Å,
 Langmuir surface 1590 m² g⁻¹



representation of the "breathing" effect in MIL-53 through water desorption and adsorption

Synthesis:

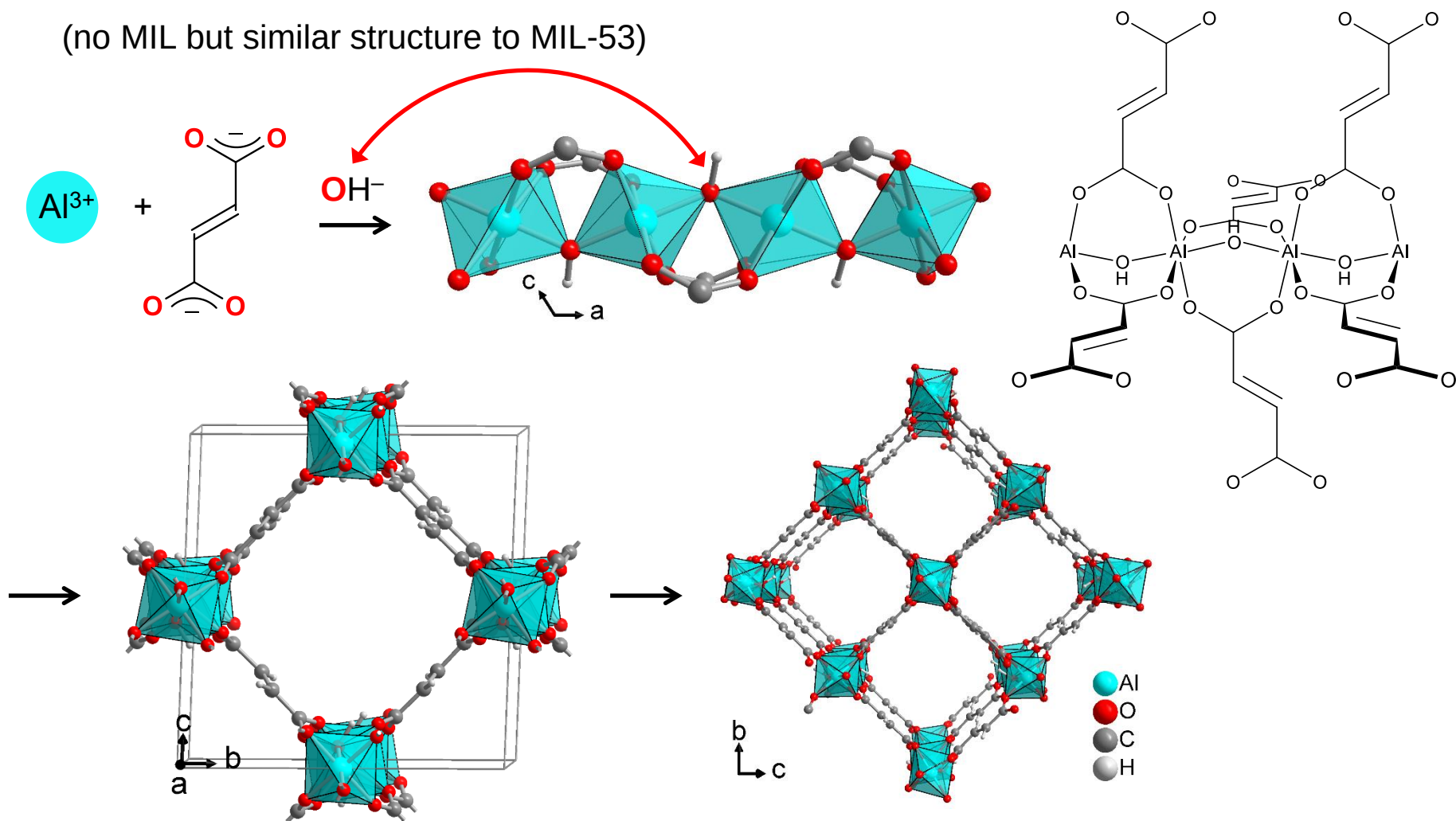
3D-[M(μ_4 -bdc)(μ -OH)] (M = Cr, Fe, Al) or MIL-53

Al(NO₃)₃ · 9 H₂O (375.1 mg, 1.00 mmol) and terephthalic acid (166 mg, 1.00 mmol) are dissolved in 5 mL of methanol and the solution is briefly homogenized by shaking. Then the solution is placed in the Teflon-liner of a 30 mL stainless-steel autoclave and heated to 125 °C for 5 h.

The product is poured from the Teflon-liner with additional dest. water. After evaporation of methanol, the product is separated by filtration and washed several times with dest. water.

MOF example: 3D-[Al(OH)(O₂C-CH=CH-CO₂)], **aluminium fumarate**

(no MIL but similar structure to MIL-53)



Patented and reported by BASF SE: Basolite™ A520

Structure: E. Alvarez, N. Guillou, C. Martineau, B.t Bueken, B. Van de Voorde, C. Le Guillouzer, P. Fabry, F. Nouar, F. Taulelle, D. de Vos, J.-S. Chang, K. H. Cho, N. Ramsahye, T. Devic, M. Daturi, G. Maurin, C. Serre, *Angew. Chem. Int. Ed.* **2015**, 54, 3664-3668

Synthesis:

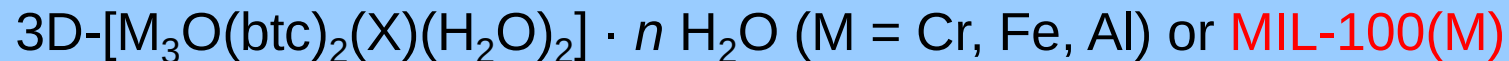
$3\text{D-}[\text{Al}(\text{OH})(\text{O}_2\text{C-CH=CH-CO}_2)],$ aluminium fumarate

5.7 mL deionized water, 0.285 g (7.14 mmol) sodium hydroxide and 0.387 g (3.33 mmol) fumaric acid were placed into a 20 mL beaker. A solution of 1.113 g aluminium sulfate octadecahydrate (1.670 mmol) in 4.8 mL deionized water was heated up to 60 °C for one hour. After that, the solution from step 1, which contains the ligand, was added with a dropping funnel to the solution of aluminium sulfate in water over a period of one hour and the complete solution was stirred for two further hours at 60 °C. The formed suspension is transferred into two centrifuge tubes and the formed precipitate is separated from the clear solution by centrifugation. The precipitate is washed two times with deionized water and then overnight in deionized water in a 50 mL round bottom flask.

The precipitate is separated again by centrifugation and dried overnight at 130 °C in a vacuum drying oven.

With DMF as solvent because of alleged better reaction control: In a 1000 mL round bottom flask, 26.9 g (108 mmol) of aluminium chloride hexahydrate (p.a., Merck) and 15.4 g (133 mmol) of fumaric acid were dissolved in 550 mL of freshly distilled dimethyl formamide. The turbid, yellowish mixture was stirred at 130 °C for 4 d. After cooling, the snow-white precipitate was filtered off and stirred twice in redist. acetone (1 h and 24 h, 200 mL each), and twice in redist. ethanol (1 h and 24 h, 200 mL each). The solid was then dried in air at 80 °C. Yield: 15.4 g (90 % based on Al).

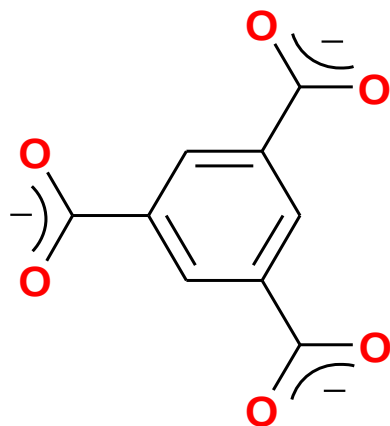
MIL example:



X= OH, F, btc = benzene-1,3,5-tricarboxylate, trimesate.

The porosity of MIL-100 originates from both 25 Å and 29 Å mesopores, which are accessible via 5.5 Å and 8.6 Å windows, respectively.

Both MIL-100s form under quite harsh synthesis conditions (Al: pH = 0.6, T = 210 °C, and Fe: pH < 1, T = 150°C).



benzene-1,3,5-tricarboxylate,
trimesate, btc



Cr: G. Férey, C. Serre, C. Mellot-Draznieks, F. Millange, S. Surblé, J. Dutour and I. Margiolaki, *Angew. Chem. Int. Ed.*, **2004**, 43, 6296-6301.

Fe: P. Horcajada, S. Surblé, C. Serre, D.-Y. Hong, Y.-K. Seo, J.-S. Chang, J.-M. Grenèche, I. Margiolaki and G. Férey, *Chem. Commun.*, **2007**, 2820-2822.

Al: C. Volkringer, D. Popov, T. Loiseau, G. Férey, M. Burghammer, C. Riekell, M. Haouas and F. Taulelle, *Chem. Mater.*, **2009**, 21, 5695-5697.

MIL example:

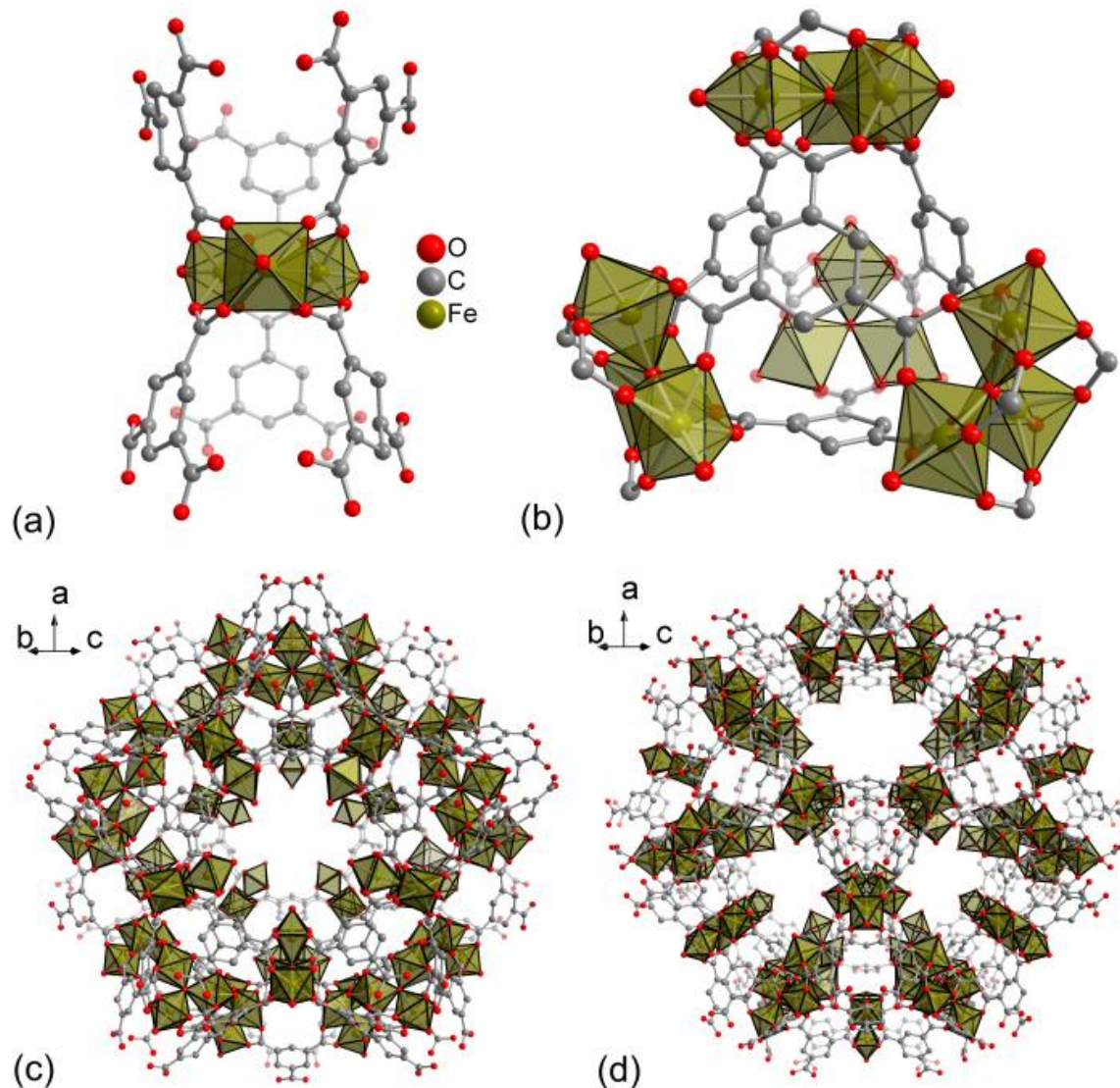


- (a) SBU,
- (b) supertetrahedra,
- (c) small S cage and
- (d) large L cage in
MIL-100(Fe)

(different objects are not drawn to scale).

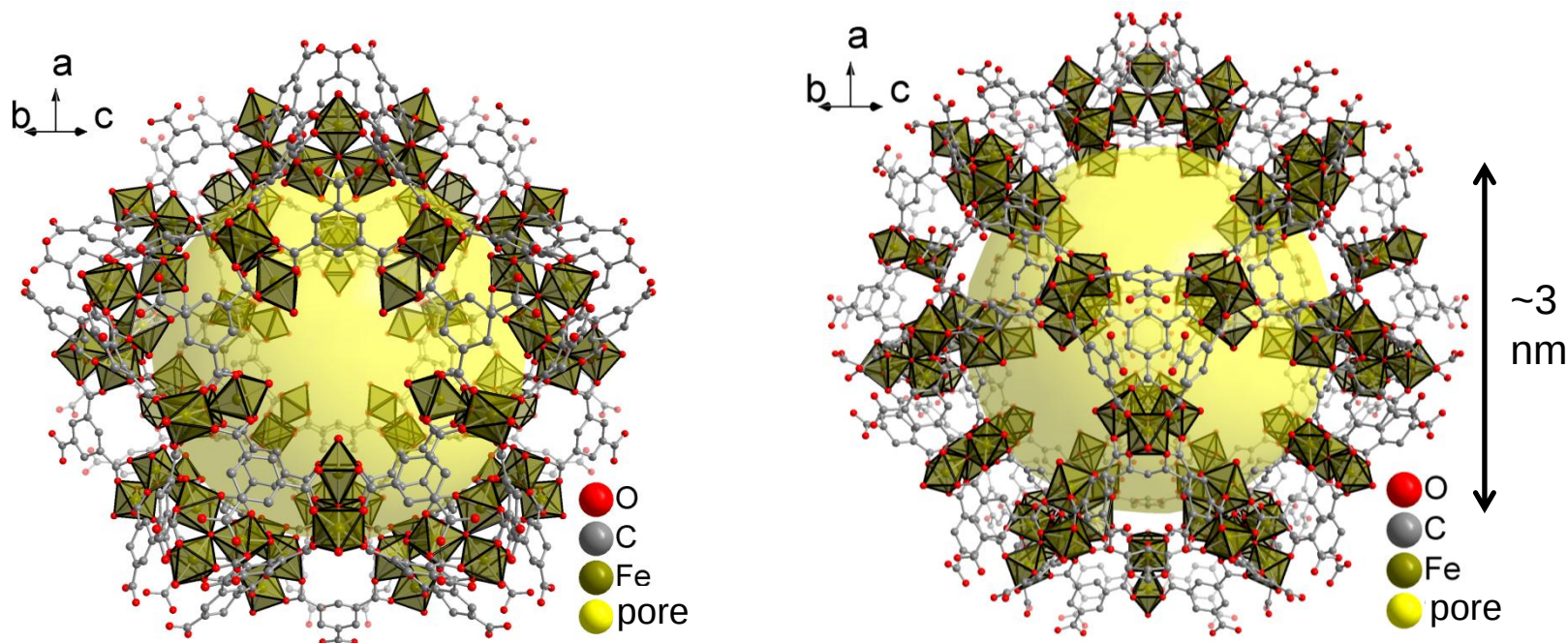
Hydrogen atoms and solvent molecules of crystallization are not shown.

MIL-100s are isostructural.



MIL example:
 $3\text{D}-[\text{M}_3\text{O}(\text{btc})_2(\text{X})(\text{H}_2\text{O})_2] \cdot n \text{H}_2\text{O}$ (M = Cr, Fe, Al) or **MIL-100(M)**

small S cage and large L cage in MIL-100(Fe)



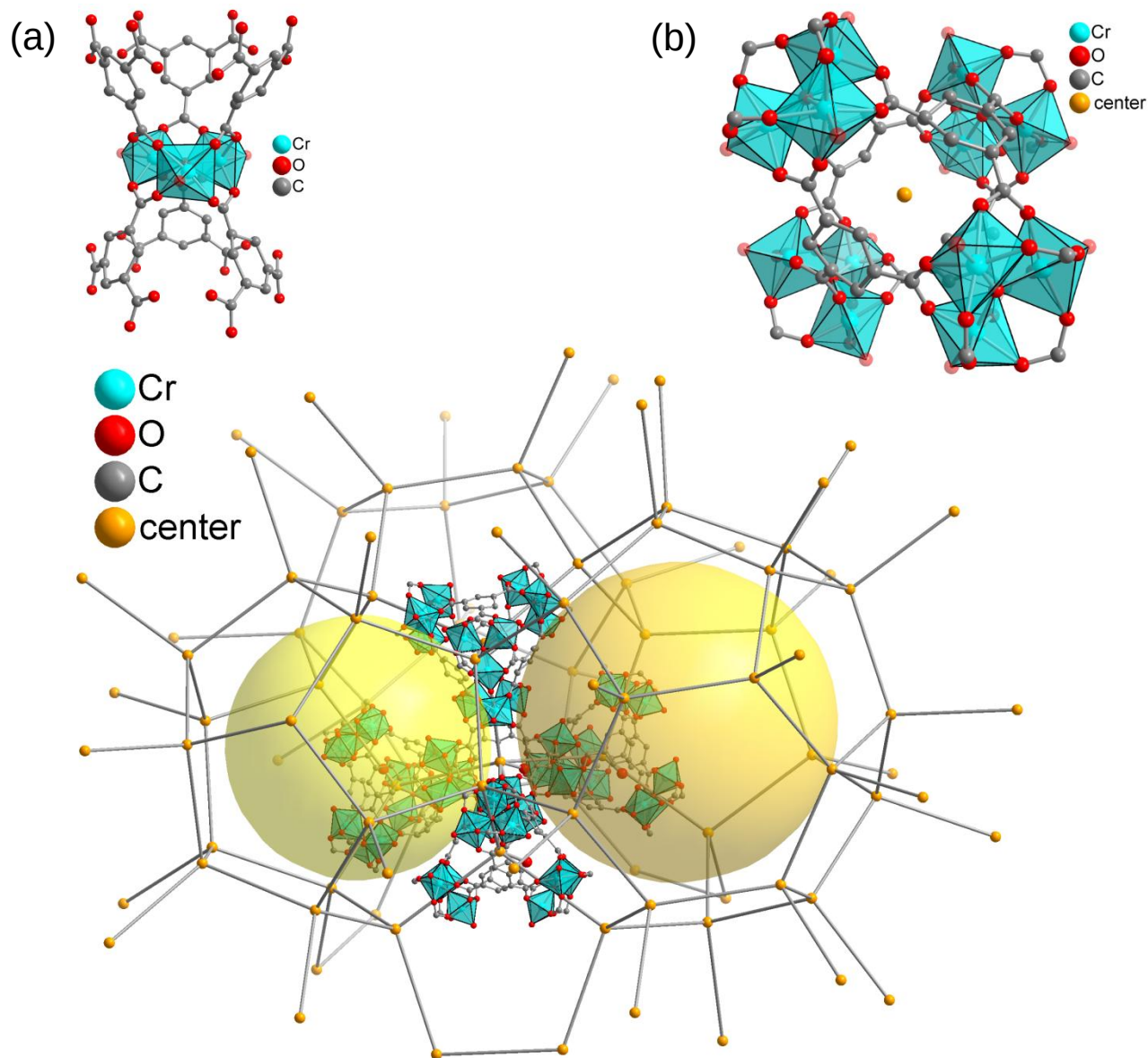
M = Fe: BET surface $S_{\text{BET}} = 1917 \text{ m}^2/\text{g}$, pore volume $V_{\text{pore}} = 1.00 \text{ cm}^3/\text{g}$;
M = Al: BET surface $S_{\text{BET}} = 1814 \text{ m}^2/\text{g}$, pore volume $V_{\text{pore}} = 1.14 \text{ cm}^3/\text{g}$
M = Cr: BET surface $S_{\text{BET}} = 1517 \text{ m}^2/\text{g}$

MIL example:
 $3\text{D}-[\text{M}_3\text{O}(\text{btc})_2(\text{X})(\text{H}_2\text{O})_2] \cdot n \text{H}_2\text{O}$ (M = Cr, Fe, Al) or **MIL-100(M)**

- (a) SBU,
- (b) supertetrahedra,
- (c) polygons for small S cage and large L cage in MIL-100(Cr) formed by connecting the centers of the corner-sharing supertetrahedra

(different objects are not drawn to scale).

MIL-100s are isostructural.



Synthesis:

$3\text{D}-[\text{M}_3\text{O}(\text{btc})_2(\text{X})(\text{H}_2\text{O})_2] \cdot n \text{H}_2\text{O}$ (M = Cr, Fe, Al) or **MIL-100(Al)**

230 mg (0.61 mmol) $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 104 mg (0.41 mmol) trimesic acid-trimethylester were placed with 0.77 mL (0.77 mmol) nitric acid ($c = 1 \text{ mol/L}$) and 2.8 mL (155.6 mmol) H_2O in a 10 mL Teflon-lined stainless-steel autoclave.

In the autoclave a hydrothermal synthesis is carried out with the following temperature program: Heat to 210 °C within 1 h; hold for 3.5 h at 210 °C; cool within 8 h to room temperature.

Purification of MIL-100(Al):

After the synthesis the precipitate is filtered and given to 8 mL of DMF in the autoclave which is heated to 150 °C for 4 h. Then the precipitate is separated again by filtration and refluxed with 100 mL de-ionized water in a 250 mL flask overnight.

Then the precipitate is filtered and dried at room temperature in air.

Synthesis:

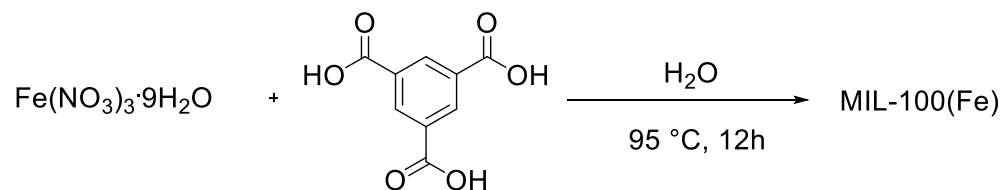


MIL-100(Fe) - microwave-assisted

A solution of trimesic acid (0.42 g, 2.00 mmol, 1 eq) and $\text{Fe}(\text{NO}_3)_3 \cdot 9 \text{H}_2\text{O}$ (1.21 g, 3.00 mmol, 1.5 eq) in deionized water (10 mL + one drop of nitric acid) is transferred into a microwave-autoclave, which is subsequently heated to 130 °C within 30 s and maintained at this temperature for 5 min 30 s (1600 W). The reacting mixture is cooled to room temperature and centrifuged. An orange solid should be recovered by decantation.

Activation of the solid consisted of a re-dispersion of the as-synthesized MIL-100(Fe) nanoparticles into deionized water (20 mL per 500 mg), followed by centrifugation and four cycles in absolute ethanol (20 mL per 500 mg) under the same conditions.

- under reflux only



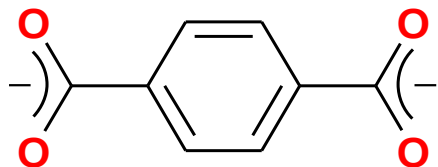
see: F. Zhang, J. Shi, Y. Jin, Y. Fu, Y. Zhong, W. Zhu, *Chem. Eng. J.*, **2015**, 259, 183-190.

yield	2.7 g
BET-surface	1484 m ² /g (Lit. 1836)

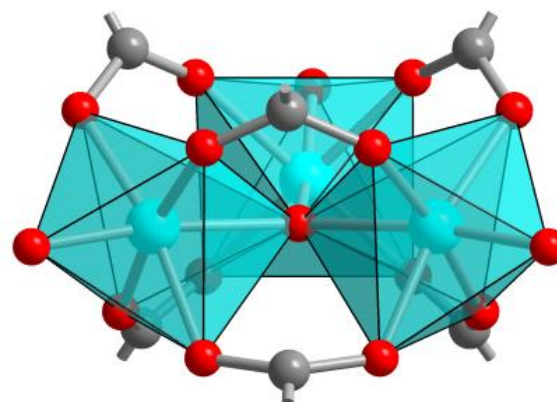
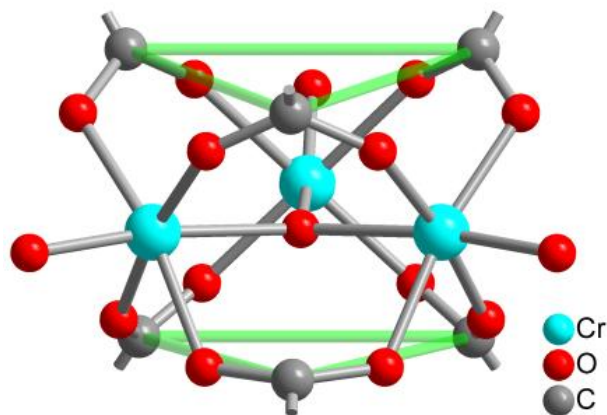
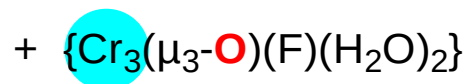


MIL example:

3D-[Cr₃(O)(bdc)₃(F,OH)(H₂O)₂] $\cdot$ ~25H₂O or MIL-101(M)

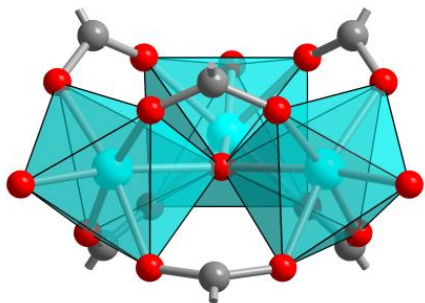


benzene-1,4-dicarboxylate,
terephthalate, bdc

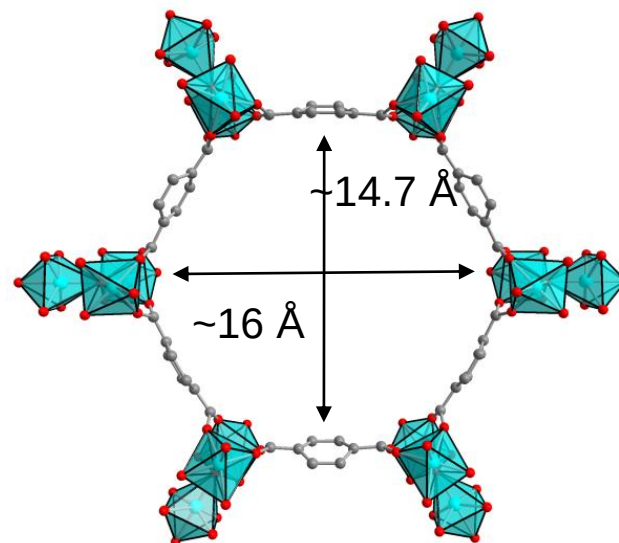
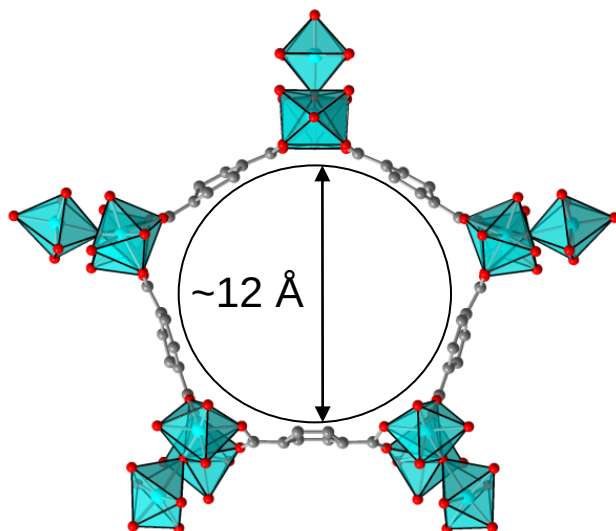


G. Férey, C. Mellot-Draznieks, C. Serre, F. Millange, J. Dutour, S Surble and I. Margiolaki, Science **2005**, 309, 2040–2042.

MIL example:
 $3\text{D}[\text{Cr}_3(\text{O})(\text{bdc})_3(\text{F},\text{OH})(\text{H}_2\text{O})_2] \cdot \sim 25\text{H}_2\text{O}$ or MIL-101

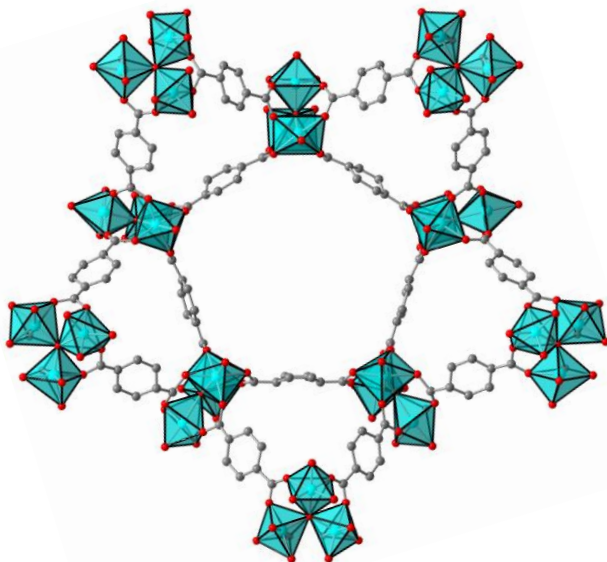


→ pentagonal windows and hexagonal windows
as largest windows in cages:

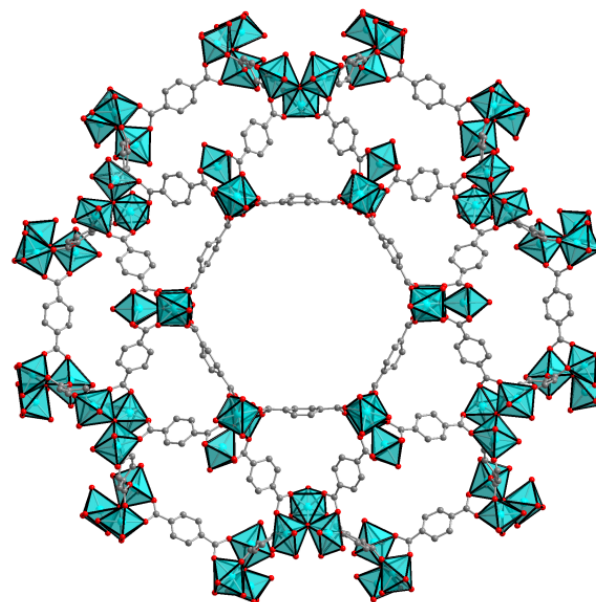


MIL example:
 $3\text{D}-[\text{Cr}_3(\text{O})(\text{bdc})_3(\text{F},\text{OH})(\text{H}_2\text{O})_2]\cdot\sim 25\text{H}_2\text{O}$ or MIL-101

pentagonal + trigonal
windows

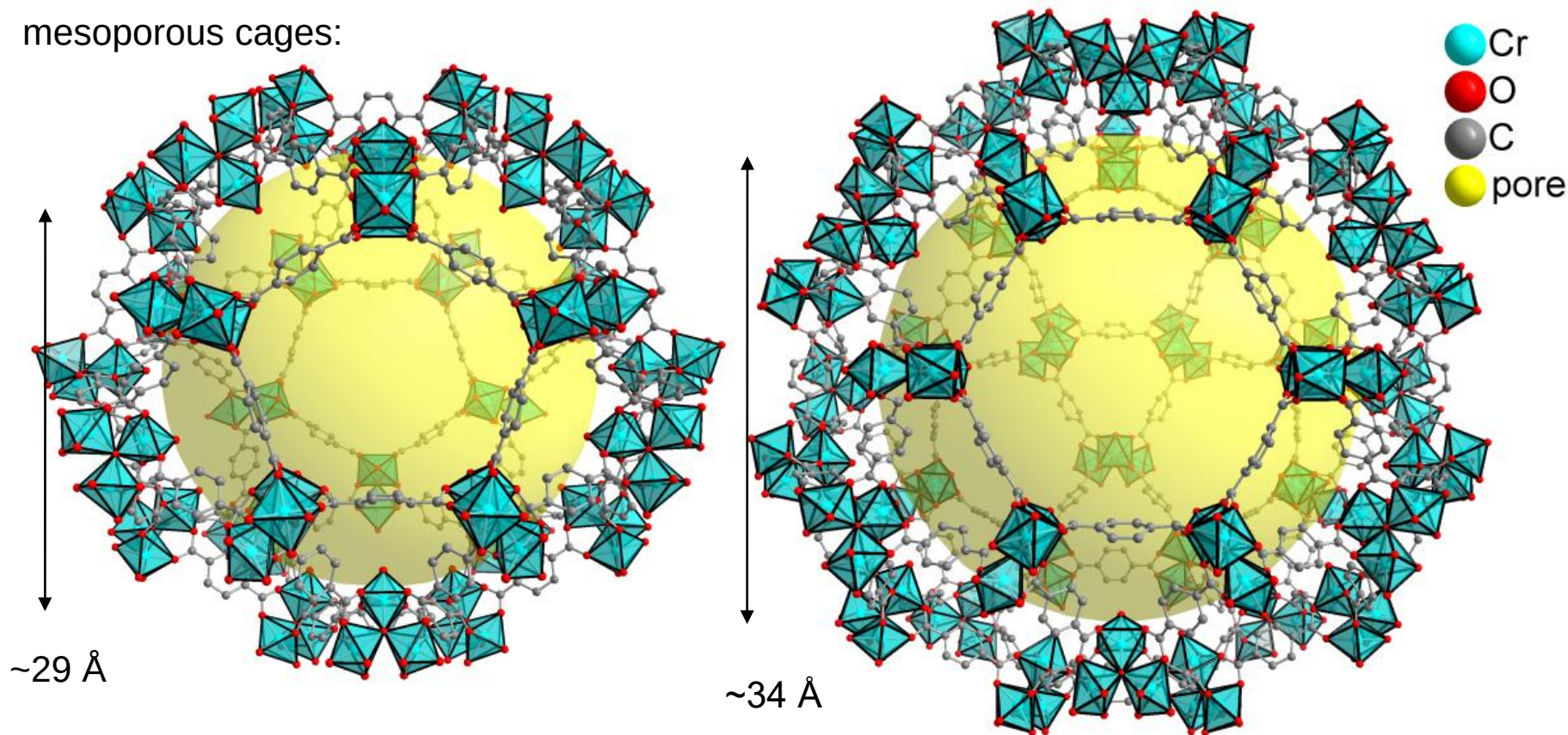


hexagonal + pentagonal + trigonal
windows



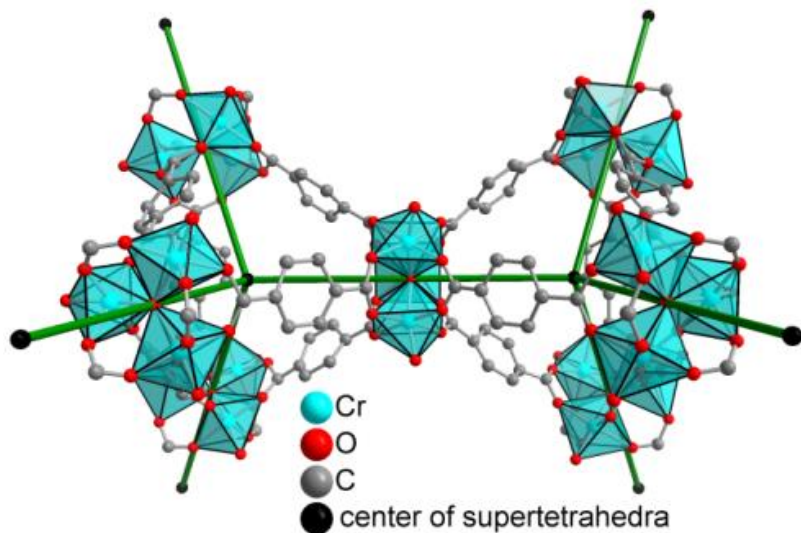
MIL example:
 $3\text{D}[\text{Cr}_3(\text{O})(\text{bdc})_3(\text{F},\text{OH})(\text{H}_2\text{O})_2]\cdot\sim 25\text{H}_2\text{O}$ or MIL-101

mesoporous cages:



Building blocks for MIL-101, $[\text{Cr}_3(\text{O})(\text{BDC})_3(\text{F},\text{OH})(\text{H}_2\text{O})_2]\cdot\sim 25\text{H}_2\text{O}$. The benzene-1,4-dicarboxylate ligands bridge between trinuclear $\{\text{Cr}_3\text{O}\}$ building units. The largest aperture windows of the mesoporous cages which build up the MTN zeolite network are pentagonal and hexagonal rings. The yellow spheres in the mesoporous cages with diameter of 29 or 34 Å, respectively, take into account the van-der-Waals radii of the framework walls (water-guest molecules are not shown). The different objects are not drawn to scale

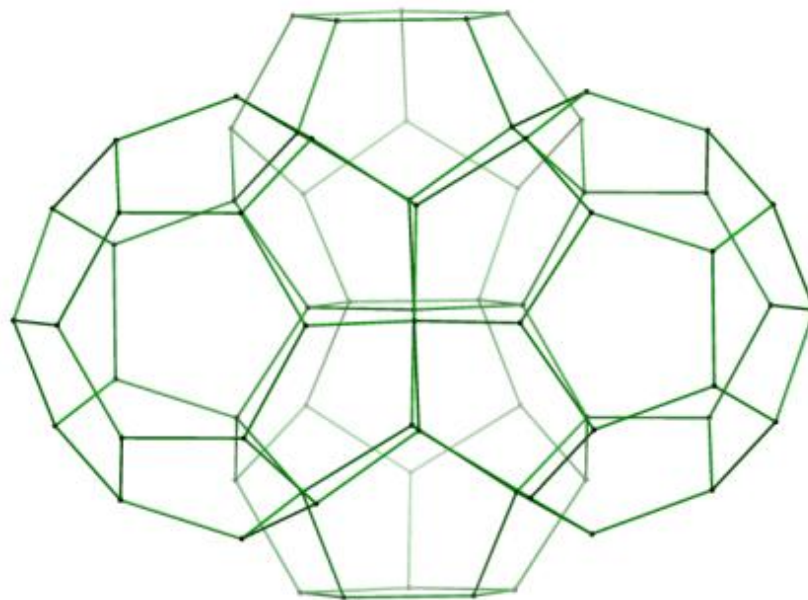
MIL example:
 $3\text{D}[\text{Cr}_3(\text{O})(\text{bdc})_3(\text{F},\text{OH})(\text{H}_2\text{O})_2] \cdot \sim 25\text{H}_2\text{O}$ or MIL-101



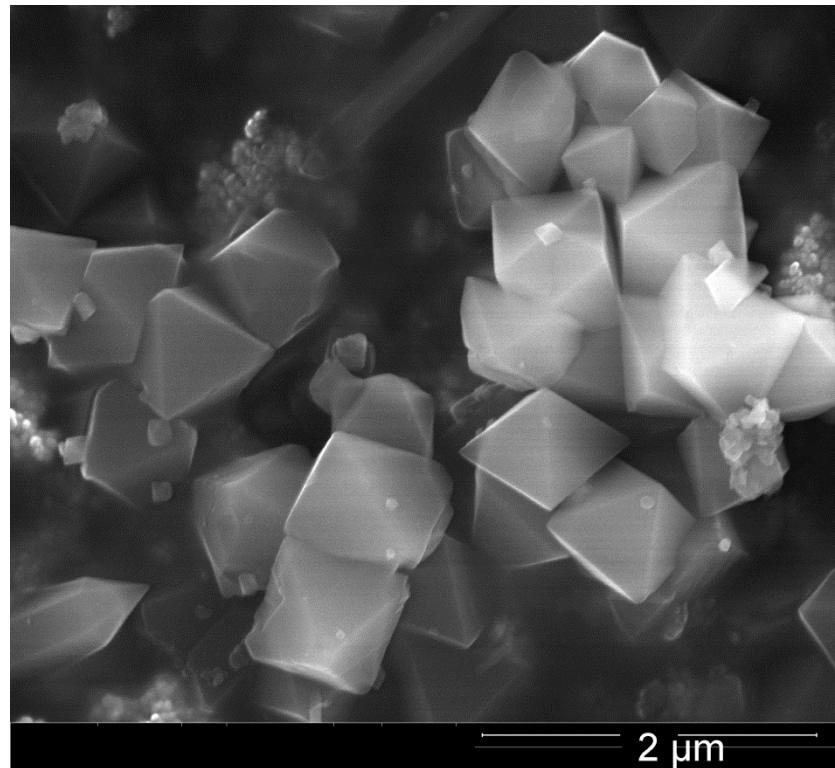
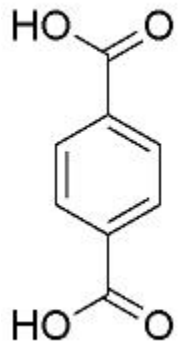
BET surface $\sim 4000 \text{ m}^2/\text{g}$

The vertex-sharing supertetrahedra

and MTN zeolite topology from two types of mesoporous cages with pentagonal and hexagonal windows

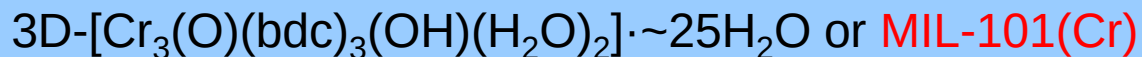


MIL example:
 $3\text{D-}[\text{Cr}_3(\text{O})(\text{bdc})_3(\text{F},\text{OH})(\text{H}_2\text{O})_2]\cdot\sim 25\text{H}_2\text{O}$ or MIL-101



Tanh: (MIL101-je_17_01modif.tif)

Synthesis (fluorine-free)



Place 401 mg $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (1 mmol) and 166 mg (1 mmol) terephthalic acid (benzol-1,4-dicarboxylic acid, H_2bdc) in a Teflon cup of an autoclave and add 4.8 ml of water.

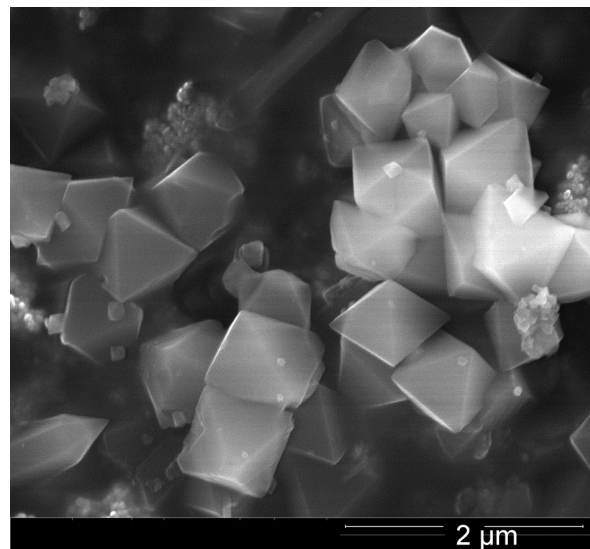
Homogenize the mixture during 5 minutes by a glass rod. Close the Teflon beaker, place it in the autoclave and seal the whole set-up. Program the oven to ramp to 220°C during 1 h and to hold this temperature for the next 8 h.

After the oven cools down (within 6 h), open the autoclave, centrifuge off the precipitate and purify it according to the following description:

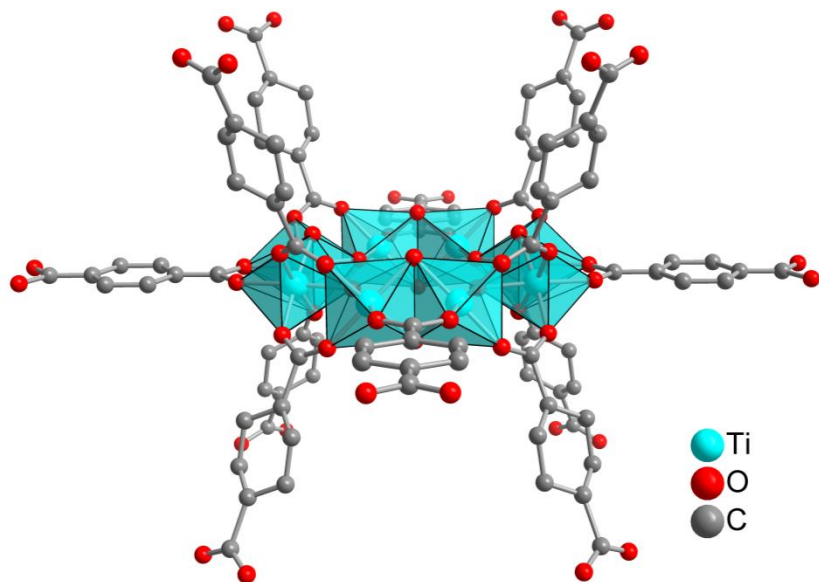
The precipitate is stirred in 20 mL DMF at 100°C for 2 h and centrifuged off again.

The precipitate is stirred in 20 mL MeOH at 50°C for 2 h and centrifuged off again.

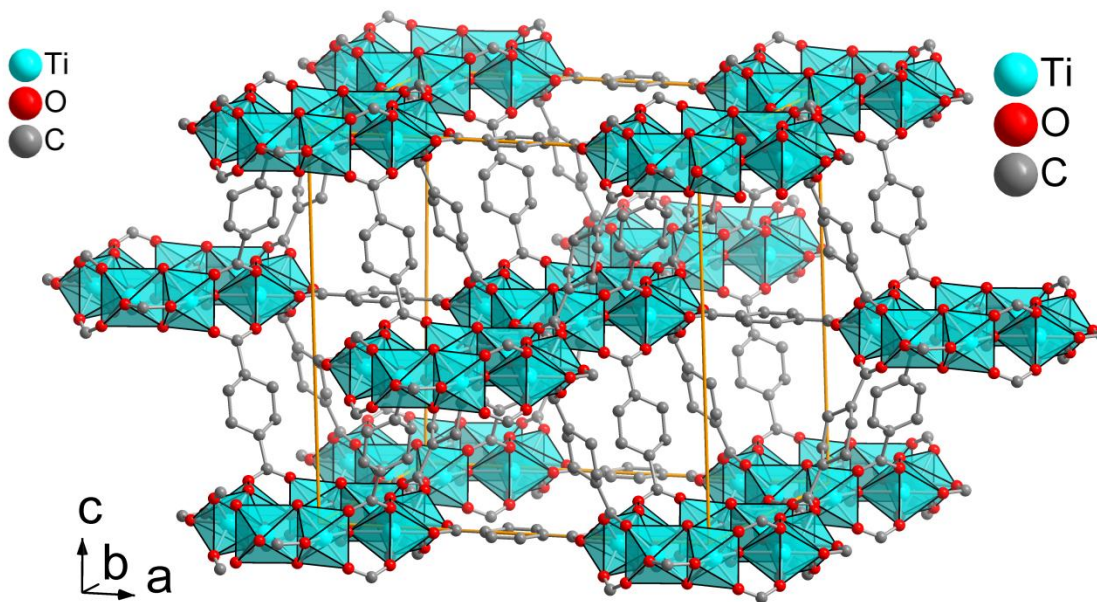
The precipitate is stirred in 20 mL of deionized water at 80°C for 2 h and centrifuged off again. Remove the supernatant and dry the compound at 100°C in the oven.



MIL example:
 $3\text{D}[\text{Ti}_8\text{O}_8(\text{OH})_4(\text{bdc})_6]$ or MIL-125(Ti)



Structure of titanium terephthalate MIL-125. The SBU is an eight-membered ring of edge- and vertex-sharing TiO_6 octahedra, which is connected to **12 neighboring SBUs** in a **body-centered cubic (bcc) packing** arrangement.



MOF-terminology

and **university acronyms**, e.g.

CAU = Christians-Albrecht-University (Kiel, Germany) (N. Stock)

NU = Northwestern University (Michigan, USA) (J. T. Hupp, O. K. Farha)

DUT = Dresden University of Technology (Germany) (S. Kaskel)

UiO = University in Oslo (**M** = Zr^{4+} , K. P. Lillerud)

compare to zeolite acronyms:

ZSM-5 - Zeolite Socony Mobil – 5, patented by Mobil Oil Company in 1975

MCM = Mobil Crystalline Materials, produced at Mobil Corporation laboratories

MCM-41 etc. = Mobil Composition of Matter No. 41

SBA-15 = Santa Barbara Amorphous type material

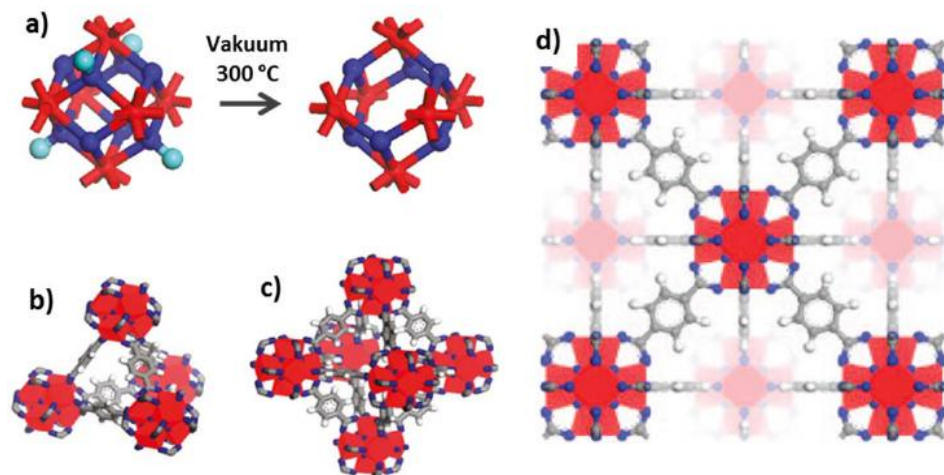
MOF-examples

3D-[Zr₆O₄(OH)₄(bdc)₆] or UiO-66

UiO = University in Oslo

inner {Zr₆O₄(OH)₄} core in which the triangular faces of the Zr₆-octahedron are alternatively capped by μ_3 -O and μ_3 -OH groups

a) weakly bound μ_3 -OH groups → SBUs can be reversibly dehydrated



zirconium = red,
oxygen = blue,
carbon = gray
hydrogen = light-blue or white

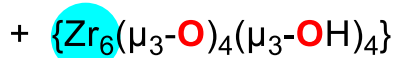
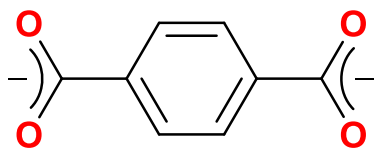
SBUs form (b) tetrahedral and (c) octahedral cages which combine to a (d) face-centered cubic packing

{Zr₆O₄(OH)₄(CO₂)₁₂} is 12 coordinated, the highest SBU-coordination reported for a MOF. 12-coordination is the coordination of metal atoms in closed-packed metal structures.

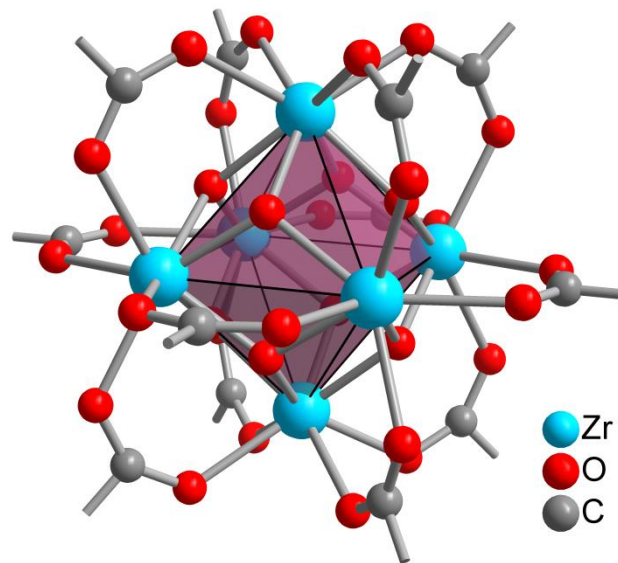
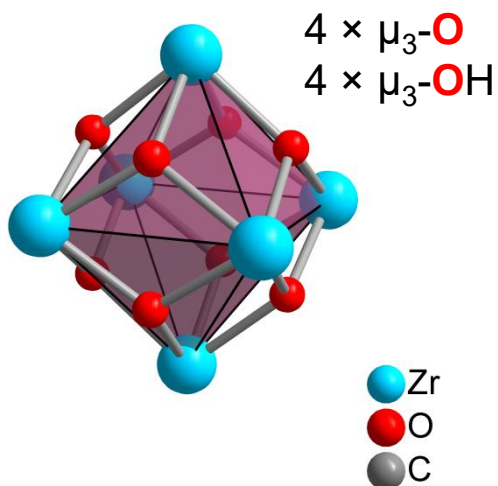
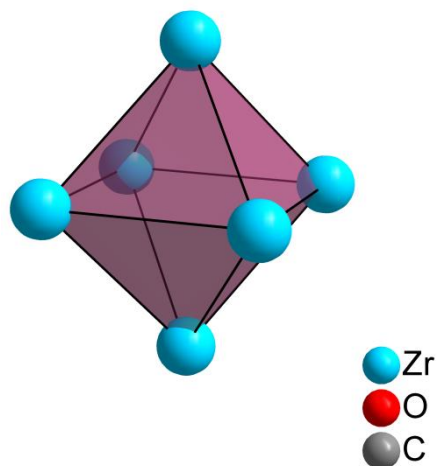
L. Valenzano, B. Civalieri, S. Chavan, S. Bordiga, M. H. Nilsen, S. Jakobsen, K. P. Lillerud, C. Lamberti, *Chem. Mater.*, **2011**, 23, 1700-1718.

J. H. Cavka, S. Jakobsen, U. Olsbye, N. Guillou, C. Lamberti, S. Bordiga, K.P. Lillerud, *J. Am. Chem. Soc.*, **2008**, 130, 13850–13851.

MOFs – just two examples: 3D-[Zr₆O₄(OH)₄(bdc)₆] or UiO-66



benzene-1,4-dicarboxylate,
terephthalate, bdc

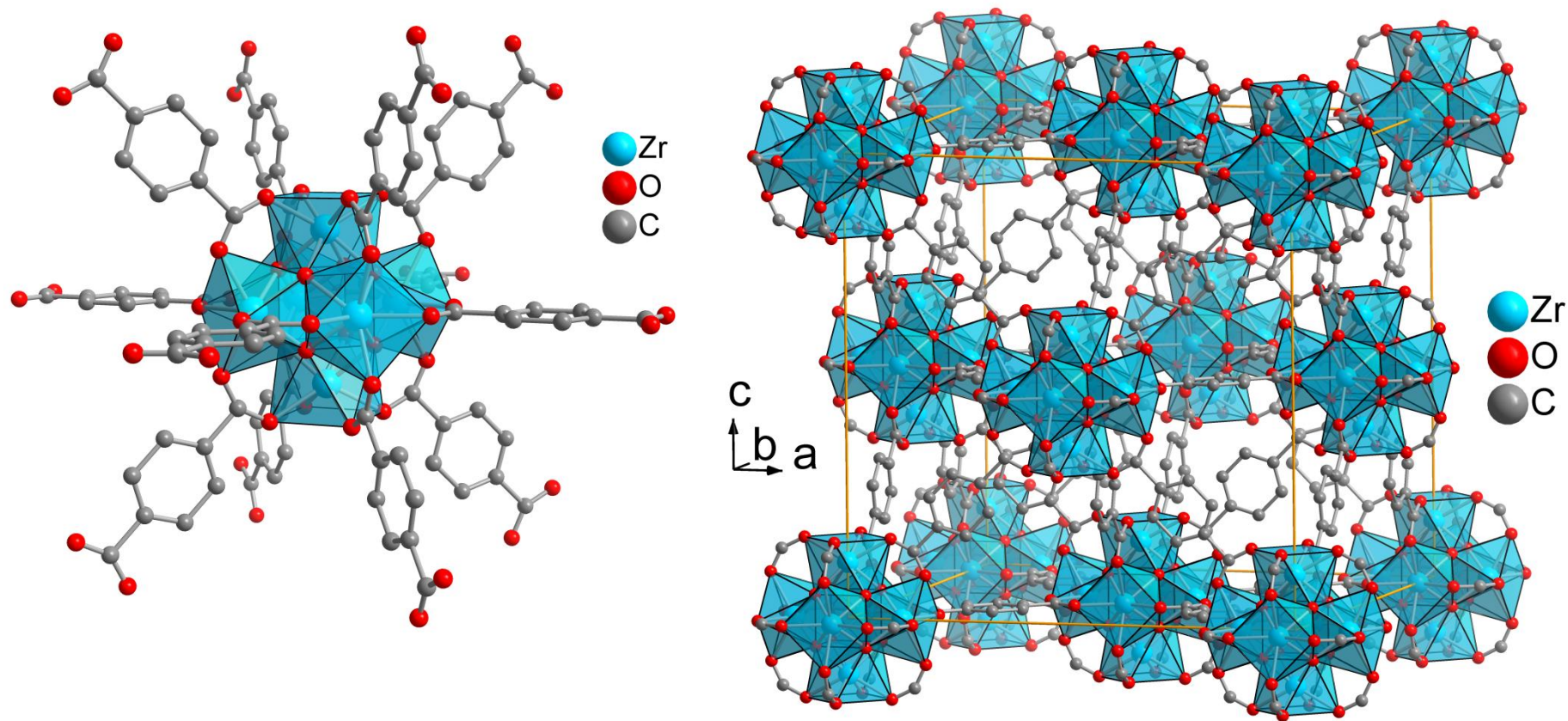


L. Valenzano, B. Civalieri, S. Chavan, S. Bordiga, M. H. Nilsen, S. Jakobsen, K. P. Lillerud, C. Lamberti, *Chem. Mater.*, **2011**, 23, 1700-1718.

J. H. Cavka, S. Jakobsen, U. Olsbye, N. Guillou, C. Lamberti, S. Bordiga, K. P. Lillerud, *J. Am. Chem. Soc.* **2008**, 130, 13850-13851.

MOF-examples

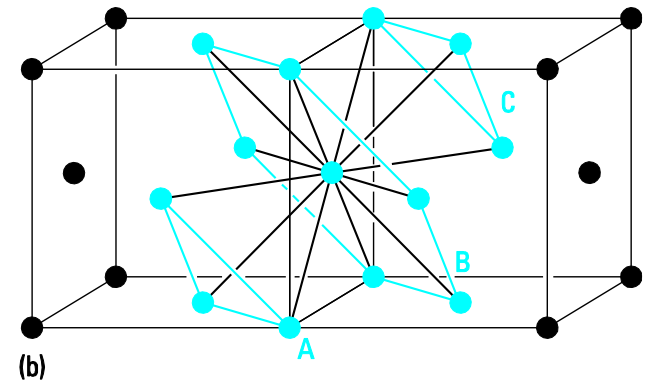
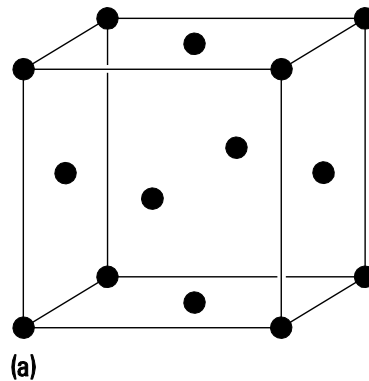
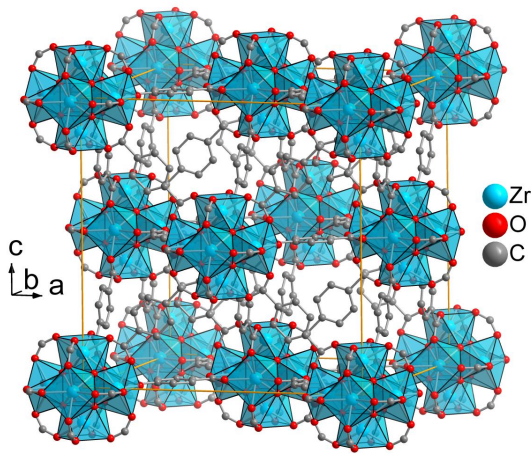
3D-[Zr₆O₄(OH)₄(bdc)₆] or **UiO-66**



Structure of zirconium terephthalate UiO-66. The SBU is an octahedral cluster of six edge-sharing ZrO₈ square-antiprism, which is connected to 12 neighboring SBUs in a face-centered cubic (fcc) packing arrangement.

face-centered cubic, fcc-packing

12-coordination is the coordination of metal atoms in closed packed metal structures. The Zr-MOF structures formed with linear ligands are therefore expanded versions of the cubic close packed (CCP) structure (= fcc).



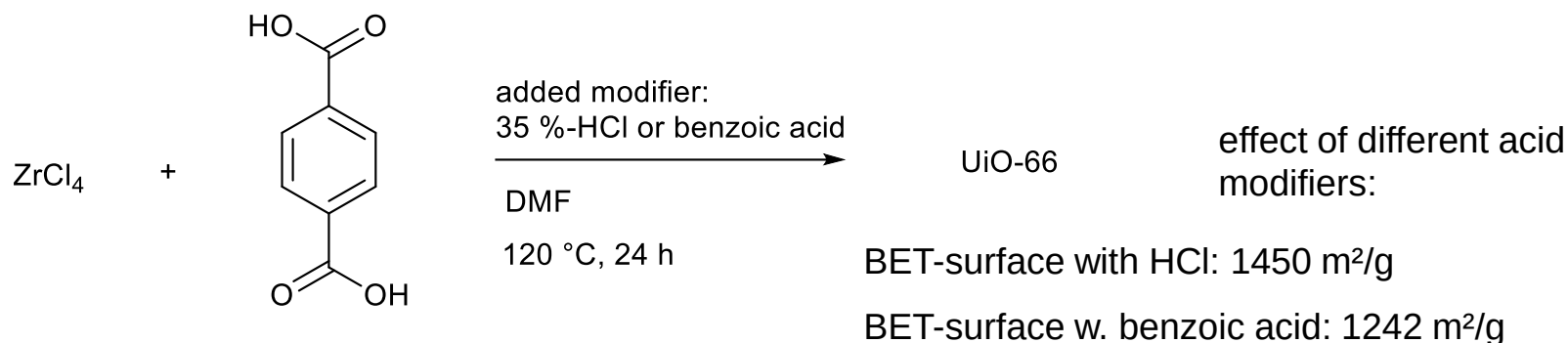
- a) face-centered cubic unit cell = cubic close packed (CCP) structure
 - b) most densely packed layers lie perpendicular to the diagonal of the unit cell.
- Each atom or SBU has 12 neighbors at equal distance.

Synthesis:

3D-[Zr₆O₄(OH)₄(bdc)₆] or UiO-66

In a 50 mL round bottom flask, ZrCl₄ (0.084 g, 0.343 mmol), CH₃COOH solution (0.618 g, 10.29 mmol) and HCl solution (0.36 g, 10 mmol) are dissolved in DMF (20 mL). H₂BDC is added to the clear solution (0.057 g, 0.343 mmol). The mixture is transferred to a 40 mL Teflon-lined stainless-steel auto-clave. The system is heated for 24 h at 120 °C.

After 24 h, the solution is cooled to room temperature and the precipitate is isolated by centrifugation. The solid is suspended in DMF (10 mL). After standing at room temperature for 2-6 h, the solid is recovered by centrifugation. The obtained powder is washed with ethanol (2·10 mL) in the same way as described before (DMF washing step). Finally, the solids are dried under reduced pressure.



Lillerud et. al., *Chem. Mater.* **2014**, 26, 4068–4071.

S. Pullen, H. Fei, A. Orthaber, S. M. Cohen, S. Ott, *J. Am. Chem. Soc.*, **2013**, 135, 16997–17003.

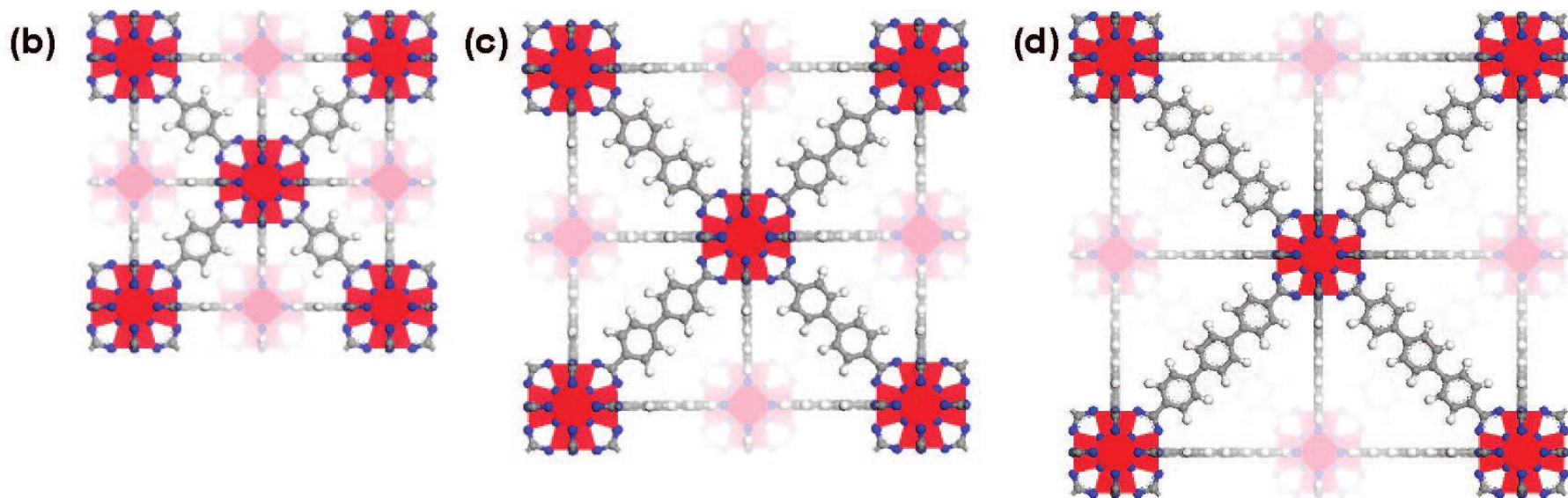
HCl: O. V. Gutov, M. G. Hevia, E. C. Escudero-Adan, A. Shafir, *Inorg. Chem.* **2015**, 54, 8396–8400.

benzoic acid: S. Gökpınar, T. Diment, C. Janiak, *Dalton Trans.* **2017**, 46, 9895-9900.

MOF-examples

3D-[Zr₆O₄(OH)₄(O₂C-aryl-CO₂)₆] or UiO-66 to UiO-68

The Zr-MOF structure can be made with increasing length of the linkers (scale chemistry = isorecticular chemistry).



- (b) Zr-MOF with 1,4-benzene-dicarboxylate (bdc) as linker, UiO-66,
(c) Zr-MOF with 4,4'-biphenyl-dicarboxylate (bpdc) as linker, UiO-67,
(d) Zr-MOF with terphenyl dicarboxylate (tpdc) as linker, UiO-68.

J. H. Cavka,, S. Jakobsen,, U. Olsbye, N. Guillou, C. Lamberti, S. Bordiga, K.P. Lillerud, *J. Am. Chem. Soc.*, **2008**, 130, 13850–13851.

MOF-terminology

and **university acronyms**, e.g.

CAU = Christians-Albrecht-University (Kiel, Germany) (N. Stock)

NU = Northwestern University (Michigan, USA) (J. T. Hupp, O. K. Farha)

DUT = Dresden University of Technology (Germany) (S. Kaskel)

UiO = University in Oslo (**M** = Zr^{4+} , K. P. Lillerud)

compare to zeolite acronyms:

ZSM-5 - Zeolite Socony Mobil – 5, patented by Mobil Oil Company in 1975

MCM = Mobil Crystalline Materials, produced at Mobil Corporation laboratories

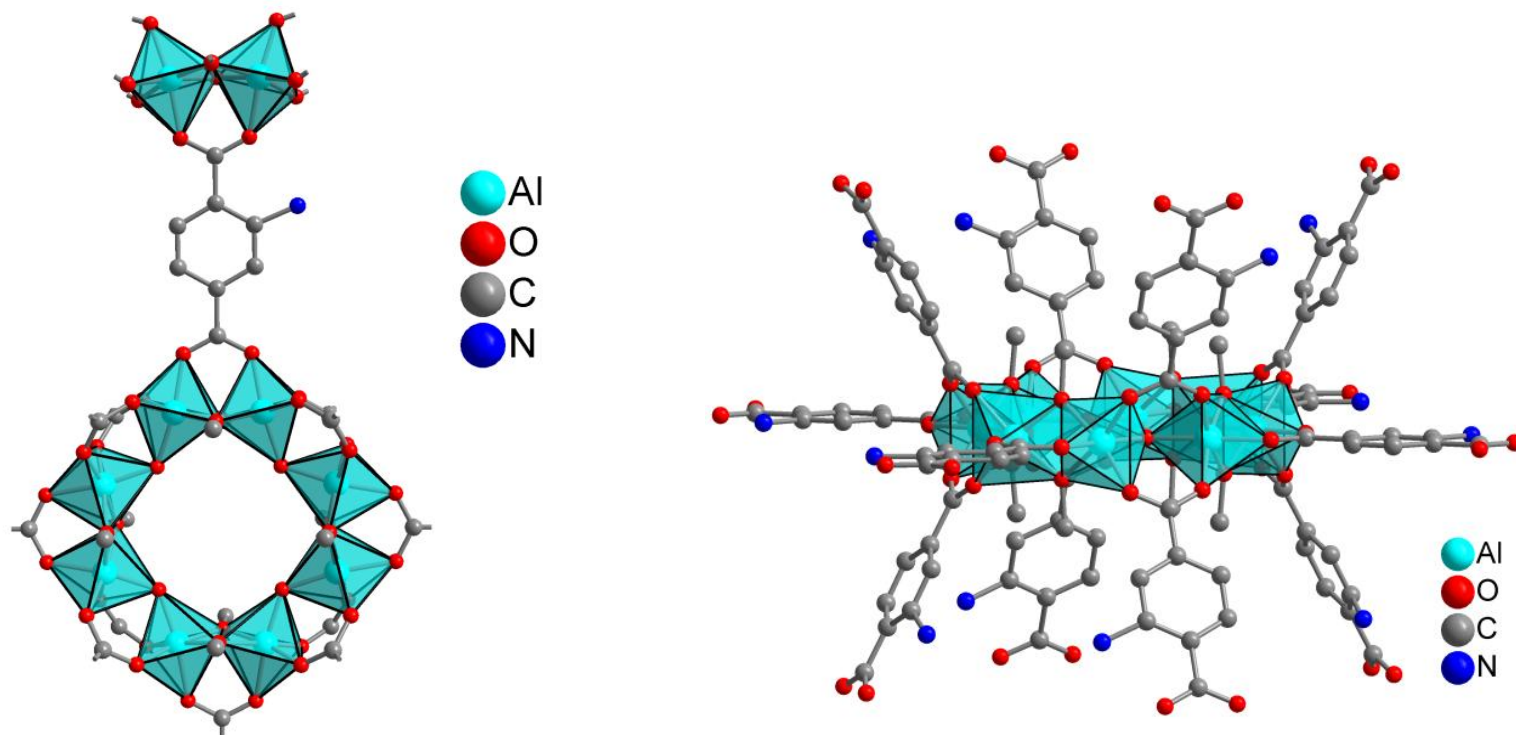
MCM-41 etc. = Mobil Composition of Matter No. 41

SBA-15 = Santa Barbara Amorphous type material

MOF-examples

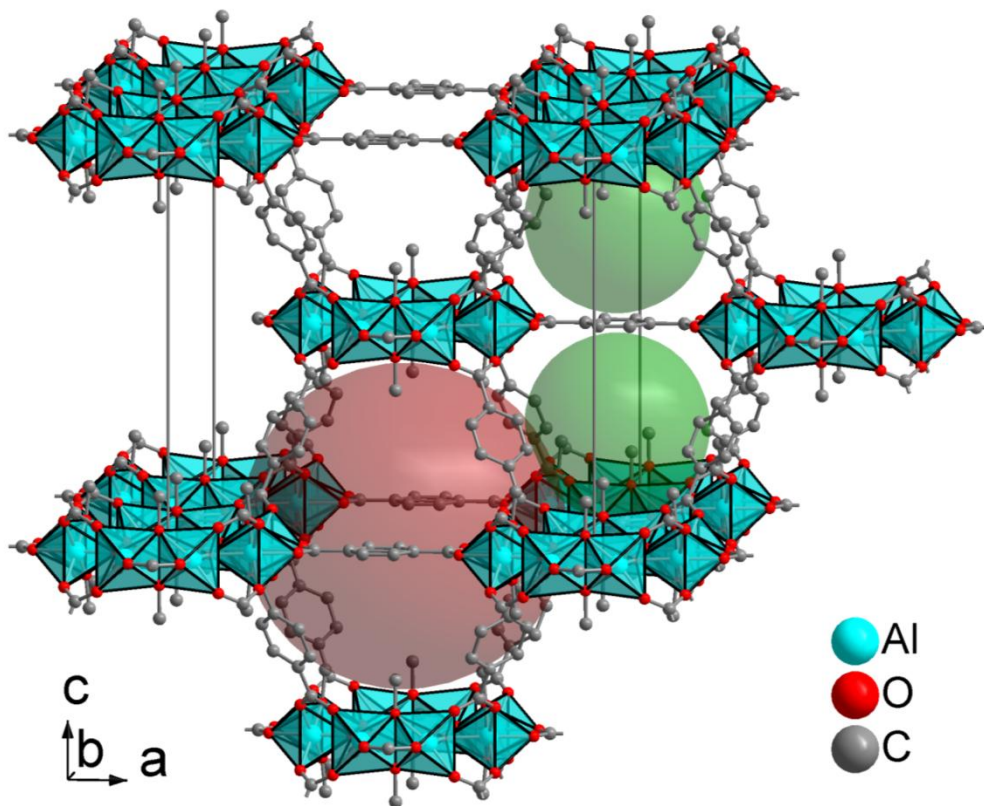
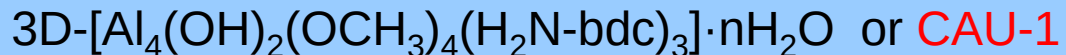
$3\text{D}-[\text{Al}_4(\text{OH})_2(\text{OCH}_3)_4(\text{H}_2\text{N-bdc})_3]\cdot n\text{H}_2\text{O}$ or **CAU-1**

from 2-amino-terephthalic acid ($\text{bdcH}_2\text{-NH}_2$) and AlCl_3 in methanol
- stable up to 580 K and built from $\{\text{Al}_8(\text{OH})_4(\text{OCH}_3)_8\}^{12+}$ units.



$\{\text{Al}_8(\text{OH})_4(\text{OCH}_3)_8\}^{12+}$ building unit in polyhedral presentation in (a) top view showing only one of the 12 bdc-NH₂ connections and (b) side view with all 12 connecting bdc-NH₂ linkers to adjacent units

MOF-examples

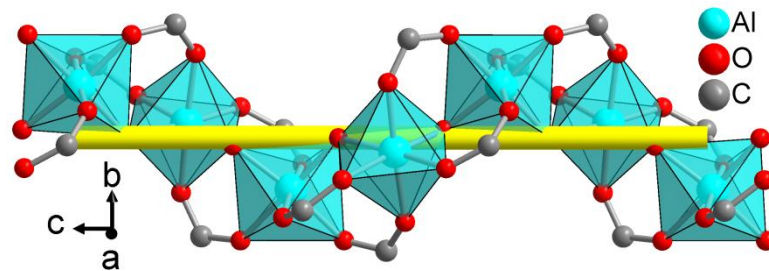
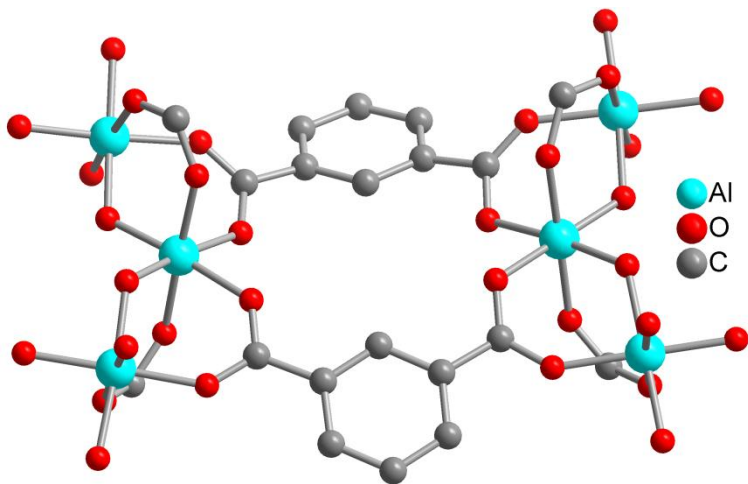


Each Al_8 -SBU is surrounded and interconnected by 12 amino-terephthalate anions to enclose **tetrahedrally-** and **octahedrally-** shaped voids with inner diameters of 0.5 and 1.0 nm, respectively, and a Langmuir surface of $1700 \text{ m}^2 \text{ g}^{-1}$

packing diagram of $3\text{D}-[\text{Al}_4(\text{OH})_2(\text{OCH}_3)_4(\text{bdc-NH}_2)_3]\cdot x\text{H}_2\text{O}$ (CAU-1) with the two main types of **octahedral (red)** and **tetrahedral (green)** voids. Crystal water molecules in the voids are not shown. The NH_2 -groups on the aromatic terephthalate ring are disordered

MOF-examples

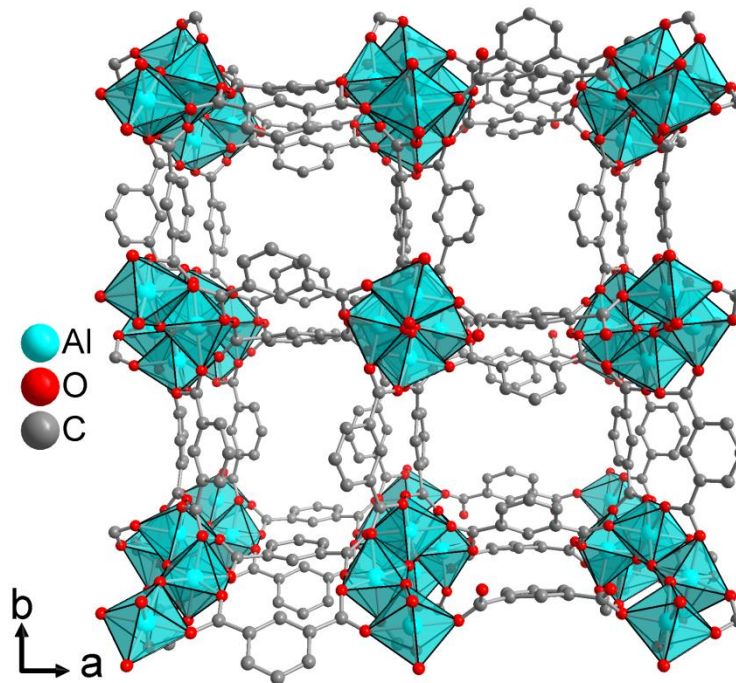
3D-[Al(OH)(benzene-1,3-dicarboxylate)]·nH₂O or **CAU-10-H**



H. Reinsch, M. A. van der Veen, B. Gil, B. Marszalek, T. Verbiest, D. de Vos and N. Stock, *Chem. Mater.*, 2013, **25**, 17-26;

pictures from:

D. Fröhlich, E. Pantatosaki, P. D. Kolokathis, K. Markey, H. Reinsch, M. Baumgartner, M. A. van der Veen, D. E. De Vos, N. Stock, G. K. Papadopoulos, S. K. Henninger, C. Janiak, *J. Mater. Chem. A* **2016**, 4, 11859-11869. DOI: 10.1039/C6TA01757F



Synthesis:

3D-[Al(OH)(benzene-1,3-dicarboxylate)]·nH₂O or CAU-10-H

CAU-10-H was synthesized from 505 g (0.75 mol) of aluminium sulfate hydrate (Al₂(SO₄)₃·18H₂O) dissolved in 2.5 L of H₂O, and 132 g (0.79 mol) isophthalic acid dissolved in 625 mL of DMF. The two solutions were combined in a 5 L round bottom flask. The combined mixture was heated under reflux for 117 h. After cooling down the precipitate was filtered, redispersed for washing in 3 L of H₂O by sonication and stirring. The dispersion was filtered again and dried for 4 days at 100 °C and after this activated for 2 days at 120°C in vacuum. The total yield was 150 g (91 %) CAU-10-H.

Surface area $S_{\text{BET}} = 564 \text{ m}^2 \text{ g}^{-1}$ and pore volume $0.26 \text{ cm}^3 \text{ g}^{-1}$.

MOF-terminology

Acronyms:

MOF = metal organic framework (Yaghi) - general, generic term
i.e., without implying any special metal or ligand.

Other terms more implications:

Yaghi-MOF: Zn^{2+} and di-carboxylate ligand (O. M. Yaghi)

MIL = Materials Institute Lavoisier/Material of Institute Lavoisier (G. Férey)
mostly $\text{M(III)} = \text{M}^{3+}$, di- or tri-carboxylate ligand, zeolite topology

PCP = porous coordination polymer (S. Kitagawa)
from the Kitagawa group, often dynamic, breathing MOFs

ZIF = zeolitic imidazolate frameworks

$\text{M} = \text{Zn}^{2+}$ and imidazolate ligands

ZIF = zeolitic imidazolate framework

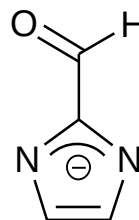
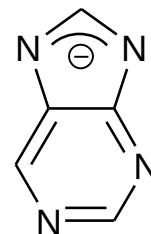
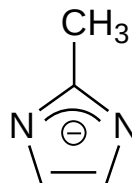
MOFs based on non-carboxylate ligands:

- other anionic ligands

- imidazolate ligands

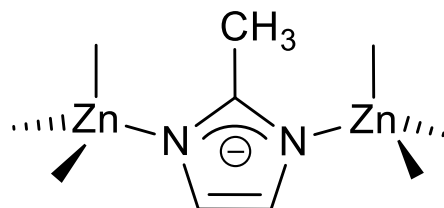
giving ZIFs

- still rigid ligands



ZIF = zeolitic imidazole framework

Zeolitic imidazolate frameworks or ZIFs are constructed from the bridging action of the anionic imidazolate ligand (or its derivatives) between zinc atoms. The imidazolate function in ZIFs does not tend to form metal clusters, which could serve as secondary building blocks (SBUs).



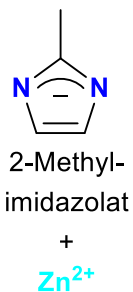
ZIFs are porous but with less open pores than other MOFs due to smaller pore aperture windows. Access in ZIFs is limited to smaller molecules.

The formation of metal clusters as SBUs may thus be, however, an important concept in the formation of MOFs with large and accessible pore systems.

Metal-Organic Frameworks (MOFs)

– Beispiel

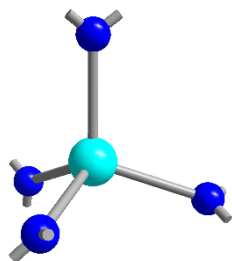
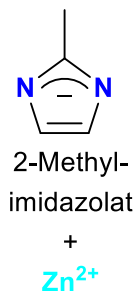
Linker



Metal-Organic Frameworks (MOFs)

– Beispiel

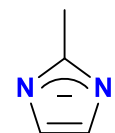
Linker



Metal-Organic Frameworks (MOFs)

– Beispiel

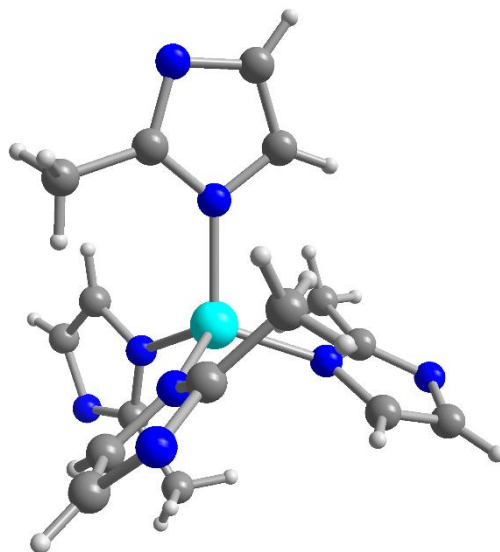
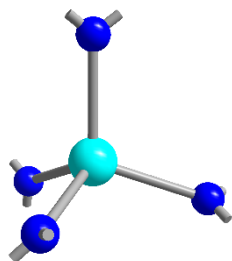
Linker



2-Methyl-
imidazolat

+

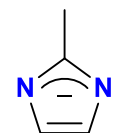
Zn^{2+}



Metal-Organic Frameworks (MOFs)

– Beispiel

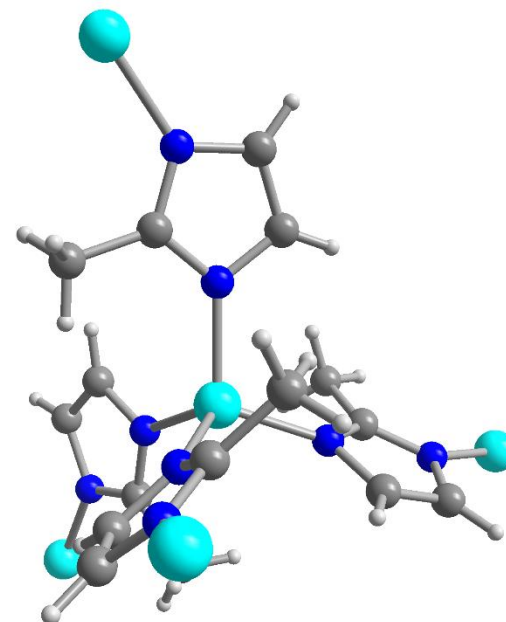
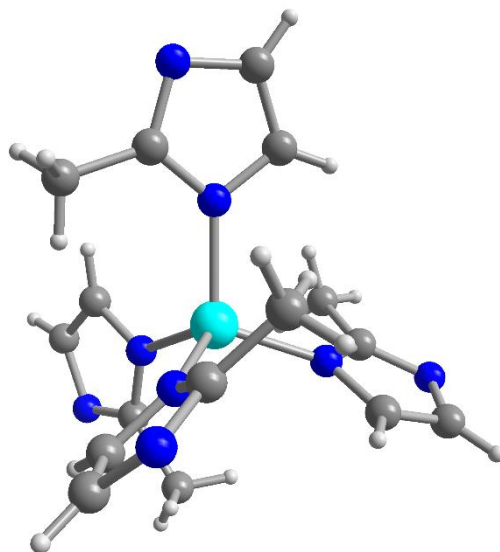
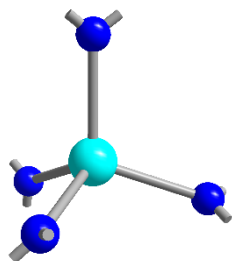
Linker



2-Methyl-
imidazolat

+

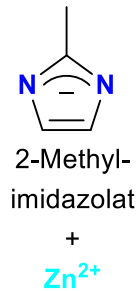
Zn^{2+}



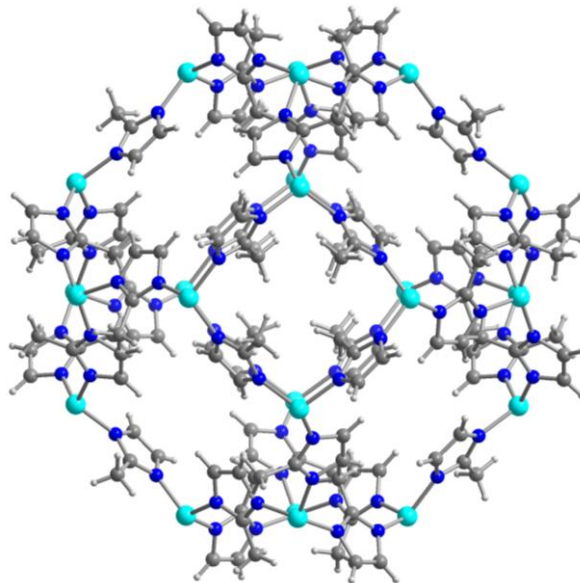
Metal-Organic Frameworks (MOFs)

– Beispiel

Linker



3D Netzwerk



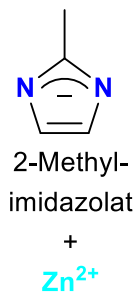
Acronym

ZIF-8

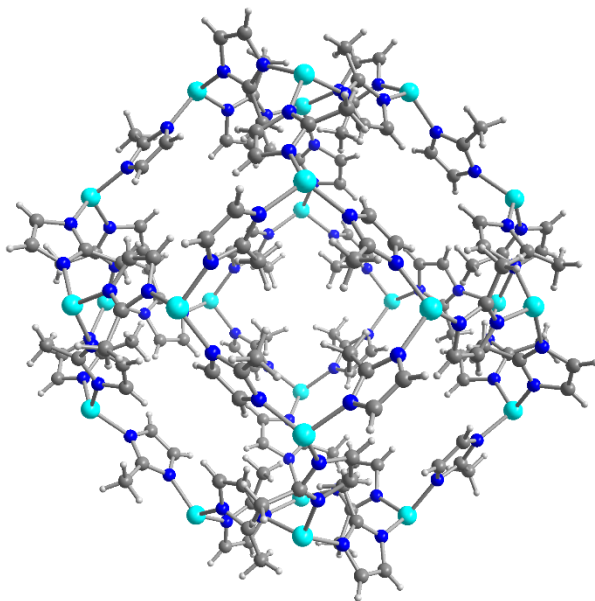
Metal-Organic Frameworks (MOFs)

– Beispiel

Linker



3D Netzwerk



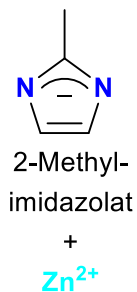
Acronym

ZIF-8

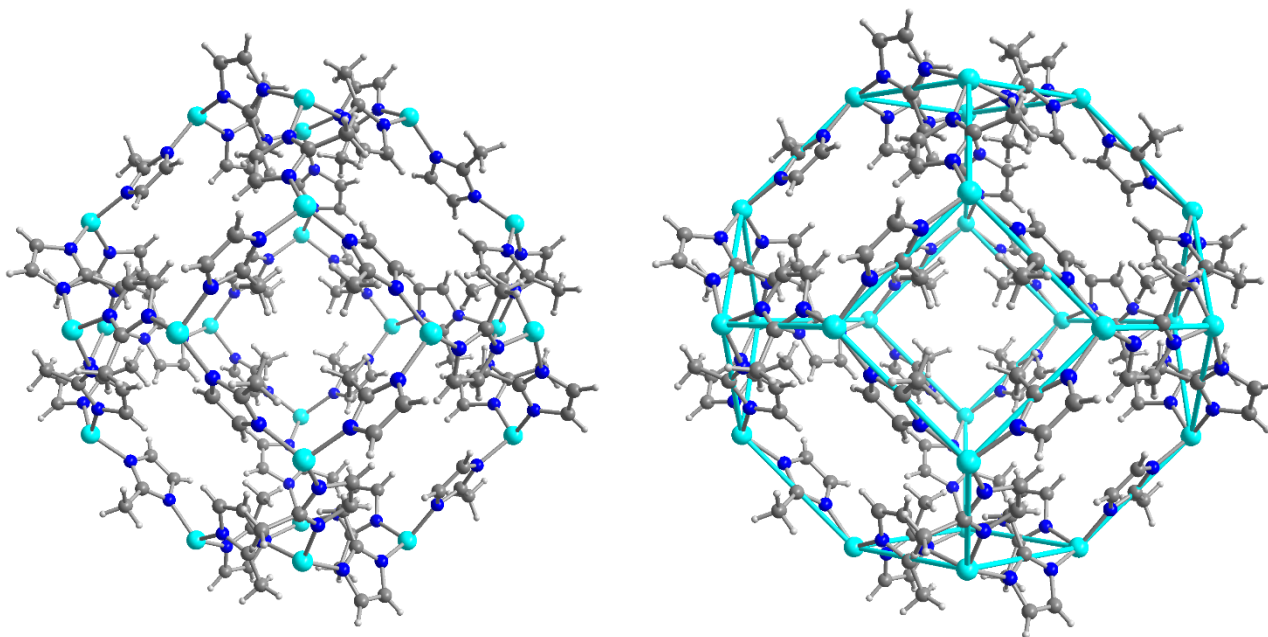
Metal-Organic Frameworks (MOFs)

– Beispiel

Linker



3D Netzwerk



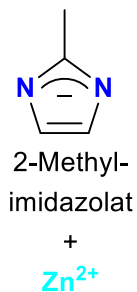
Acronym

ZIF-8

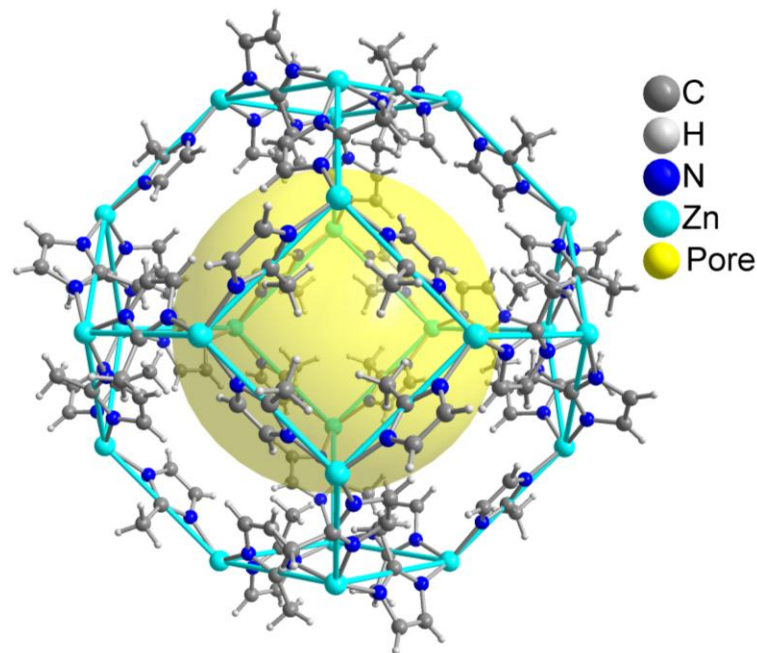
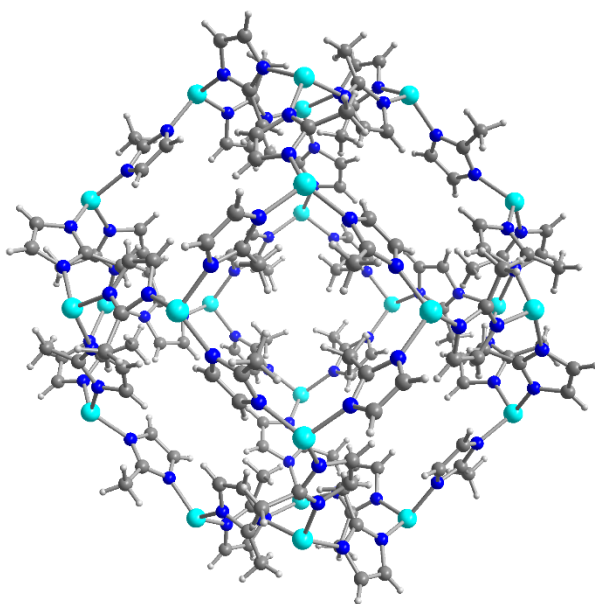
Metal-Organic Frameworks (MOFs)

– Beispiel

Linker



3D Netzwerk



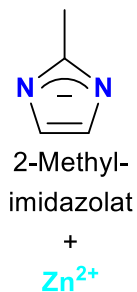
Acronym

ZIF-8

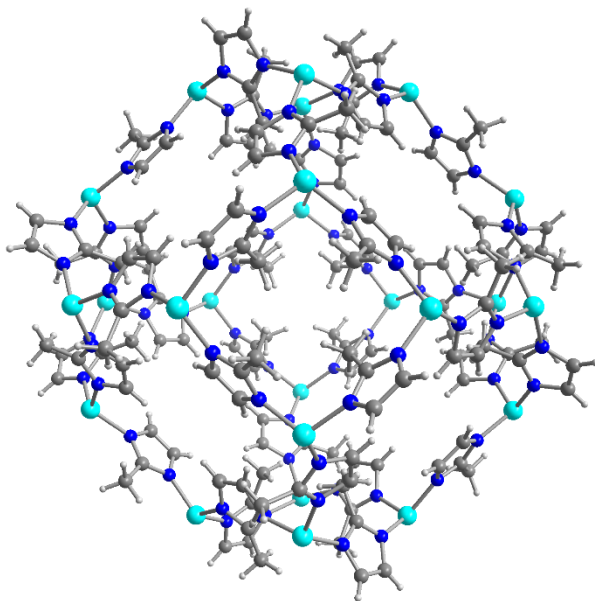
Metal-Organic Frameworks (MOFs)

– Beispiel

Linker

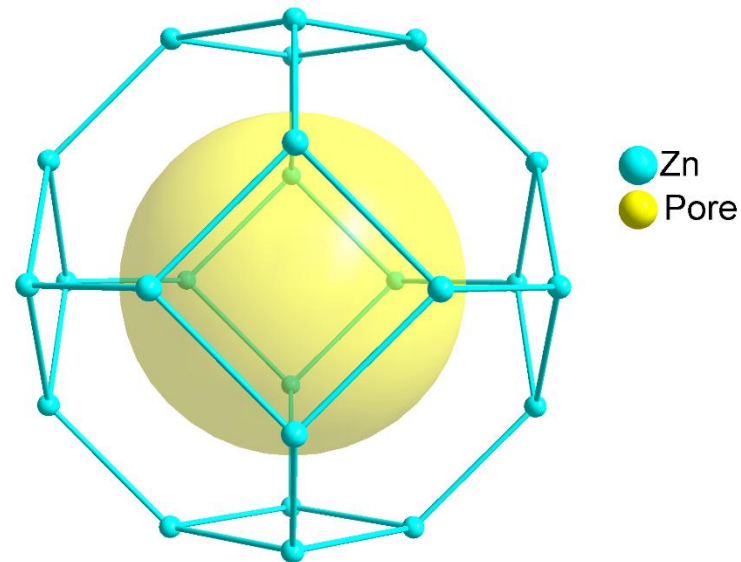


3D Netzwerk



Acronym

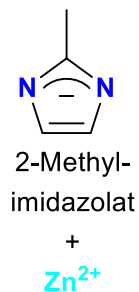
ZIF-8



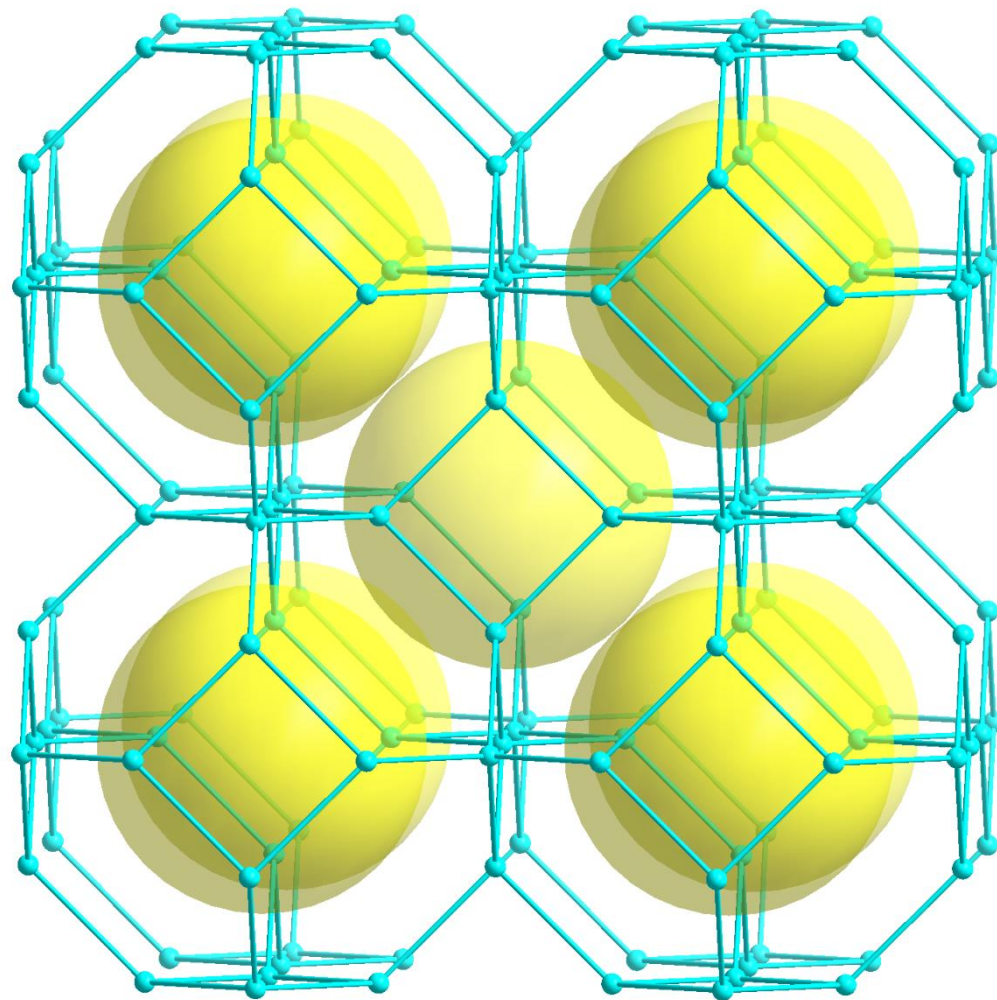
Metal-Organic Frameworks (MOFs)

– Beispiel

Linker



3D Netzwerk



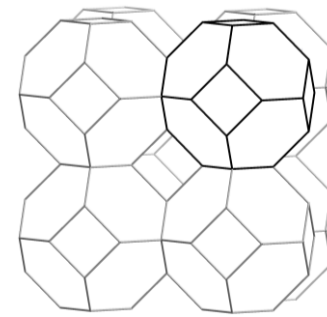
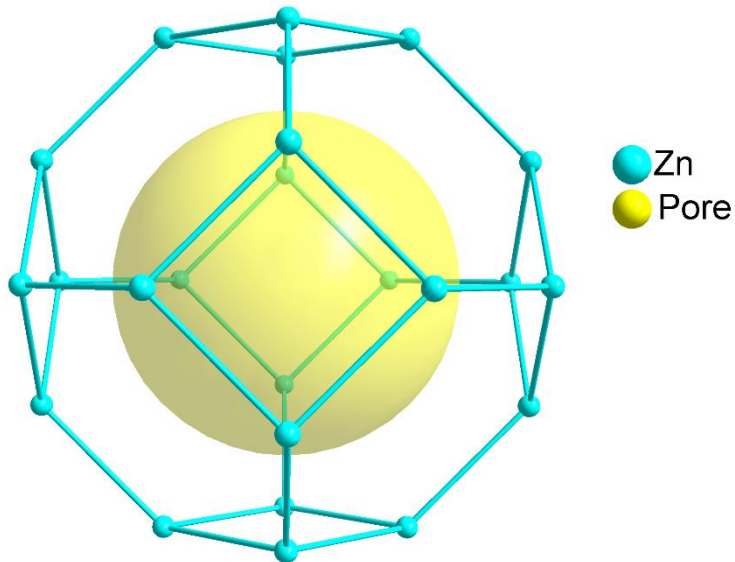
Acronym

ZIF-8

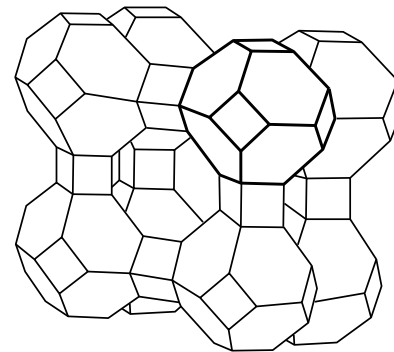
ZIF = zeolitic imidazolate framework

natürliche Zeolithe:

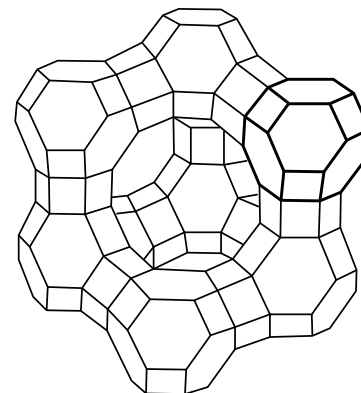
β -Käfig



Sodalith



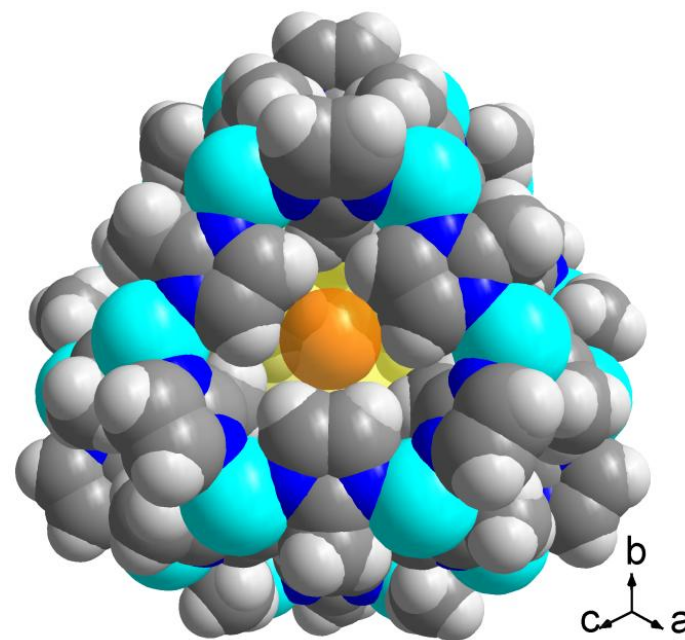
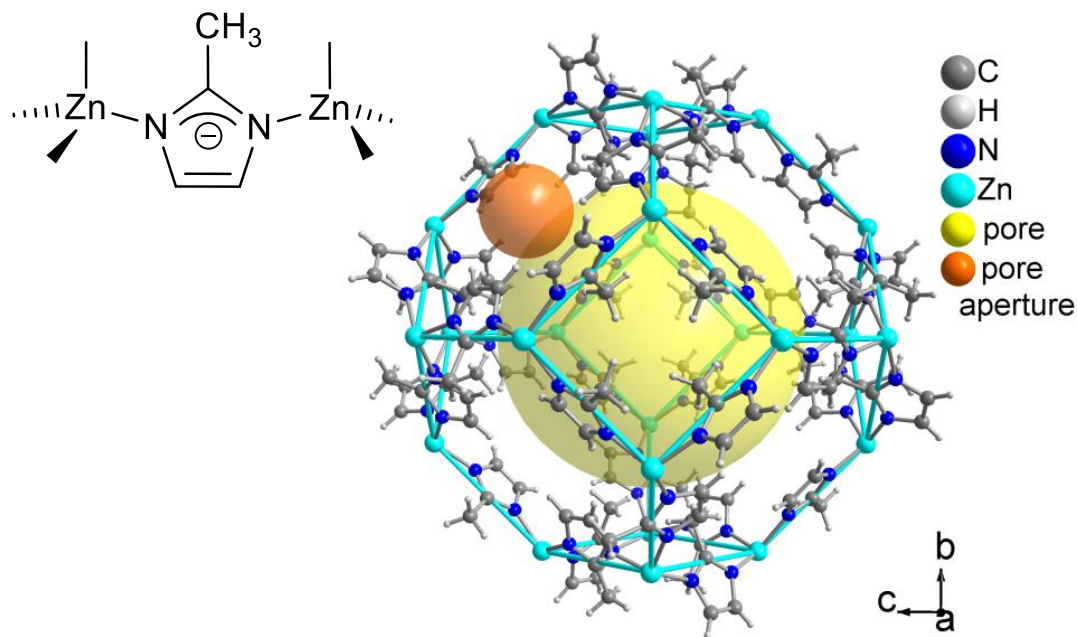
Zeolith A



(c) Faujasit

ZIF = zeolitic imidazolate framework
 $3\text{D}[\text{Zn}(\text{mim})_2] \cdot 2\text{H}_2\text{O} = \text{ZIF-8}$

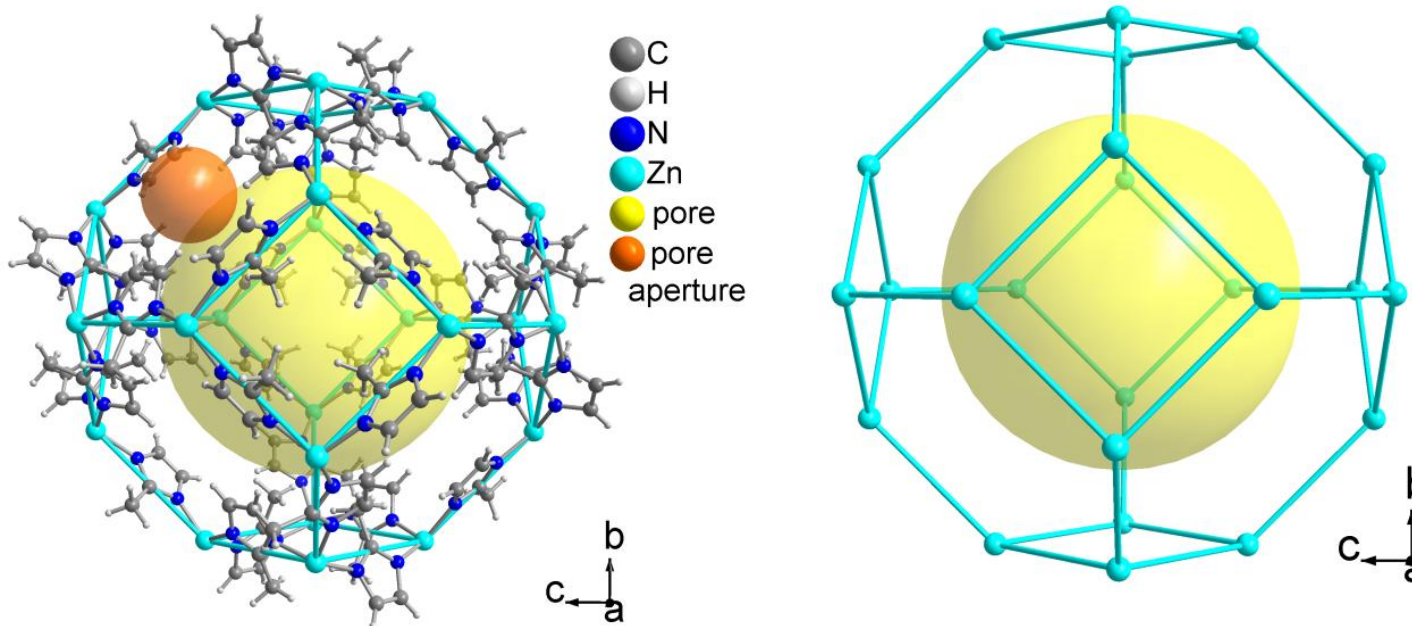
The ZIF-8, $[\text{Zn}(\text{2-methylimidazolate})_2]$ pore aperture has a diameter of 3.4 Å allowing it to readily absorb small molecules such as H_2 and CO_2 . Topologically ZIF-8 and ZIF-90 form sodalite frameworks. ZIF-8 is one of the most hydrolytically stable MOFs.



X.-C. Huang, Y.-Y. Lin, J.-P. Zhang, X.-M. Chen, *Angew. Chem. Int Ed.* **2006**, 45, 1557–1559.

K. S. Park, Z. Ni, A. P. Cote, J. Y. Choi, R. Huang, F. J. Uribe-Romo, H. K. Chae, M. O'Keeffe, O. M. Yaghi, *Proc. Natl. Acad. Sci. U.S.A.* **2006**, 103, 10186–10191.

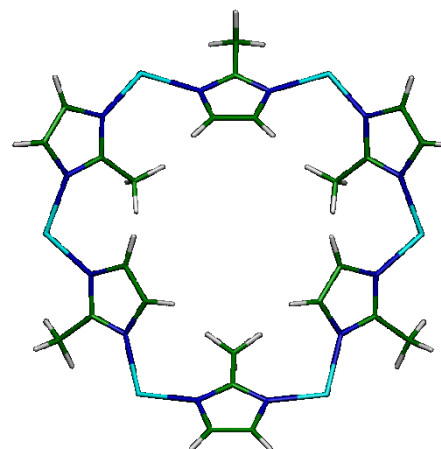
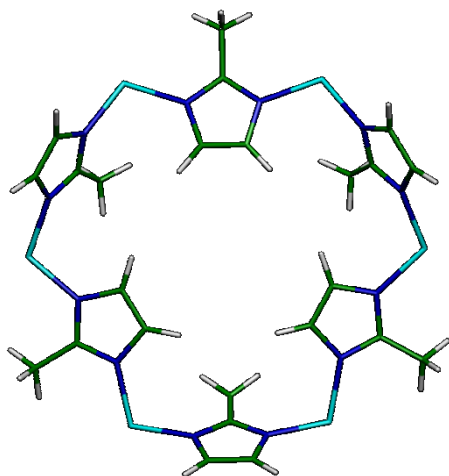
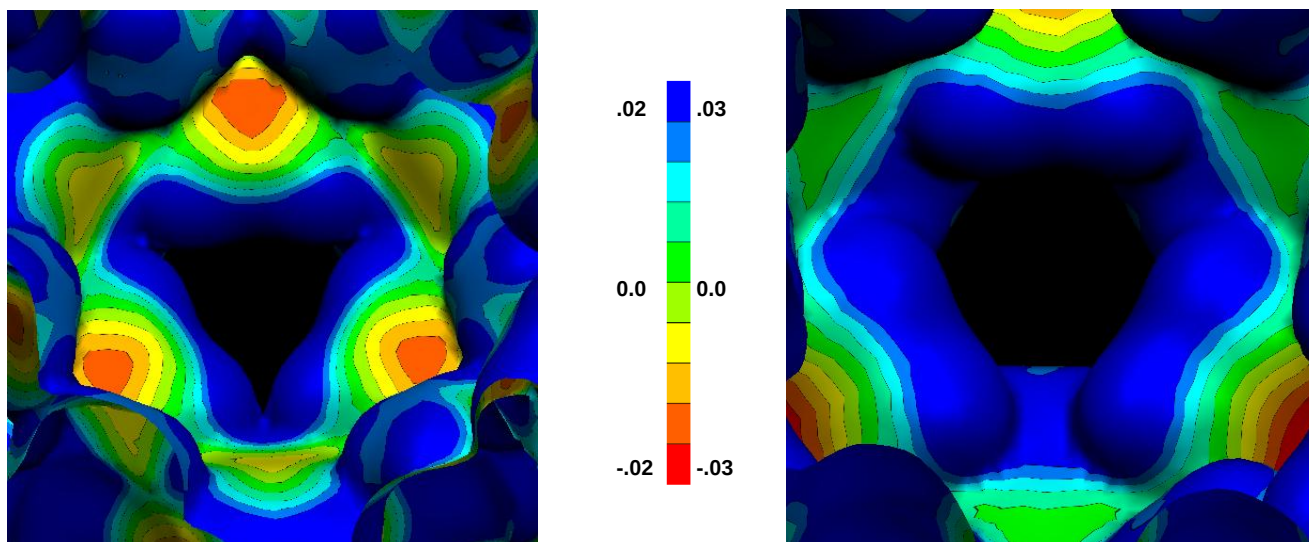
ZIF = zeolitic imidazolate framework
 $3\text{D}[\text{Zn}(\text{mim})_2] \cdot 2\text{H}_2\text{O} = \text{ZIF-8}$



Section of the crystal packing diagram of ZIF-8. 2-Methylimidazolate ligands bridge between single zinc atoms and span the edges of a sodalite cage which is depicted by the blue topological lines connecting the Zn atoms. The **yellow sphere with a diameter of 12 Å** highlights the inner pore of the sodalite cage, the orange sphere with a **diameter of 3.4 Å the pore aperture** window of the six-membered rings. Both spheres take into account the van-der-Waals radii of the framework atoms. BET surface area is about 1600 m²/g.

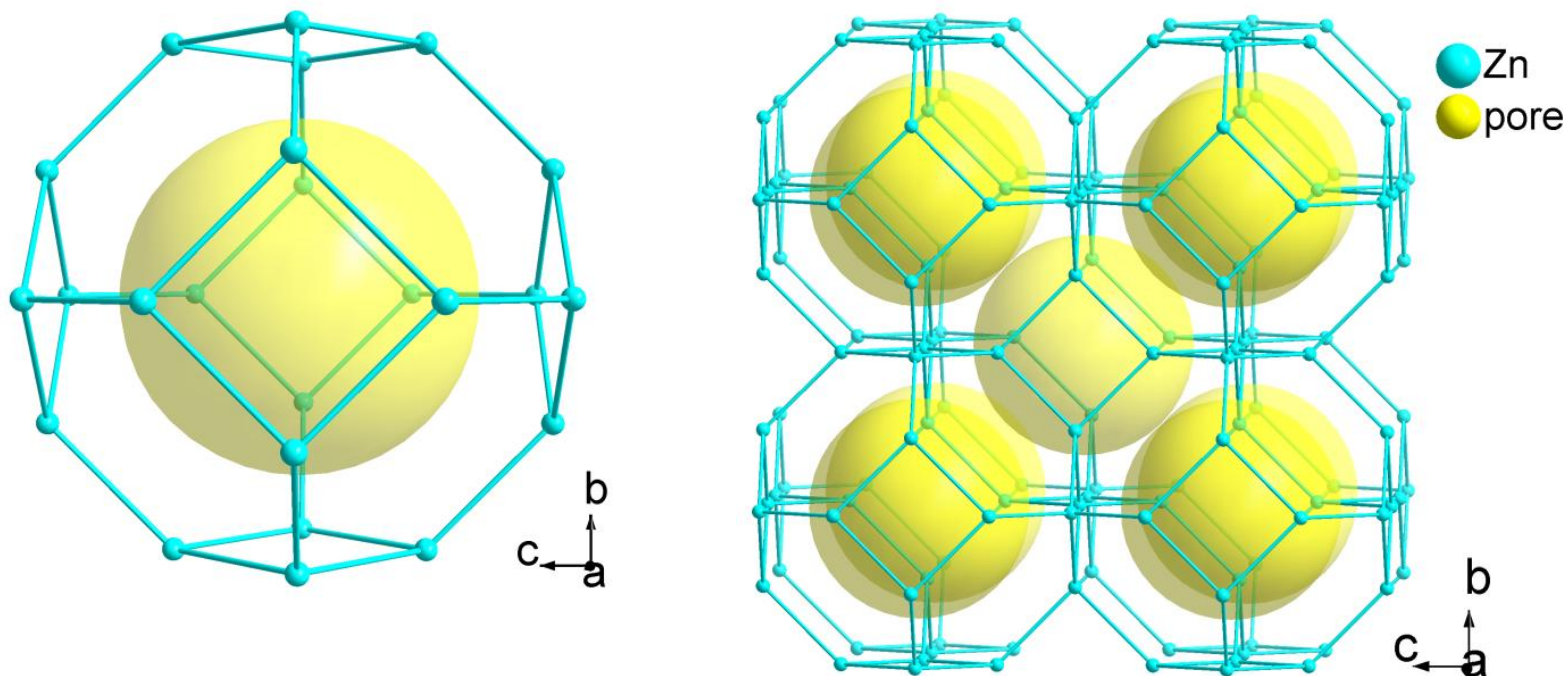
(Electrostatic) framework flexibility in $3\text{D-}[\text{Zn}-(\text{mim})_2] \cdot 2\text{H}_2\text{O} = \text{ZIF-8}$

Charge density distribution study suggests exceptional electrostatic flexibility of the ZIF-8 framework, where small conformational changes lead to a significantly different electrostatic potential (EP) distribution, a feature which could be important for the function and dynamics of the ZIF-8 framework.



S. B. Novaković, G. A. Bogdanović, C. Heering, G. Makhoulfi, D. Francuski, C. Janiak, *Inorg. Chem.* **2015**, 54, 2660-2670. DOI: 10.1021/ic5028256

ZIF = zeolitic imidazolate framework
 $3\text{D}[\text{Zn}(\text{mim})_2] \cdot 2\text{H}_2\text{O} = \text{ZIF-8}$

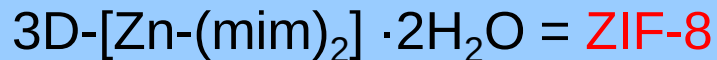


one and nine β -cages - depicted by the blue topological lines connecting the Zn atoms.

X.-C. Huang, Y.-Y. Lin, J.-P. Zhang, X.-M. Chen, *Angew. Chem. Int Ed.* **2006**, 45, 1557–1559.

K. S. Park, Z. Ni, A. P. Cote, J. Y. Choi, R. Huang, F. J. Uribe-Romo, H. K. Chae, M. O'Keeffe, O. M. Yaghi, *Proc. Natl. Acad. Sci. U.S.A.* **2006**, 103, 10186–10191.

Synthesis in microcrystalline form

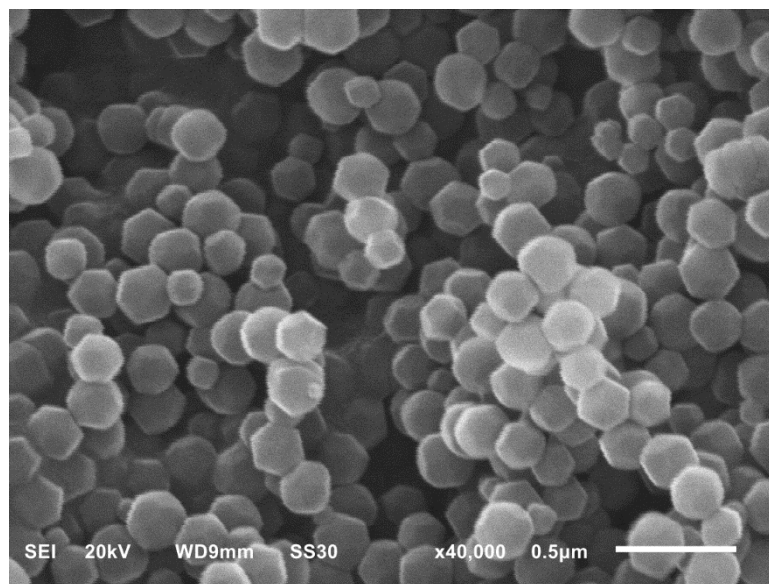
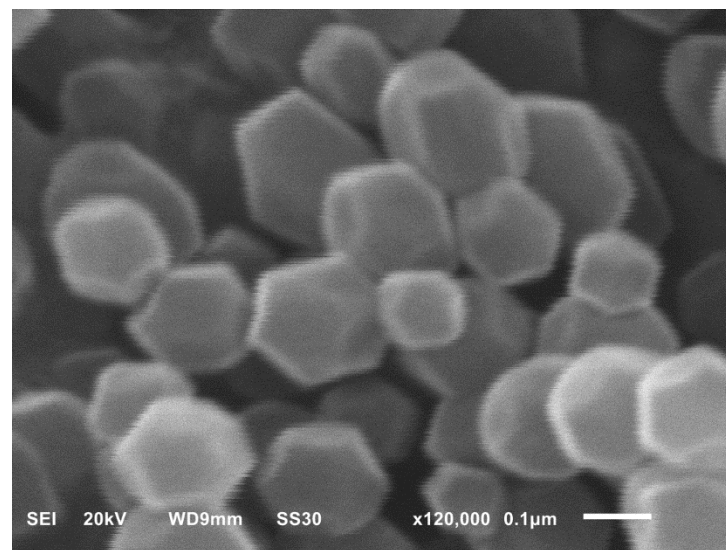
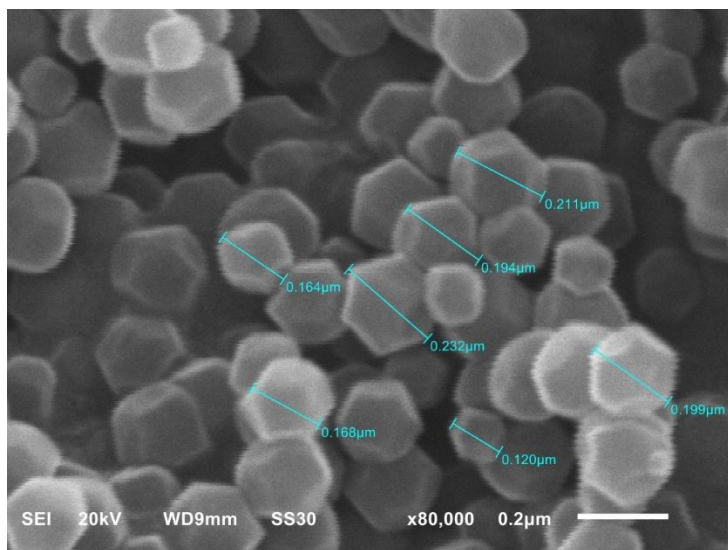


297 mg (0.998 mmol) of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ is placed in a 250 mL Erlenmeyer flask equipped with a magnetic stirrer bar. 20 mL of Methanol is added and the mixture is stirred at r.t. until formation of a clear homogenous solution. In the same way a second solution of 328 mg (4.00 mmol) of 2-Methylimidazol in 20 mL of Methanol is prepared in a 50 mL beaker. The clear second solution is poured quickly under efficient stirring to the Erlenmeyer flask containing the first solution and after complete homogenization the stirring is switched off and the Erlenmeyer flask is closed by the stopper.

After 48 h (note the exact time) the solution with the precipitate is transferred into two centrifuge tubes (ensure that the weight of the tubes is always the same after filling to prevent damage to the centrifuge!) and the formed precipitate is separated from the milky colloidal dispersion by centrifugation. The supernatant is removed by accurate decantation, using a Pasteur pipette for a maximal removal of the solution. The precipitate is washed three times with 10 mL of MeOH by repeated dispersion and centrifugation of the precipitate. The product is dried in vacuum at r.t.

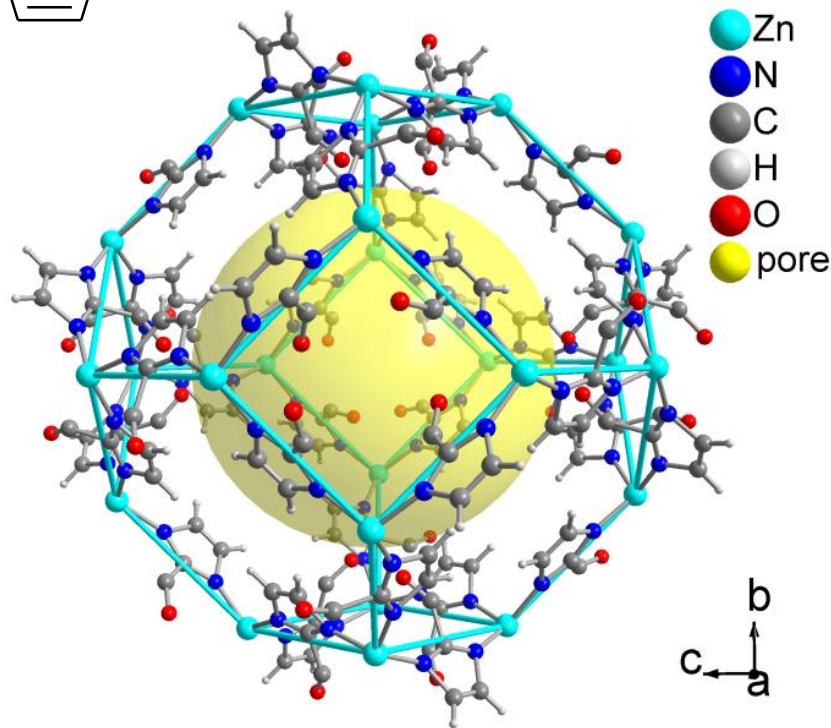
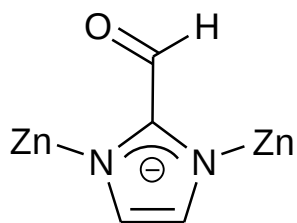
Synthesis in microcrystalline form

$3\text{D}[\text{Zn}-(\text{mim})_2] \cdot 2\text{H}_2\text{O} = \text{ZIF-8}$



ZIF = zeolitic imidazolate framework
 $3\text{D-}[\text{Zn}(\text{2-carboxyaldehyde imidazolate})_2] = \text{ZIF-90}$

$[\text{Zn}(\text{2-carboxyaldehyde imidazolate})_2]$, ZIF-90 has the **same sodalite framework topology as ZIF-8**. Only the 2-methylimidazolate ligand is replaced by a 2-carboxyaldehyde imidazolate ligand



Section of the crystal packing diagram of ZIF-90. 2-Carboxyaldehyde imidazolate ligands bridge between single zinc atoms and span the edges of a sodalite cage which is depicted by the blue topological lines connecting the Zn atoms. The (CHO) H-atom is not shown. The **yellow sphere** with a **diameter of 12 Å** highlights the inner pore of the sodalite cage, the **hexagonal pore aperture** window is as seen in ZIF-8 with a **diameter of 3.4 Å**

MOFs: Prototypical examples

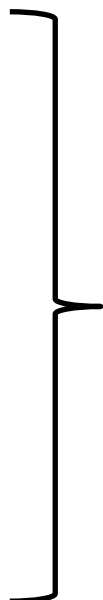
-HKUST-1
-MOF-5
-IR-MOF-*n*

-MIL-53
-Al-fumarate
-MIL-100
-MIL-101
-MIL-125

-UiO-66 (-68)

-CAU-1
-CAU-10-H

-ZIF-8



-di- or tri-carboxylate ligands:
--benzene-1,4-dicarboxylate
--benzene-1,3,5-tricarboxylate

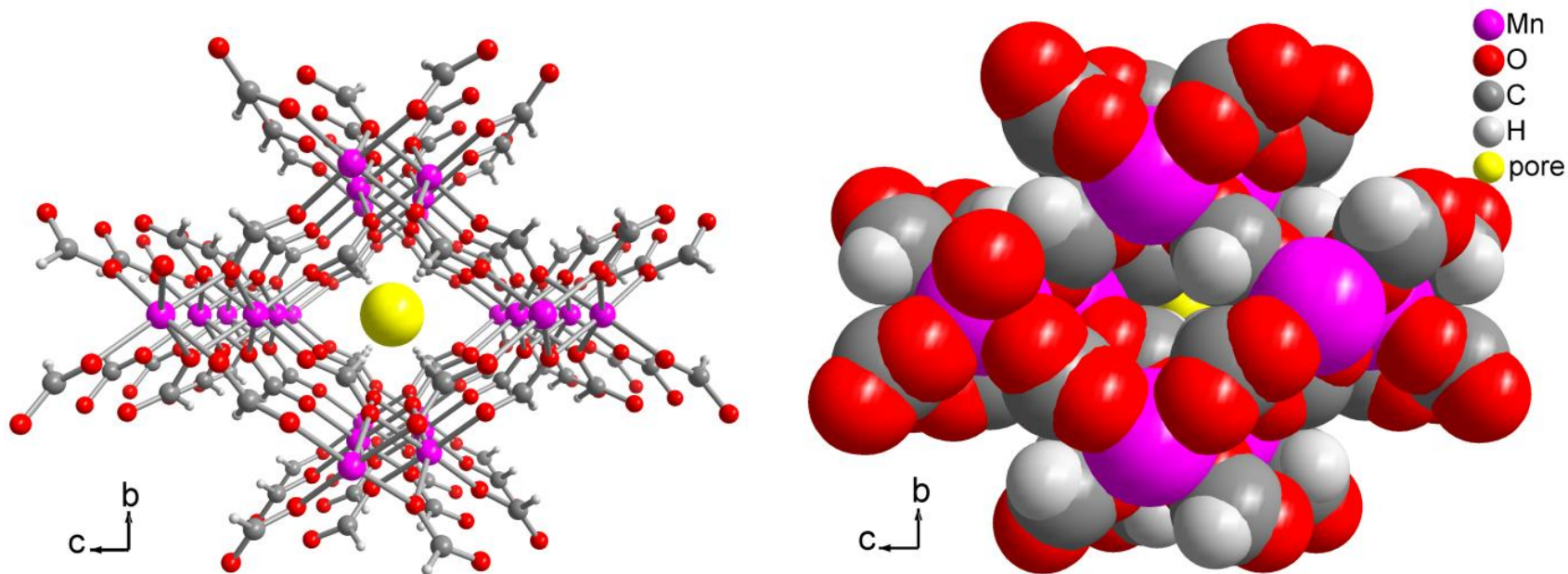
- 2-amino-benzene-1,4-dicarboxylate
- benzene-1,3-dicarboxylate

-imidazolate ligands

other ligands?

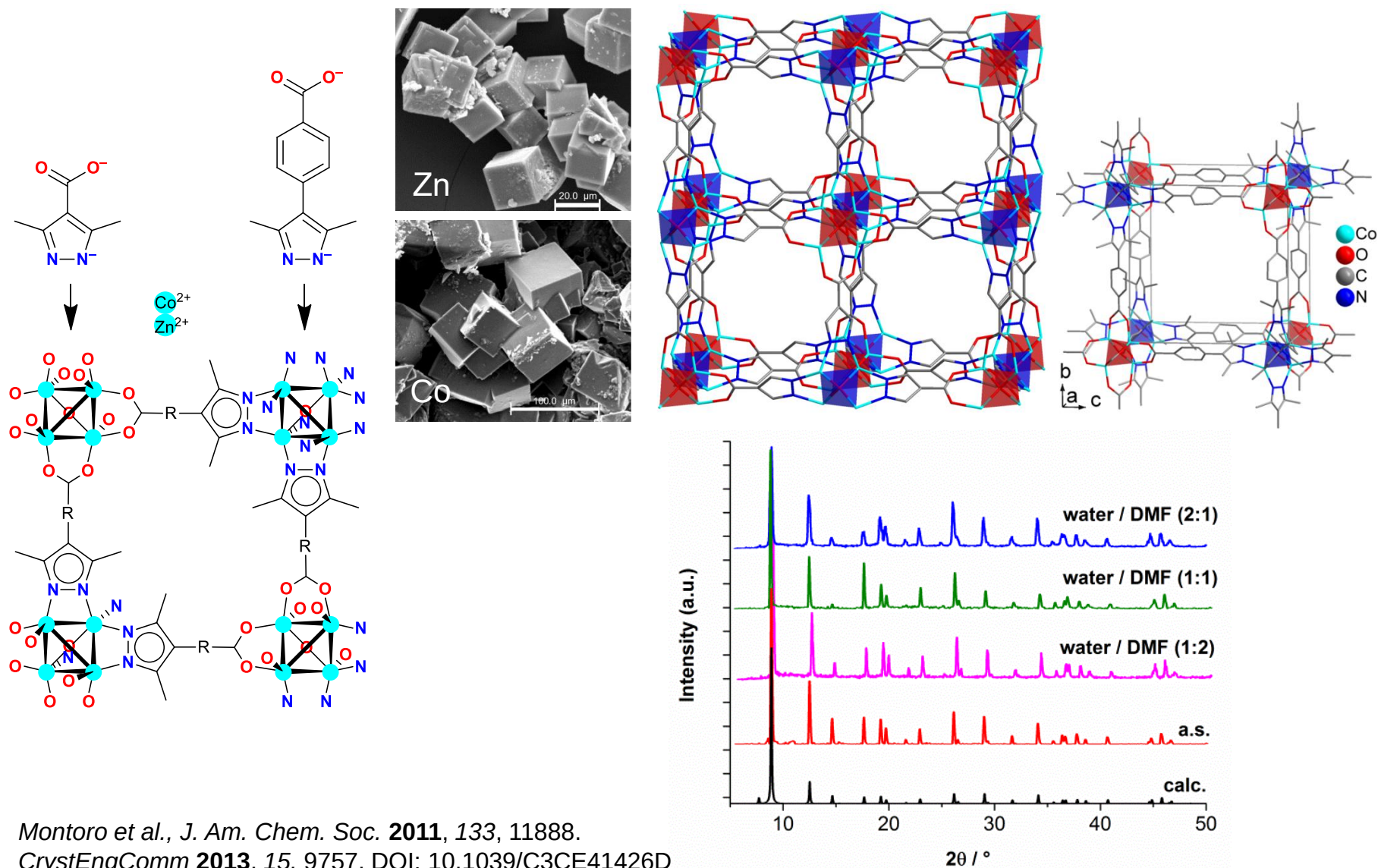
MOF example with **monocarboxylate ligand** $3\text{D}-[\text{Mn}(\text{HCOO})_2]$

- very few examples
- with formiate ligand



Section of the crystal packing diagram of channel network of $\text{Mn}(\text{HCOO})_2$. Each formiate ligand bridges between three manganese atoms. The yellow sphere with a radius of 1 Å takes into account the van-der-Waals radii of the framework walls.

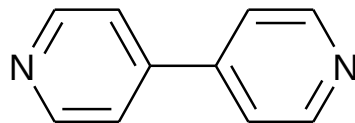
Bifunctional pyrazolate–carboxylate ligands for isorecticular cobalt and zinc MOF-5 analogs



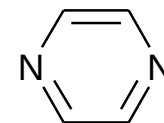
Montoro et al., *J. Am. Chem. Soc.* **2011**, 133, 11888.
CrystEngComm **2013**, 15, 9757. DOI: 10.1039/C3CE41426D

MOFs based on neutral pyridyl ligands?

-neutral ligands



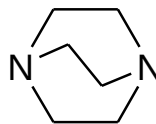
4,4'-bipyridine
4,4'-bipy



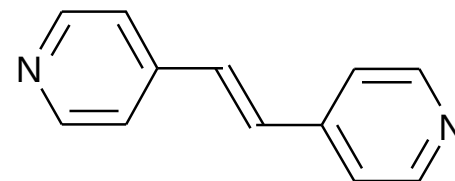
pyrazine
pyz

-nitrogen donor ligands

-pyridine/pyridyl-type ligands

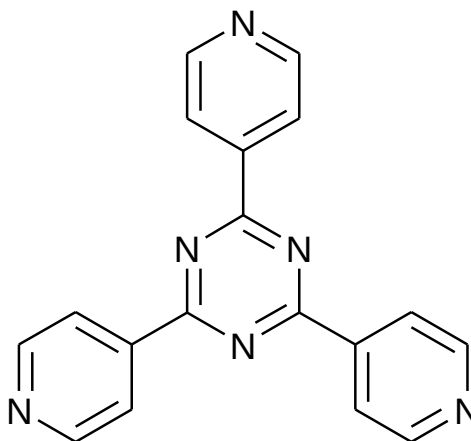


1,4-diaza-bicyclo[2.2.2]octane
dabco



bis(pyridin-4-yl)-1,2-ethene
bpe

-still rigid ligands



2,4,6-tris(4-pyridyl)-1,3,5-triazine
tpt

Electroneutrality of metal-ligand compound

examples:

– organic bridging ligand

– anionic

or

– neutral

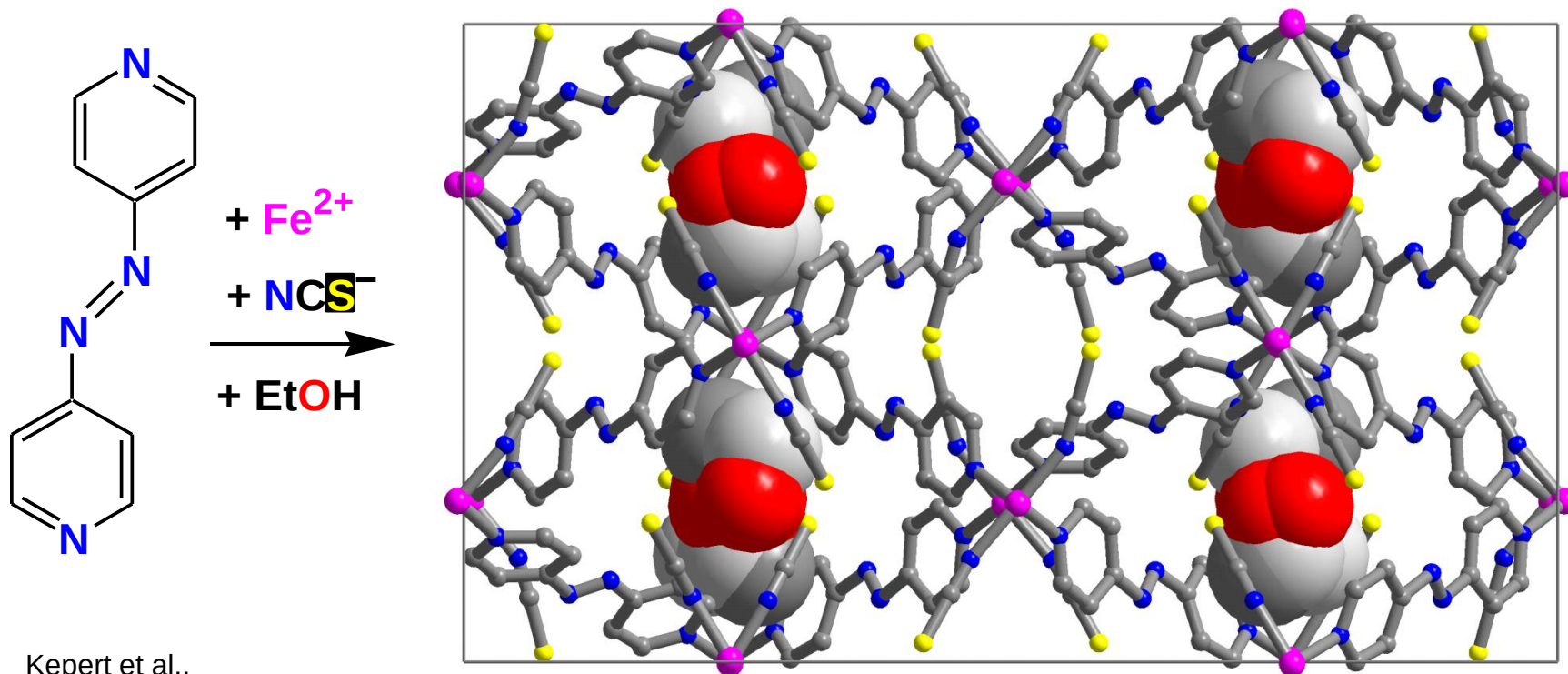
benzene-1,4-dicarboxylate (bdc)

4,4'-bipy, pyrazine

neutral ligand $\Rightarrow$ counter-anions needed for electroneutrality

- terminal anionic ligands on metal
- anions in holes of network
- additional bridging anion ligands

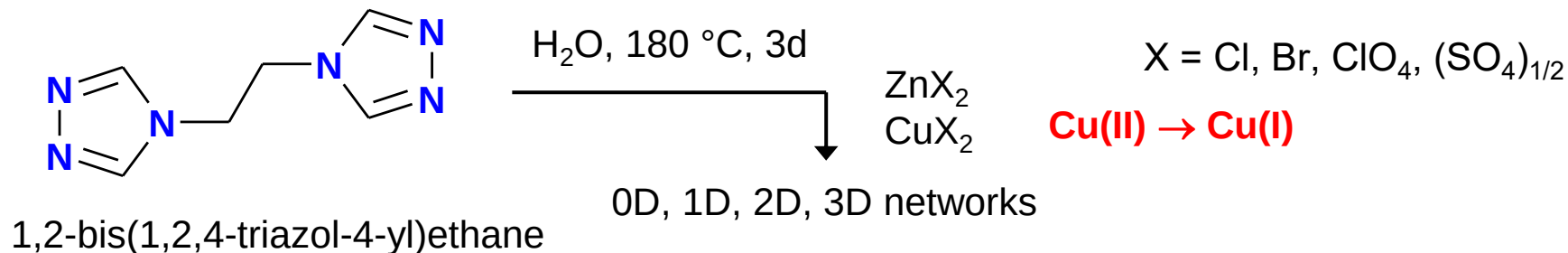
MOF example with neutral ligand - terminal anionic ligand



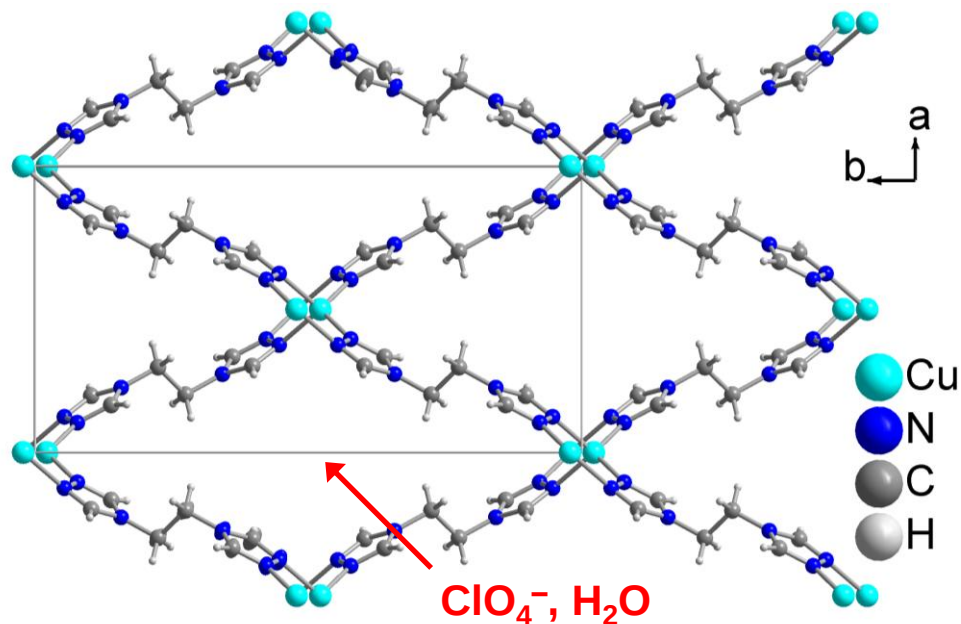
Kepert et al.,
Science **2002**, 298, 1762

Building blocks and packing diagram of the $2\text{D}-[\text{Fe}(\text{NCS})_2(4,4'\text{-azobipyridine})_2] \cdot 1/2\text{EtOH}$ framework. An interpenetration of the 2D nets (running along the viewing direction and differentiated by red and green bonds) creates the 3D MOF-structure. The EtOH guest molecules (in space-filling mode) are disordered. H-atoms are not shown

"MOF" example with neutral ligand - anions in holes of network



Disordered perchlorate ions and crystal water in channels are not shown for clarity.



- no porosity



MOF examples with neutral ligand - additional bridging anionic ligands

= section of **mixed-ligand MOFs**

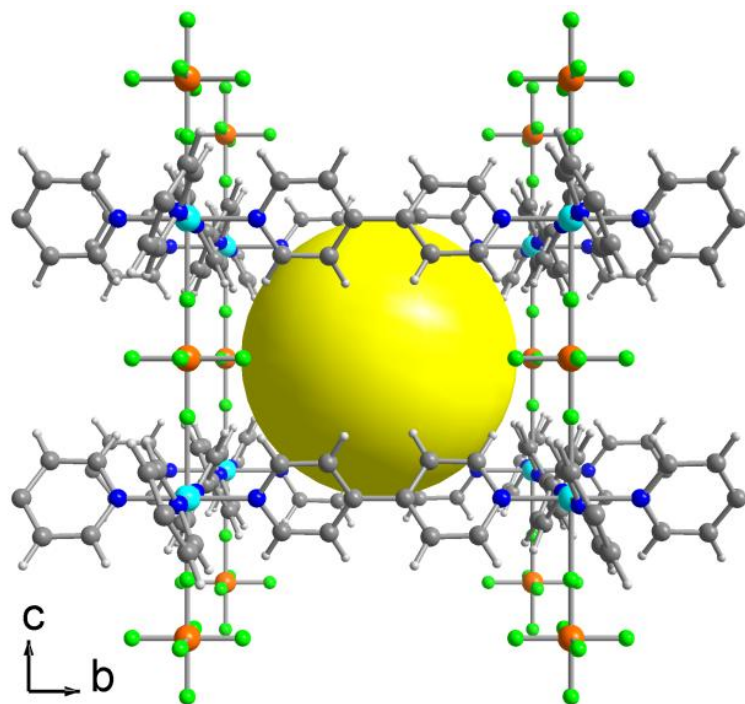
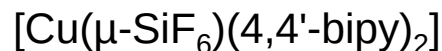
A MOF does not have to contain only one type of linker. Plenty of examples are known where two (or more) linkers build-up the framework structure. The following compounds are examples to this.

At the same time these examples are presented as cases where the MOF contains a neutral organic ligand and, hence, needs an anionic ligand which can also be bridging.

These bridging anionic ligands can be
-inorganic or
-organic.

MOF example with neutral ligand - additional bridging anionic ligands

- mixed-ligand MOF

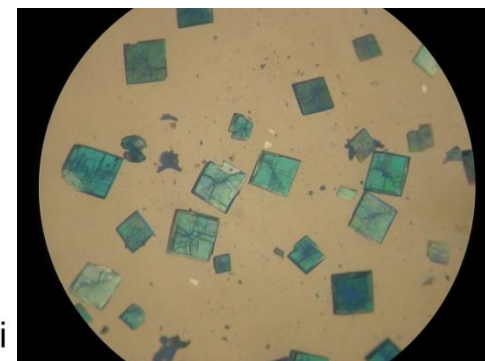
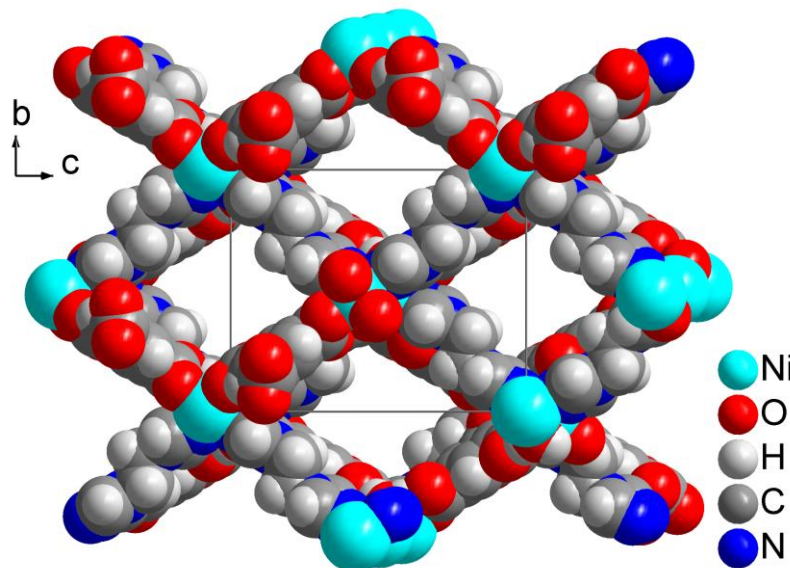
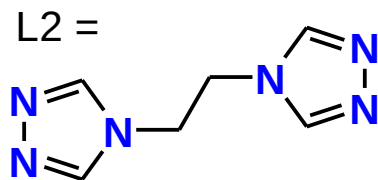
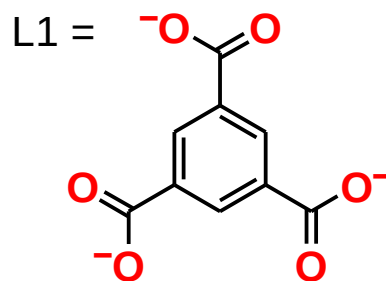


The neutral 3D-framework of $[\text{Cu}(\mu\text{-SiF}_6)(4,4'\text{-bipy})_2] \cdot 8\text{H}_2\text{O}$ (4,4'-bipy = 4,4'-bipyridine) has **square channels of $8 \times 8 \text{ \AA}$** . The three-dimensionality results from bridging of 2D- $\{\text{Cu}(4,4'\text{-bipy})_2\}$ nets by the SiF_6 dianions.

The activated, porous MOF is stable in the absence of the initially-incorporated water molecules of crystallization. Methane adsorption experiments showed that at pressures above 5 bar the framework $[\text{Cu}(\mu\text{-SiF}_6)(4,4'\text{-bipy})_2]$ can **take up significantly more methane than zeolite 5A**, which has the highest methane adsorption capacity of all zeolites.

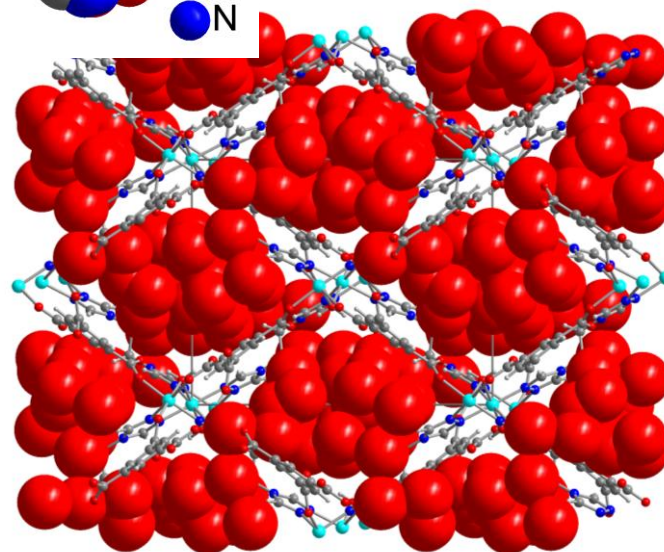
MOF example with neutral ligand - additional bridging anionic ligands

- **mixed-ligand MOF** $3D-\{[Ni_3(\mu_3-L1)_2(\mu_4-L2)_2(\mu-H_2O)_2]\cdot\sim 22H_2O\}$



potential solvent volume
 1621 Å³/unit cell
 = 52%
 (H₂O ~40 Å³)

MOF for reversible
water sorption

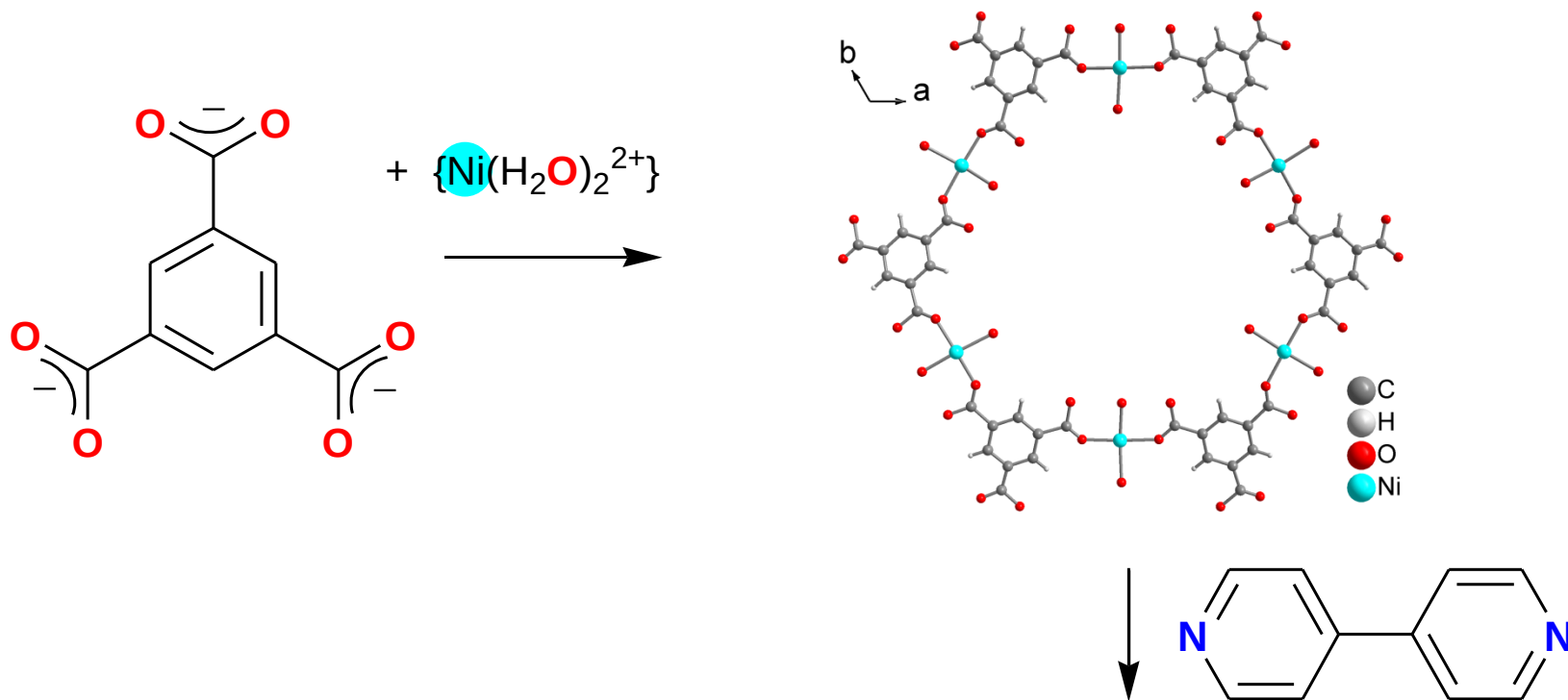


MOF example with neutral ligand

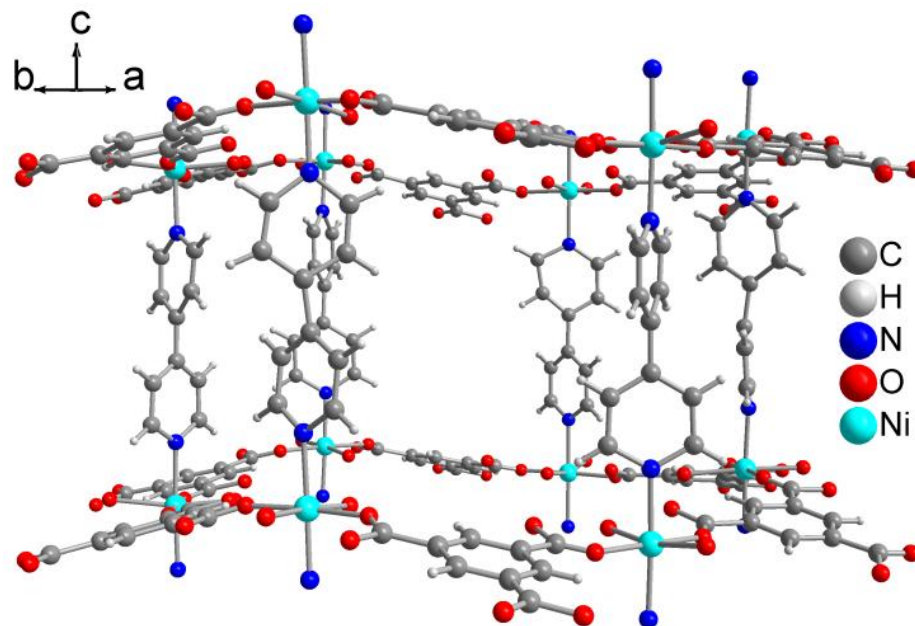
- additional bridging anionic ligands

- mixed-ligand MOF

3D- $[\text{Ni}_2(\text{btc})_4(\text{bipy})_6]$ is an example where benzene-1,3,5-tricarboxylate (btc) and 4,4'-bipyridine construct a MOF with **74% porous volume after solvent-guest removal** (structure is schematically constructed stepwise).

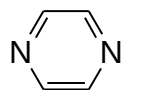


Schematic presentation of the construction of 3D- $[\text{Ni}_2(\text{btc})_4(\text{bipy})_6]$ by two different bridging ligands. The (6,3)-nets from $\{\text{Ni}(\text{H}_2\text{O})_2^{2+}\}$ and benzene-1,3,5-tricarboxylate (btc) lie in the *ab*-plane which are interconnected by 4,4'-bipyridine along the perpendicular *c*-axis. Guest molecules in the voids (74% of cell volume) are not shown.

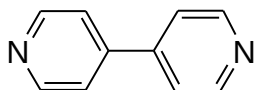


MOFs – mostly based on rigid linkers:

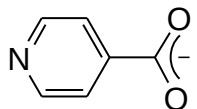
Examples of linear, trigonal and tetrahedral ligands



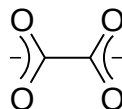
pyrazine



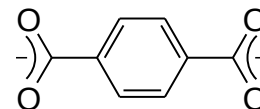
4,4'-bipyridine



isonicotinate



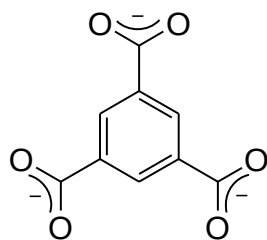
oxalate



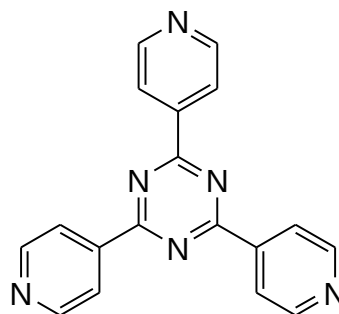
benzene-1,4-dicarboxylate,
terephthalate

linear rigid

(also
linear flexible
possible)



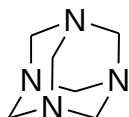
benzene-1,3,5-tricarboxylate,
trimesate



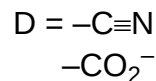
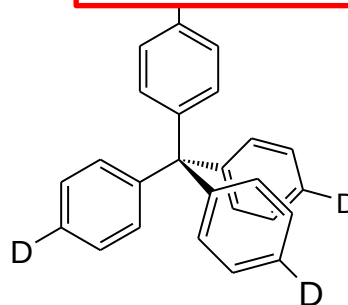
2,4,6-tris(4-pyridyl)-
1,3,5-triazine

flexible linkers?

trigonal



hexamethylene-
tetramine

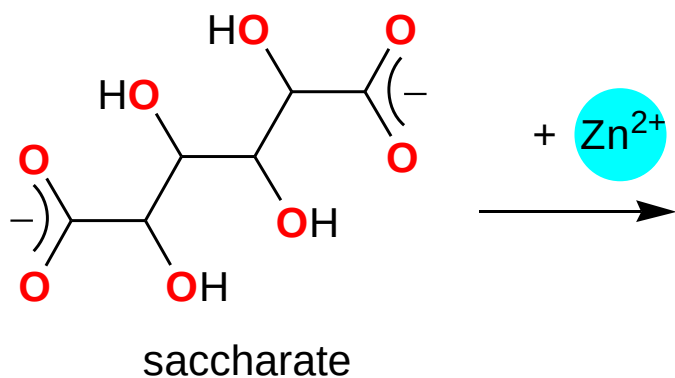


tetra(4-cyanophenyl)methane
tetra(4-carboxyphenyl)methane

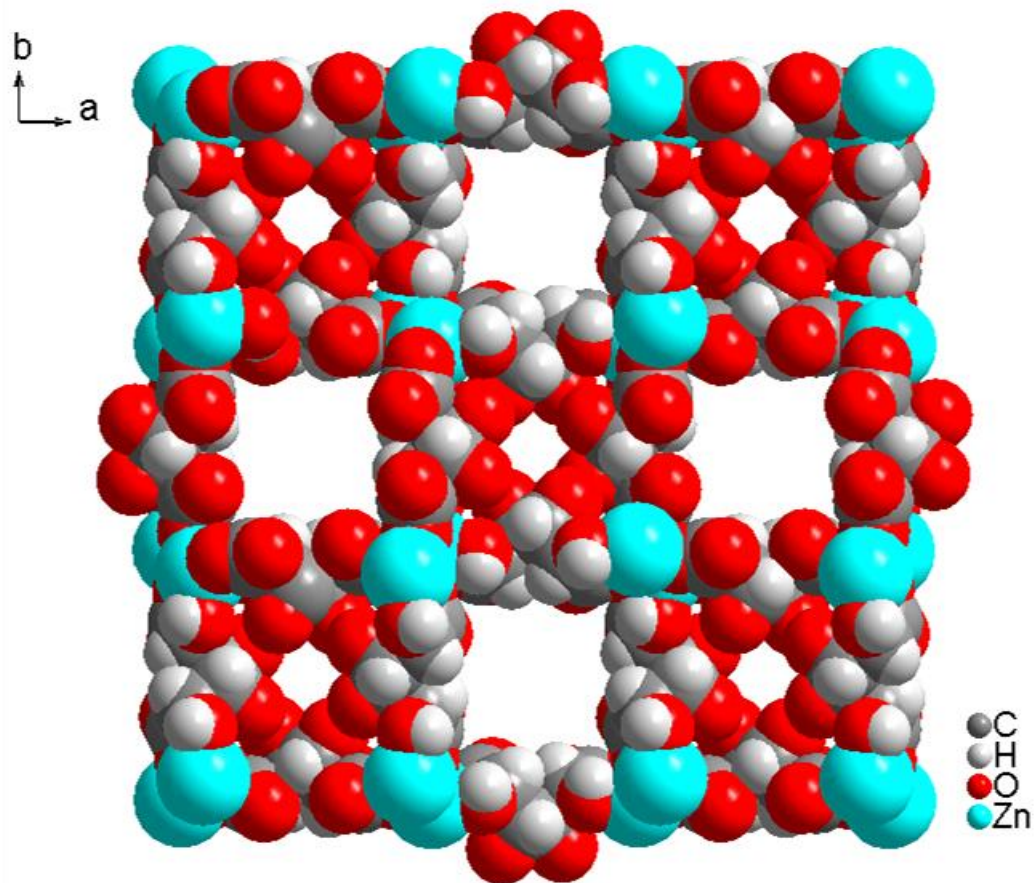
tetrahedral

Oxtoby et al., *PNAS* **2002**, 99, 4905; Kitagawa, Masaoka, *Coord. Chem. Rev.* **2003**, 246, 73;
Barnett, Champness, *Coord. Chem. Rev.* **2003**, 246, 145; Zheng et al., *Coord. Chem. Rev.* **2003**, 246, 185;
Ye et al., *Coord. Chem. Rev.* **2005**, 249, 545; Zhang, Chen, *Chem. Commun.* **2006**, 1689

MOFs with flexible ligands

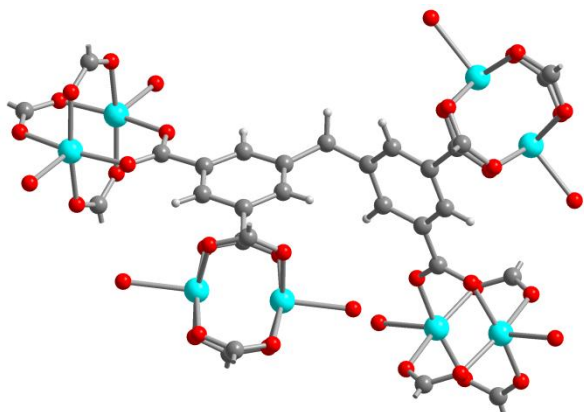


Space-filling illustration of the checker game pattern of 3D- $[\text{Zn}(\mu_4\text{-saccharate})]$ with its alternating **large hydrophobic and small hydrophilic channels**, when viewed along the tetragonal axis. The crystal water in the channels is not shown.

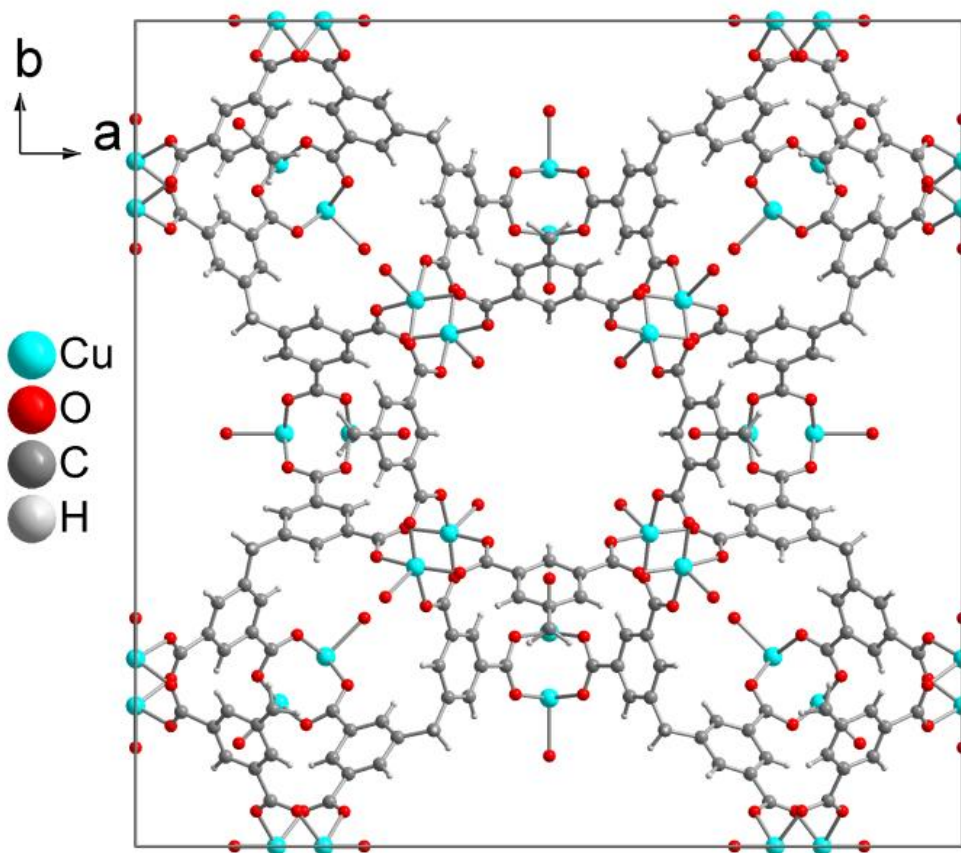


MOFs with flexible ligands

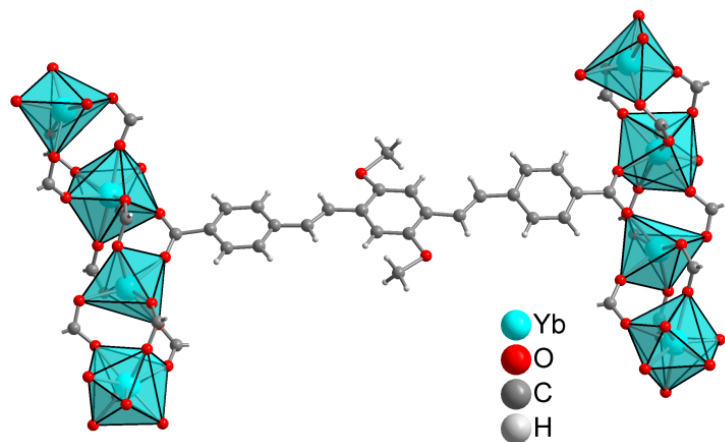
In the MOF 3D-[Cu₆(μ₈-5,5'-methylenedi(isophthalato))(H₂O)₆], termed PCN-12 (PCN=porous coordination network) the Cu₂ paddle-wheel unit as in HKUST-1 (see above) is again found.



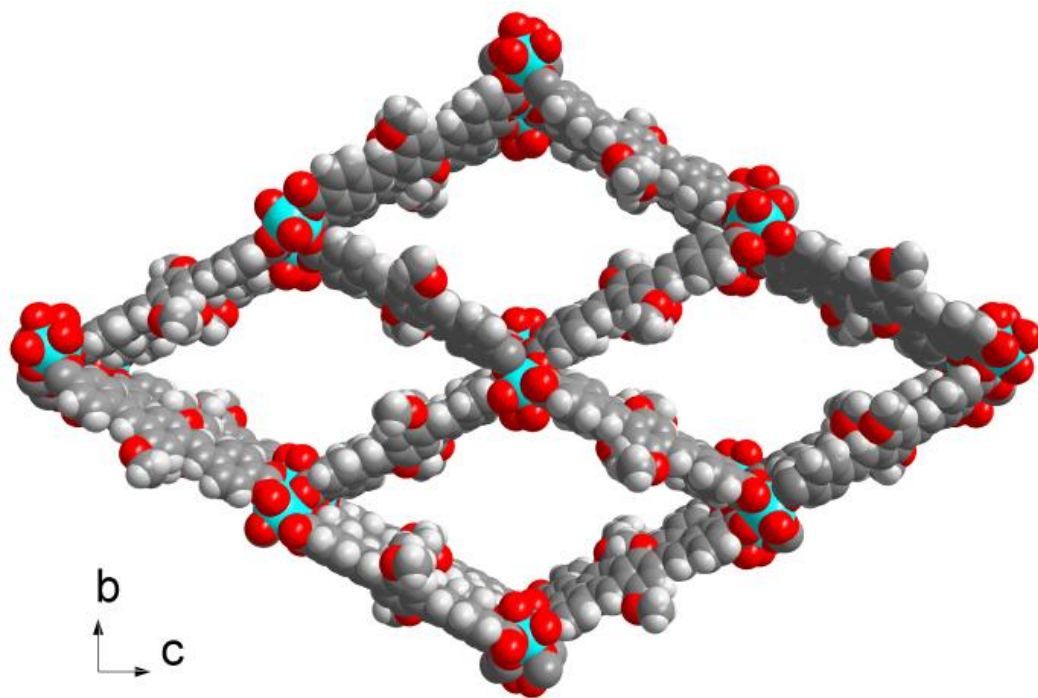
{Cu₂} paddle-wheel SBUs with one μ₈-bridging 5,5'-methylenedi-(isophthalato) ligand and packing diagram with the tetragonal unit cell of (PCN-12, $a = b = 32.87 \text{ \AA}$, $c = 22.63 \text{ \AA}$). DMSO guest molecules are not shown.



MOFs with flexible ligands



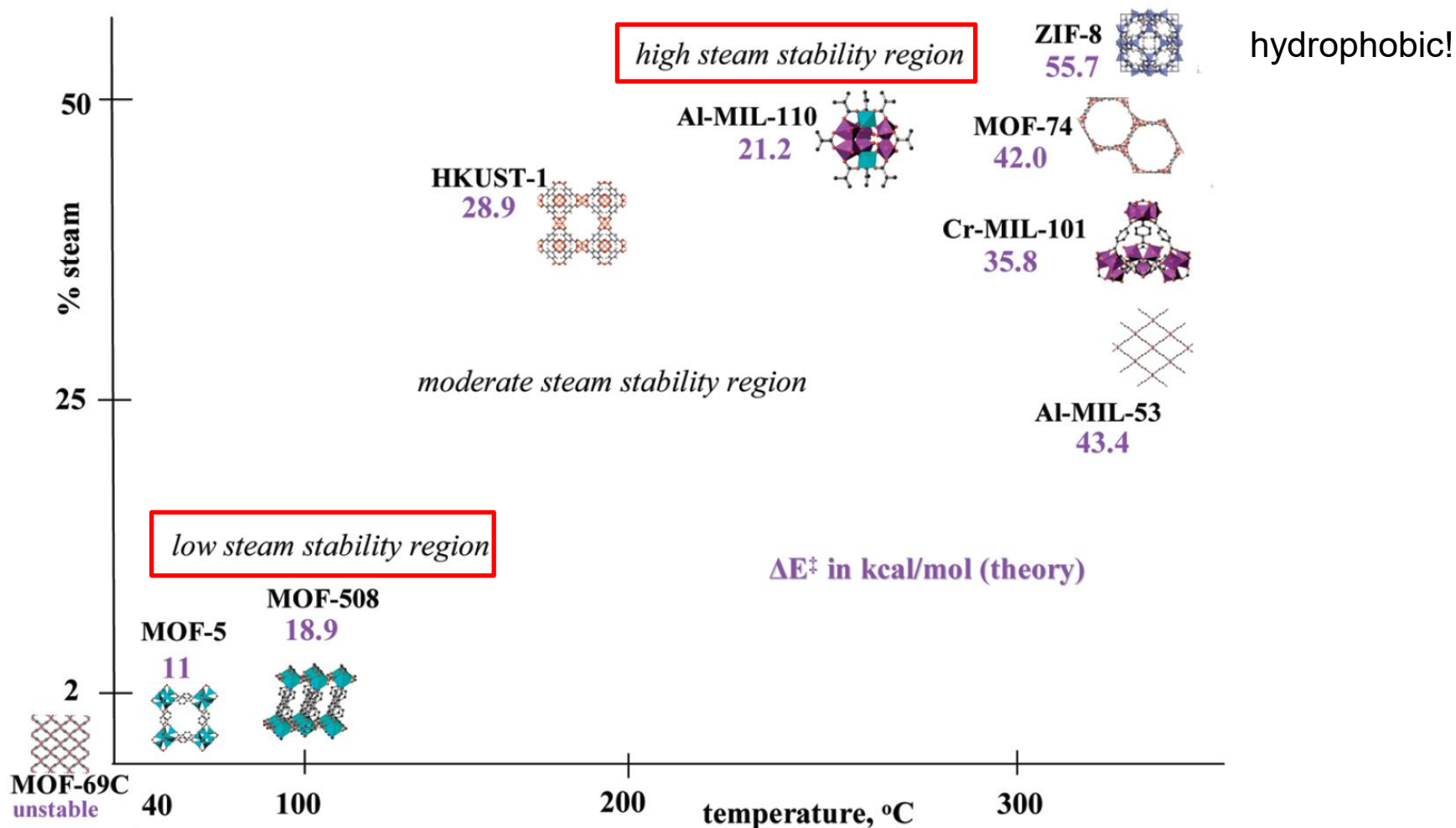
Building unit of 3D-
[Yb₂(pvdc)₃(H₂O)₂]·(dmf)₆(H₂O)_{8.5}
(named Yb-PVDC-1,
pvdc²⁻ = 4,4'-[(2,5-dimethoxy-1,4-
phenylene)di-2,1-ethenediyl]bis-
benzoate, dmf = dimethylformamide)
showing the ligand bridging action
between chains of hexa- and octa-
coordinated Yb³⁺; packing diagram in
space-filling representation of the
luminescent framework (solvent guest
molecules in the channels omitted)



The MOFs have been prepared with two or three
different lanthanoids which emit in the near-infrared.
Variation of the amount of lanthanoid starting salts in
the synthesis tailor the emission spectrum of the
framework. Er_xYb_{1-x}-PVDC-1 emits a unique light
spectrum that reads like a colored barcode.

Remember: Water stability of MOFs

- an inherent problem (as water is almost everywhere)



Remember: Water stability of MOFs

- an inherent problem (as water is almost everywhere)

	3 days												2 months												Steam test	
	pH=0		pH=4		pH=12		H ₂ O ₂		H ₂ O		air		pH=0		pH=4		pH=12		H ₂ O ₂		H ₂ O		air			
	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂
MIL-101 (Cr)																										
NH ₂ -MIL-101 (Al)																										
MIL-53 (Al)																										
NH ₂ -MIL-53		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP
UiO-66																										
NH ₂ -UiO-66																										
UiO-67																										
CuBTC																										
ZIF-8																										

(indicated
MOFs are
described later)

Powder X-ray diffractogram
and surface area, S_{BET}

 completely
 partially
destroyed

K. Leus, T. Bogaerts, J. De Decker, H. Depauw,
K. Hendrickx, H. Vrielinck, V. Van Speybroeck,
P. Van Der Voort,
Micropor. Mesopor. Mater. **2016**, 226, 110-116

Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis - new file
6. Selected applications